

Embryology of the Kidney

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KEY POINTS

- The development of the kidney relies on reciprocal signaling and inductive interactions between neighboring cells.
- Epithelial cells that comprise the tubular structures of the kidney are derived from two distinct cell lineages: the ureteric epithelia lineage that branches and gives rise to collecting ducts and the nephrogenic mesenchyme lineage that undergoes mesenchyme to epithelial transition to form connecting tubules, distal tubules, the loop of Henle, proximal tubules, parietal epithelial cells, and podocytes.
- Nephrogenesis and nephron endowment requires an epigenetically regulated balance between nephron progenitor self-renewal and epithelial differentiation.
- The timing of incorporation of nephron progenitor cells into nascent nephrons predicts their positional identity within the highly patterned mature nephron.
- Stromal cells and their derivatives coregulate ureteric branching morphogenesis, nephrogenesis, and vascular development.
- Endothelial cells track the development of the ureteric epithelia and establish the renal vasculature through a combination of vasculogenic and angiogenic processes.
- Collecting duct epithelia have an inherent plasticity enabling them to switch between principal and intercalated cell identities.

MAMMALIAN KIDNEY DEVELOPMENT

ANATOMIC OVERVIEW OF THE MAMMALIAN KIDNEY

The kidney is a sophisticated, highly vascularized organ that plays a central role in overall body homeostasis. In humans, the kidneys filter as much as 180 liters of blood per day, receiving as much as ~20% of the total cardiac output. Renal filtration of blood removes metabolic waste products (e.g., urea, ammonia, and by-products of bile from the liver) as urine while concomitantly adjusting the levels of water, electrolytes, and pH of tissue fluids. Additionally, the kidneys regulate blood pressure via the renin-angiotensin-aldosterone system, secrete erythropoietin that stimulates erythrocyte production, and contribute to the activation of vitamin D to control calcium and phosphate balance.

The filtration function of the kidneys is accomplished by basic units called nephrons (Fig. 1.1). Humans on average have 1 million nephrons per adult kidney but the range of total nephrons is highly variable across human populations.⁴ Each mouse kidney may contain up to 12,000–16,000 nephrons depending on the strain.⁵ This wide range in nephron number is influenced by genetic background, fetal nutrition and environment, and maturity at birth.^{6,7} Nephron endowment can be clinically important as markedly reduced nephron numbers raises the susceptibility risk to hypertension and chronic kidney disease.^{1–3,8,9} At the core of the nephron is the renal corpuscle or glomerulus (see Fig. 1.1). The glomerulus consists of a porous and highly convoluted capillary bed composed of highly fenestrated glomerular endothelial cells. These glomerular capillaries are circumscribed by morphologically elaborate and interdigitating cells called podocytes. These capillaries are further structurally supported by pericytes

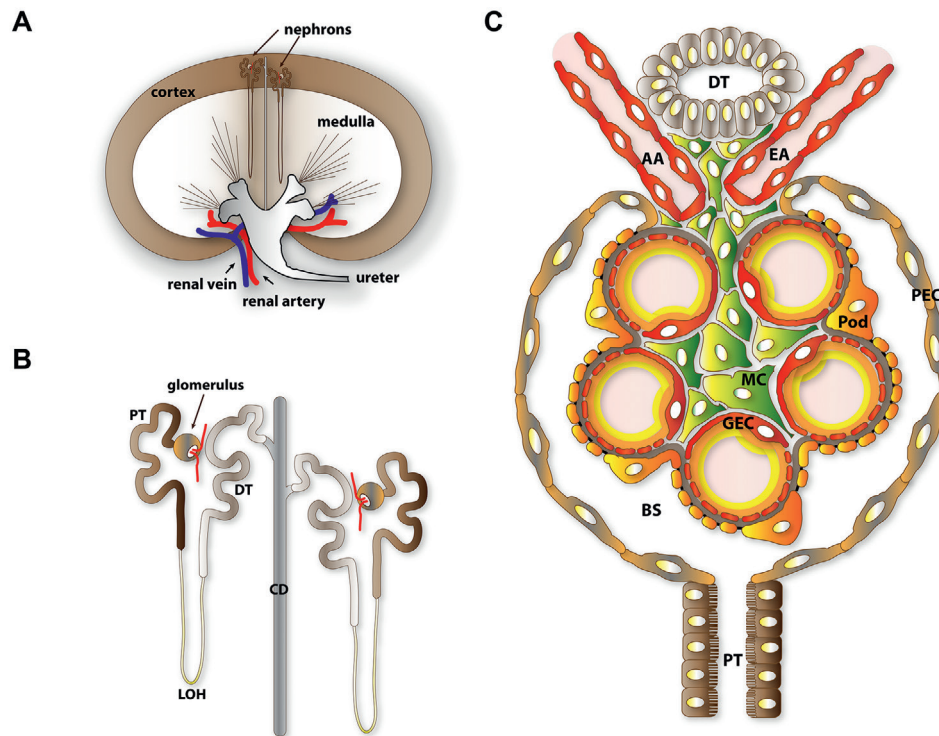


Fig. 1.1 Anatomic organization of the kidney. (A) Spatial distribution of nephron within the metanephric kidney. Glomeruli, the filtration compartments of the nephrons, are found in the cortex. (B) Segmental structure of nephrons. The vascularized glomerulus is found at the proximal end and is connected through a series of renal tubules where urinary filtrate composition is refined through resorption and secretion. (C) Cellular organization of the glomeruli. AA, Afferent arteriole; BS, Bowman space; CD, Collecting duct; DT, distal tubule; EA, efferent arteriole; GEC, glomerular endothelial cell; LOH, loop of Henle; MC, mesangial cell; PEC, parietal epithelial cell; Pod, podocyte; PT, proximal tubule. Reproduced with permission from Scott RP, Quaggin SE. The cell biology of renal filtration. *J Cell Biol.* 2015;209:100–210.

called mesangial cells. Blood filtration occurs through this capillary tuft, generating primary urine that collects within the Bowman capsule, an enclosure formed by parietal epithelial cells. From the Bowman capsule, urine drains through a series of tubules starting with the proximal tubules, the loop of Henle, the distal tubules, and the collecting ducts. These tubules are responsible for dynamic resorption and secretion processes that help recycle filtered small molecules; they also adjust water, electrolyte, and acid–base balance by fine-tuning the composition of the final urine output before it exits the ureter and is excreted via the bladder. Supporting the main functions of the nephrons are interstitial fibroblasts and a heterogeneous network of extraglomerular vasculature.

DEVELOPMENT OF THE UROGENITAL SYSTEM

The vertebrate kidney derives from the intermediate mesoderm of the urogenital ridge, a structure found along the posterior wall of the abdomen in the developing fetus.^{10,11} Mammalian kidneys develop in three successive stages, generating three distinct excretory structures known as the pronephros, the mesonephros, and the metanephros (Fig. 1.2). The pronephros and mesonephros are vestigial structures in mammals and degenerate before birth; the metanephros is the definitive mammalian kidney. The early stages of kidney development are required for the development of the adrenal glands and gonads that also form within the urogenital ridge. Furthermore, many of the signaling pathways and genes that play important roles in the metanephric kidney appear to play parallel roles during the development of the pronephros

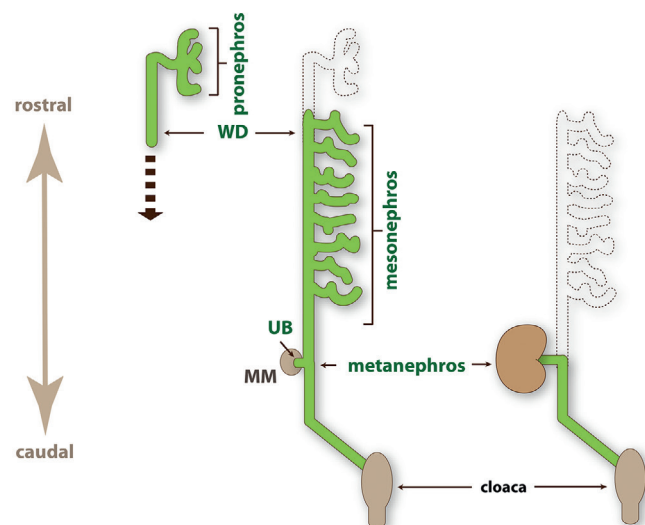


Fig. 1.2 Three stages of mammalian kidney development. The pronephros and mesonephros develop in a rostral-to-caudal direction and the tubules are aligned adjacent to the wolffian or nephric duct (WD). The metanephros develops from an outgrowth of the distal end of the wolffian duct known as the ureteric bud epithelium (UB) and a cluster of cells known as the metanephric mesenchyme (MM). The pronephros and mesonephros are vestigial structures in mice and humans and are regressed by the time the metanephros is well developed.

and mesonephros. The pronephros consists of pronephric tubules and the pronephric duct (also known as the precursor to the wolffian duct) and develops from the rostral-most region of the urogenital ridge at 22 days of gestation in humans and 8 days post coitum (embryonic stage E8) in mice (Table 1.1). Throughout the rest of this chapter, most timelines of kidney development are with reference to the mouse. The pronephros serves as the principal excretory organ of the larval stages of fishes and amphibians. The mesonephros develops caudal to the pronephric tubules in the midsection of the urogenital ridge. The mesonephros becomes the functional excretory apparatus in lower vertebrates (adult fish and amphibians) and may perform a filtering function during embryonic life in mammals. Prior to its degeneration, endothelial, peritubular myoid, and steroidogenic cells from the mesonephros migrate into the adjacent adrenogonadal primordia, which ultimately form the adrenal gland and gonads.¹² Abnormal mesonephric migration leads to gonadal dysgenesis, a fact that underscores the intricate association between these organ systems during development and explains the common association of gonadal and renal defects in congenital syndromes.^{13,14}

DEVELOPMENT OF THE METANEPHROS

The metanephros is the third and final stage, representing the definitive adult kidney of higher vertebrates. It results from a series of reciprocal inductive interactions that occur between the metanephric mesenchyme (MM) and

the epithelial ureteric bud (UB) at the caudal end of the urogenital ridge. The UB is first visible as an outgrowth at the distal end of the wolffian duct approximately between the fourth and fifth week of gestation in humans or E10.5 in mice. The MM becomes histologically distinct from the surrounding mesenchyme and is found adjacent to the UB. Upon invasion of the MM by the UB, signals from the MM cause the UB to branch into a T-tubule (at around E11.5 in mice) and then to undergo iterative dichotomous branching, giving rise to the urinary collecting duct system (Fig. 1.3). Simultaneously, the UB sends reciprocal signals to the MM, which is induced to condense along the surface of the bud. Following condensation, a subset of MM cells aggregates adjacent and inferior to the tips of the branching UB. These collections of cells, known as pretubular aggregates, undergo mesenchymal-to-epithelial conversion to become the renal vesicle (Fig. 1.4).

URETERIC BRANCHING MORPHOGENESIS

The collecting duct system is composed of hundreds of tubules through which the filtrate produced by the nephrons is conducted out of the kidney, to the ureter, and then to the bladder. Water and salt resorption and excretion, ammonia transport, and H^+ ion secretion required for acid–base homeostasis also occur in the collecting ducts, under different regulatory mechanisms, and using different transporters and channels than are active along tubular portions of the nephron. The collecting ducts are all derived from the original

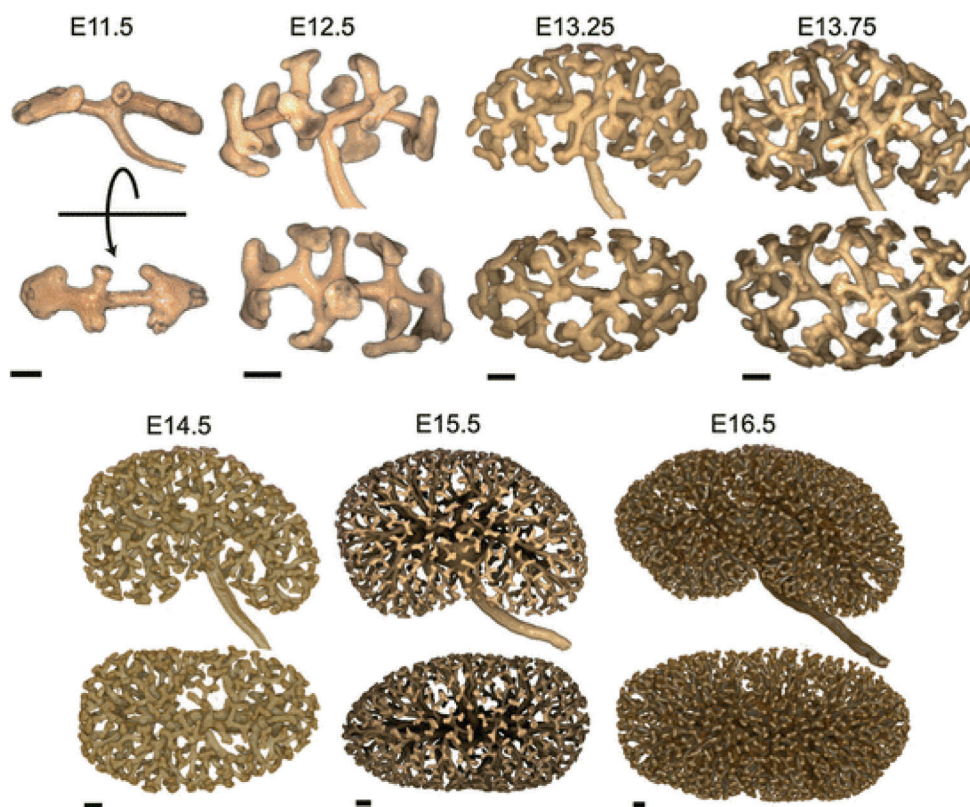


Fig. 1.3 Ureteric branching morphogenesis. Rapid reiterative branching of the UB within a 5-day period in mice as imaged with a pan-cytokeratin antibody using optical projection tomography. By E16.5, the renal pelvis, formed by the widening and coalescence of the earliest branches of the ureteric tree, is already apparent. Reproduced with permission from Short KM, Smuth I. Imaging, analyzing and interpreting branching morphogenesis in the developing kidney. *Results Probl Cell Differ.* 2017;60:233–256.

Table 1.1 Timelines of Human and Mouse Kidney Development

Stage/Event	Human ^a	Mouse ^b
Pronephros		
• Emergence	22nd day	E9
• Disappearance by	25th day	E10
Mesonephros		
• Emergence	24th day	E10
• Disappearance by	16th week	E14
Metanephros		
• Ureteric bud induction	28th–32nd day	E10.5
• Nephrogenesis	44th day	E13
• Glomerulogenesis	8th–9th week	E14
• Cessation of nephrogenesis	36th week	P3
Gestation (Total Length)	40 weeks	19–21 days

^aHuman timelines refer to gestational periods.

^bMouse timelines are indicated as either embryonic days post coitum (E) or postnatal (P).

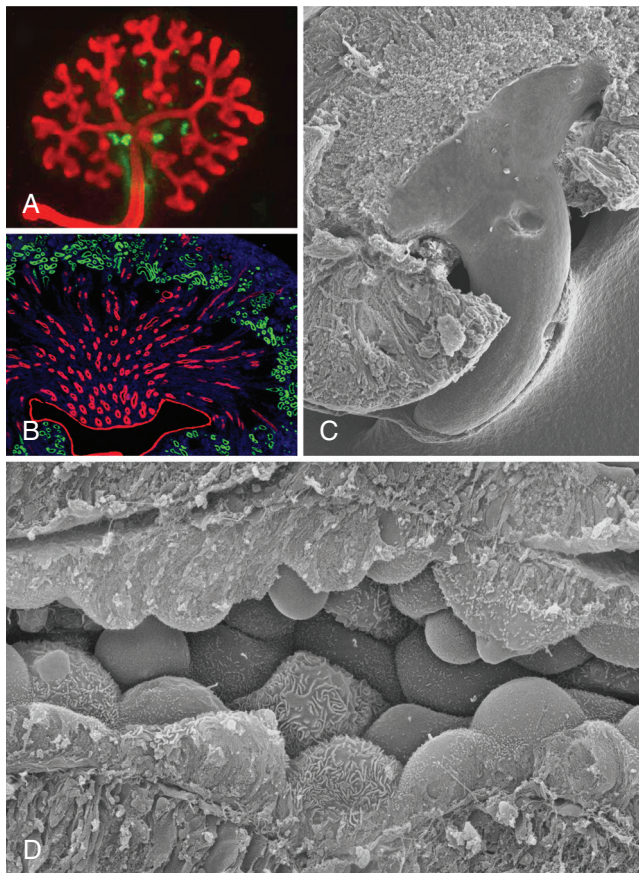


Fig. 1.4 The collecting duct system. The branching ureteric epithelia gives rise to the collecting duct system. (A) E12.5 mouse embryonic kidney explant grown for 2 days and (B) neonatal mouse kidney section stained for the ureteric epithelium and collecting ducts (pan-cytokeratin, red) and proximal tubules (Lotus lectin, green). (C) Scanning electron micrograph of a hemisected adult mouse kidney showing the funnel-shaped renal papilla. (D) Scanning electron micrograph of a collecting duct showing smooth principal cells and reticulated intercalated cells.

UB (Fig. 1.5). Whereas each nephron is an individual unit separately induced and originating from a distinct pretubular aggregate, the collecting ducts are the product of branching morphogenesis from the UB. Considerable remodeling is involved in forming collecting ducts from branches of the UB.¹⁵ The branching is highly patterned, with the first several rounds of branching being somewhat symmetric, followed by additional rounds of asymmetric branching, in which a main trunk of the collecting duct continues to extend toward the nephrogenic zone, while smaller buds branch as they induce new nephrons within the nephrogenic zone. Originally, the UB derivatives are branching within a surrounding mesenchyme. Ultimately, they form a funnel-shaped structure in which cone-shaped groupings of ducts or papillae sit within a funnel or calyx that drains into the ureter. The mouse kidney has a single papilla and calyx, whereas a human kidney has 8 to 10 papillae, each of which drains into a minor calyx, with several minor calyces draining into a smaller number of major calyces.

DEVELOPMENT OF THE NEPHRON

The renal vesicle undergoes patterned segmentation and proceeds through a series of morphologic changes that include gradual recruitment of mesenchymal progenitors to form the glomerulus and components of the nephrogenic tubules from the proximal convoluted tubule, the loop of Henle, and the distal tubule. The renal vesicles undergo differentiation, passing through morphologically distinct stages starting from the comma-shaped and proceeding to the S-shaped body, capillary loop, and mature stage, each step involving precise proximal-to-distal patterning and structural transformations (see Fig. 1.4). Remarkably, this process is repeated 600,000 to 1 million times in each developing human kidney as new nephrons are sequentially born at the tips of the UB throughout fetal life.

The glomerulus develops from the most proximal end of the renal vesicle that is furthest from the UB tip.^{16,17} Distinct cell types of the glomerulus can first be identified in the S-shaped stage, where presumptive podocytes appear as a columnar-shaped epithelial cell layer. A vascular cleft develops and separates the presumptive podocyte layer from more distal cells that will form the proximal tubule. Parietal epithelial cells differentiate and flatten to form the Bowman capsule, a structure that surrounds the urinary space and is continuous with the proximal tubular epithelium. Concurrently, endothelial cells migrate into the vascular cleft. Together with podocytes, the endothelial cells produce the glomerular basement membrane, a major component of the mature filtration barrier. Initially the podocytes are connected by intercellular tight junctions at their apical surface.¹⁸ As glomerulogenesis proceeds, the podocytes flatten and spread out to cover the increased surface area of the growing glomerular capillary bed. They develop microtubular-based primary processes and actin-based secondary foot processes.^{19–21} Foot processes of neighboring podocytes interdigitate and elongate. As podocytes mature, intercellular epithelial tight junctions linking become restricted to the basal aspect of the podocyte, relocate from the cell body to the foot processes, and are eventually replaced by a modified adherens junction-like structure known as the slit diaphragm.^{18,22} The slit diaphragms are signaling hubs serving as the final layer of the glomerular filtration barrier.²³ Mesangial cell ingrowth follows

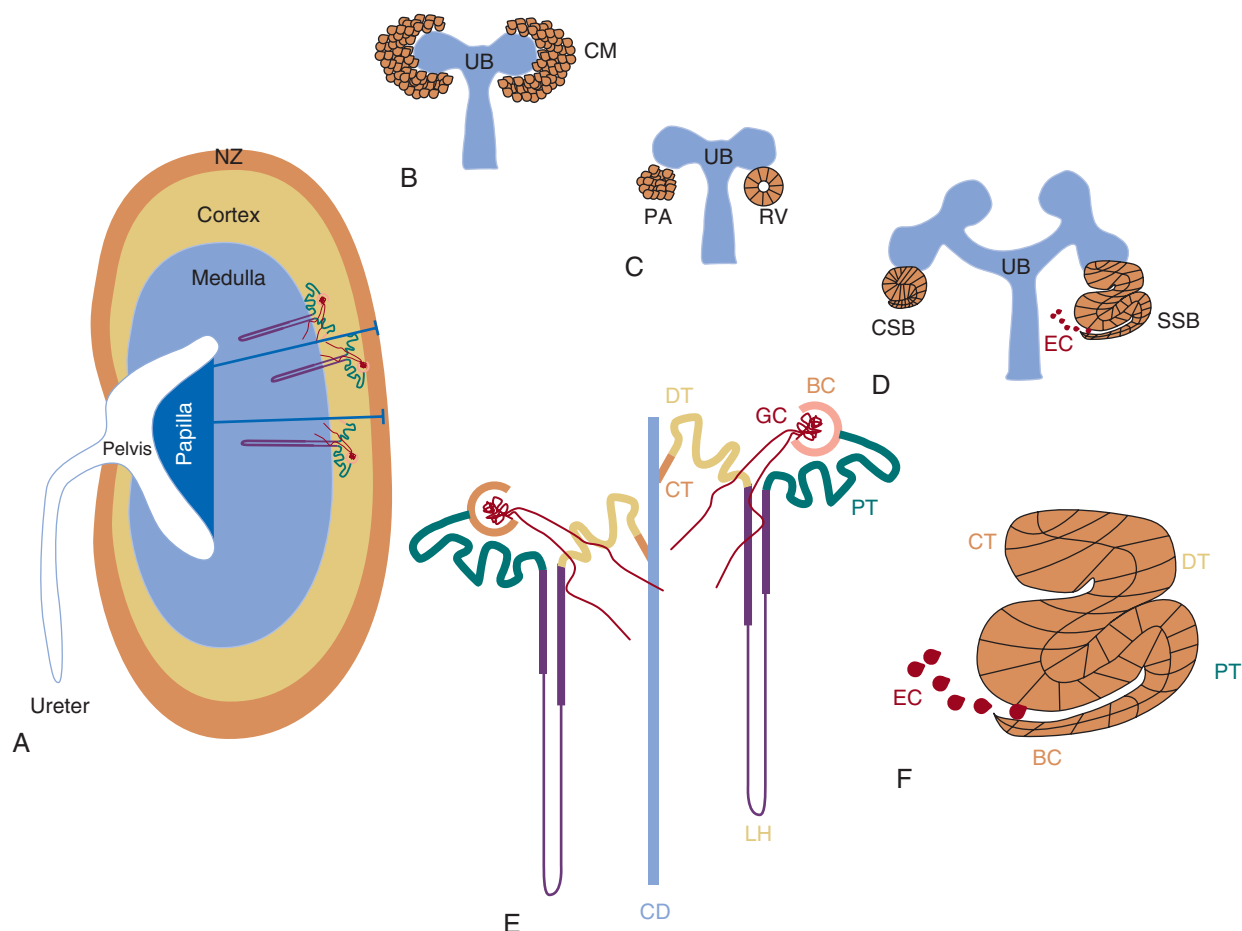


Fig. 1.5 Overview of nephrogenesis. (A) Gross kidney histoarchitecture. (B–E) As described in the text, reciprocal interaction between the ureteric bud and metanephric mesenchyme results in a series of well-defined morphologic stages leading to formation of the nephron, including the branching of the UB epithelium and the epithelialization of the metanephric mesenchyme into a highly patterned nephron. (F) Distinctive segmentation of the S-shaped body defines the patterning of the nephron. *BC*, Bowman capsule; *CD*, collecting duct; *CM*, cap mesenchyme; *CSB*, comma-shaped body; *CT*, connecting tubule; *DT*, distal tubule; *EC*, endothelial cells; *LH*, loop of Henle; *NZ*, nephrogenic zone; *PA*, pretubular aggregate; *PT*, proximal tubule; *SSB*, S-shaped body; *UB*, ureteric bud.

the migration of endothelial cells and is required for development and patterning of the capillary loops that are found in normal glomeruli. The endothelial cells also flatten considerably and capillary lumens are formed due to apoptosis of a subset of endothelial cells.²⁴ At the capillary loop stage, glomerular endothelial cells develop fenestrae, which are semipermeable transcellular pores common in capillary beds exposed to high hemodynamic flux.

In the mature stage glomerulus, the podocytes, fenestrated endothelial cells, and intervening glomerular basement membrane comprise the filtration barrier that separates the urinary from the blood space. Together, these components provide a size- and charge-selective barrier that permits free passage of small solutes and water but prevents the loss of larger molecules such as proteins. The mesangial cells are found between the capillary loops where they are required to provide ongoing structural support to the capillaries and possess smooth-muscle cell-like characteristics that have the capacity to contract, which may account for some of the dynamic properties of the glomerulus. The tubular portion of the nephron becomes segmented in a proximal–distal order, into the proximal convoluted tubule, the descending and ascending loops of Henle, and distal convoluted tubule.

The distal tubule is contiguous with the collecting duct, a derivative of the UB. Imaging and fate mapping studies reveal that this interconnection results from the invasion of the UB by cells from the distal segments of nascent nephrons (around the S-shaped body stage).²⁵

Although all segments of the nephron are present at birth and filtration occurs prior to birth, maturation of the tubule continues in the postnatal period. Increased expression levels of transporters, switches in transporter isoforms, alterations in paracellular transport mechanisms, and permeability and biophysical properties of tubular membranes have all been observed to occur postnatally.²⁶ These observations emphasize the importance of considering the developmental stage of the nephron in interpretation of renal transport and may explain the age of onset of symptoms in inherited transport disorders.

THE NEPHROGENIC ZONE

After the first few rounds of branching of the UB, and the concomitant induction of nephrons from the MM, the kidney subdivides into two major compartments: an outer region called the cortex and an inner region called the medulla. The glomeruli and proximal and distal tubules localize within

the cortex, together with the distal part of the nephron that connects directly to the collecting ducts. The loop of Henle and the rest of the collecting duct network comprise the epithelial structures found in the medulla. Kidney growth and nephrogenesis occurs in a radial fashion with new branches of the ureteric tree and newer nephrons added at the outermost periphery of the developing cortex which is called the nephrogenic zone. The nephrogenic zone is morphologically identifiable as a narrow band beneath the renal capsule where the branching UB tips are found together with nascent nephrons (pretubular aggregates, renal vesicles, comma-shaped bodies, and S-shaped bodies) and self-renewing nephron progenitors. Within the developing kidney, the most mature nephrons are found in the innermost layers of the cortex, and the most immature nephrons in the most peripheral regions. The nephrogenic zone, therefore, represents an active site of nephrogenesis. The nephrogenic zone progressively thins out with the gradual depletion of nephrogenic precursors and disappears once the remaining nephron progenitors have completely epithelialized.

RENAL STROMA AND INTERSTITIAL CELL POPULATIONS

For decades in classic embryologic studies of kidney development, emphasis has been placed on the reciprocal inductive signals between MM and UB. However, in recent years, interest in the stromal cell as a key regulator of nephrogenesis has arisen.^{17,27–29} Stromal cells also derive from the MM but are not induced to condense by the UB. Two distinct populations of stromal cells have been described: cortical stromal cells exist as a thin layer beneath the renal capsule while medullary stromal cells populate the interstitial space between the collecting ducts and tubules (Fig. 1.6). Cortical stromal cells also surround the MM condensates and provide signals required for UB branching and patterning of the developing kidney. Disruption or loss of these stromal cells leads to impairment of UB branching, reduction in nephron number, disrupted nephron patterning with failure of cortical–medullary boundary formation, and maldevelopment of the renal vasculature. A reciprocal signaling loop from the UB exists to properly pattern stromal cell populations. Loss of these UB-derived signals leads to a buildup of stromal cells beneath the capsule that are several layers thick. As nephrogenesis proceeds, stromal cells differentiate into peritubular interstitial cells and pericytes that are required for vascular remodeling and the production of extracellular matrix responsible for proper nephric formation.²⁹ These cells migrate from their position around the condensates to areas between the developing nephrons within the medulla.

THE RENAL VASCULATURE

The microcirculations of the kidney include the specialized glomerular capillary system responsible for production of the ultrafiltrate, and the vasa recta bundles and peritubular capillaries involved in the countercurrent mechanism for urine osmoregulation (Fig. 1.7). Vasculogenesis and angiogenesis have been described as two distinct processes in blood vessel formation (Fig. 1.8). Vasculogenesis is the de novo differentiation of previously nonvascular endothelial cell precursors into structures that resemble capillary beds, whereas angiogenesis refers to sprouting from these early beds to form mature vessel structures, including arteries,

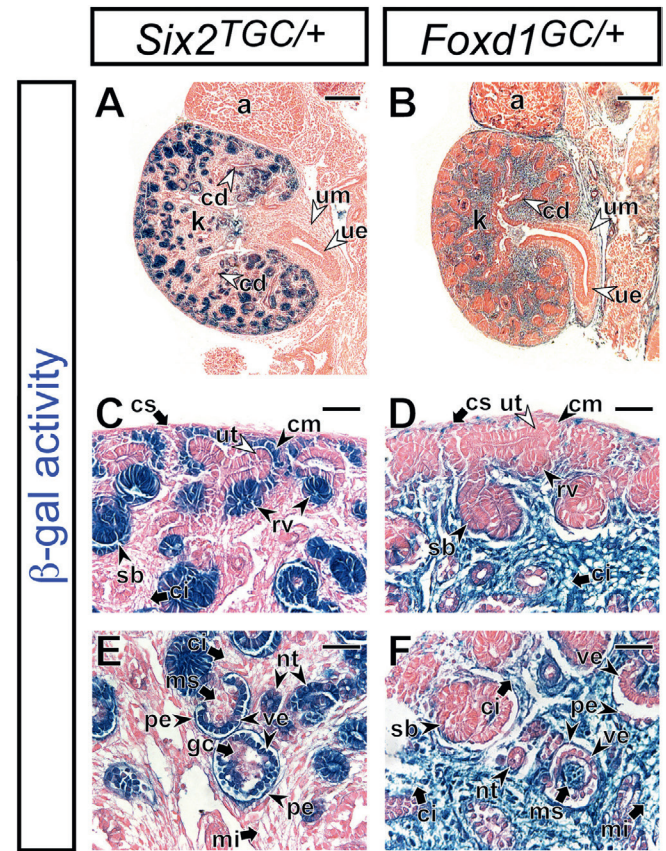


Fig. 1.6 The nephrogenic and stromal mesenchyme. Lineage tracing analysis in E14.5 mouse kidneys showing nephrogenic derivatives of *Six2*-expressing condensing mesenchyme (A, C, and E) and the *Foxd1*-expressing stromal mesenchyme (B, D, and F), stained for β-galactosidase activity (blue). a, Adrenal gland; cd, collecting duct; ci, renal cortical interstitium; cm, cap mesenchyme; cs, renal cortical stroma; gal, galactosidase; gc, glomerular capillary; k, kidney; mi, renal medullary interstitium; ms, mesangium; nt, nephrogenic tubule; pe, parietal epithelium; rv, renal vesicle; sb, S-shaped body; ue, ureter epithelium; um, ureter mesenchyme; ut, ureteric tip; ve, visceral epithelium (podocyte). Reproduced with permission from Kobayashi A, Mugford JW, Krautzberger Am, et al. Identification of a multipotent self-renewing stromal progenitor population during mammalian kidney organogenesis. *Stem Cell Reports*. 2014;3:650–662.

veins, and capillaries. Both processes are involved in development of the renal vasculature. At E11.5 in mice, the UB is tracked by a primitive vessel that elaborates in synchrony with both UB branching and nephrogenesis. A rich capillary network is identifiable by E12.5 while the presence of endothelial cell-containing glomeruli becomes apparent at E14.5.

Transplantation experiments support a model where endogenous endothelial progenitors within the MM give rise to renal vessels in situ through angiogenesis, although the origin of large blood vessels is still not clear.^{30–35} At E13, capillaries form networks around the developing nephric tubules and by E14, the hilar artery and first-order interlobar renal artery branches can be identified. These branches will form the corticomedullary arcades and interlobular arteries that branch from these arcades. Further branching produces the glomerular afferent arterioles. From E13.5 onward, endothelial cells migrate into the vascular cleft of developing glomeruli, where they undergo differentiation to form the

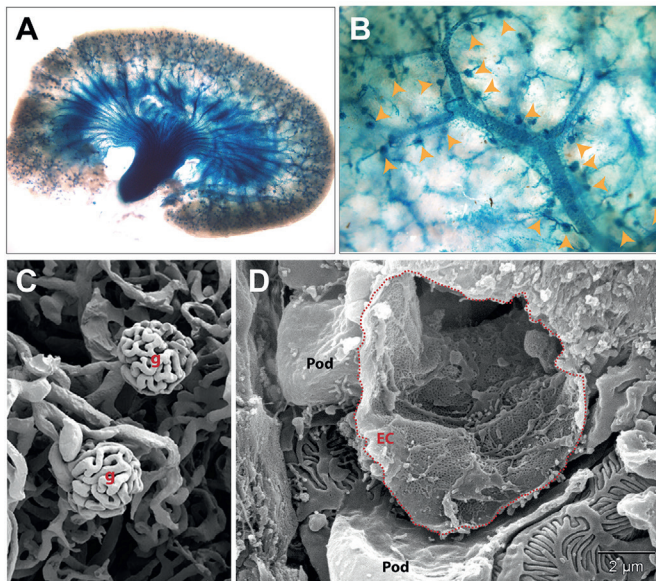


Fig. 1.7 The renal vasculature. (A) Visualization of the renal vascular network in a reporter mouse strain expressing prokaryotic β -galactosidase through the promoter of the vascular-specific phosphatase gene *Ptpnb*. (B) Higher magnification of the renal cortex in (A) showing endothelial cell distribution in glomeruli (yellow arrowheads), arterioles, peritubular capillaries, and arcuate arteries. (C) Corrosion resin cast of the renal vasculature revealing the highly convoluted assembly of the glomerular capillaries (g). (D) Scanning electron micrograph of a glomerulus with an exposed endothelial lumen (dashed outline) revealing fenestrations. EC, Endothelial cell; Pod, podocytes. Corrosion cast electron micrograph courtesy Fred Hossler, Department of Anatomy and Cell Biology, East Tennessee State University.

glomerular capillary loops. The efferent arterioles carry blood away from the glomerulus to a system of fenestrated peritubular capillaries that are in close contact with the adjacent tubules and receive filtered water and solutes reabsorbed from the filtrate.³⁶ These capillaries have few pericytes. In comparison, the vasa rectae, which surround the medullary tubules and are involved in urinary concentration, are also fenestrated but have more pericytes. They arise from the efferent arterioles of deep glomeruli.³⁷ The peritubular capillary system surrounding the proximal tubules is well developed in the late fetal period, whereas the vasa rectae mature 1 to 3 weeks postnatally.

MODEL SYSTEMS TO STUDY KIDNEY DEVELOPMENT

THE KIDNEY ORGAN CULTURE SYSTEM

From the late 1950s, the method of growing mouse embryonic kidneys as floating cultures on top of filters (Fig. 1.9), a technique pioneered by Clifford Grobstein and improved by Lauri Saxen, accelerated the advancement of the kidney developmental biology field. This classic method, which remains widely in use to this day, has the advantages that the kidney explants are cultivated within an easily manipulated, controlled environment, and there is a possibility of visualizing the pattern of kidney growth by real-time fluorescence microscopy. Although vascularization and functional

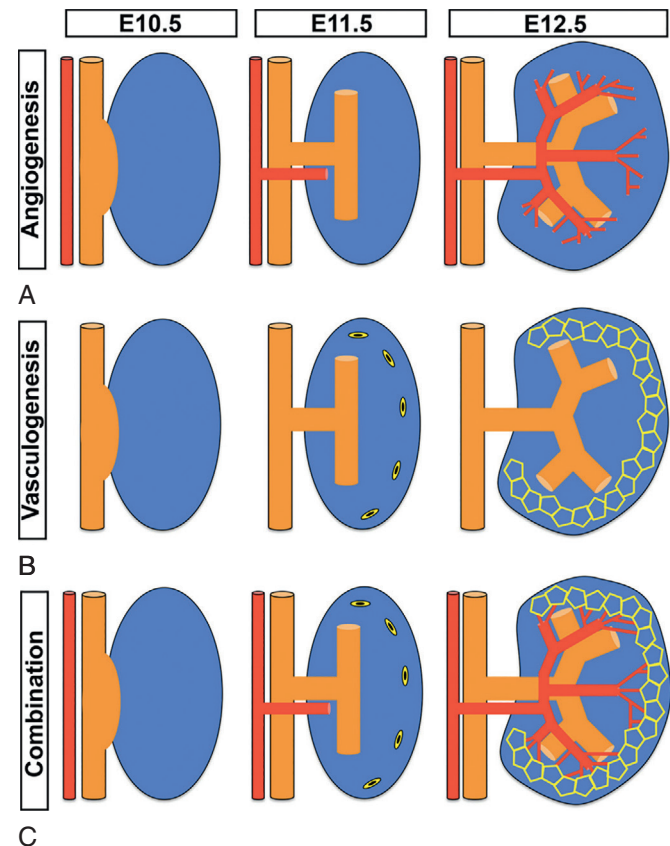


Fig. 1.8 Angiogenesis and vasculogenesis in renal vascular development. Schematic overview of early development of the renal vasculature. (A) Angiogenesis generates major blood vessels through sprouting and branching of pioneer vessels (red) that follow the branching ureteric bud (brown). (B) Scattered endothelial progenitor cells (yellow) are distinctly present as early as E11.5 at the periphery of the developing metanephric kidney (blue). These sporadic endothelial cells coalesce and organize into a primitive capillary plexus (yellow) by E12.5. (C) Major vessels formed via angiogenesis and capillaries that arise by vasculogenesis become interconnected to establish the elaborate renal vascular network. Adapted from Stolz DB, Sims-Lucas S. Unwrapping the origins and roles of the renal endothelium. *Pediatr Nephrol.* 2015;30:865–872.

maturation is largely restricted in embryonic kidney explants, in vitro cultured kidneys display remarkable recapitulation of ureteric branching and epithelialization and segmental patterning of the MM (Fig. 1.10). Historically, kidney explant cultures provided crucial proof for the principle of reciprocal tissue induction in organogenesis. It was used to demonstrate that the UB and the MM exchange inductive cues, driving branching morphogenesis of the UB and epithelialization of the MM.^{10,38}

As originally shown by Grobstein, Saxen, and colleagues, the two major components of the metanephric kidney, the MM and the UB, could be separated from each other, and the isolated mesenchyme could be induced to form nephron-like tubules by a selected set of other embryonic tissues, the best example of which is the embryonic neural tube.^{10,38} When the neural tube is used to induce the separated mesenchyme, there is terminal differentiation of the mesenchyme into tubules, but not significant tissue expansion. In contrast, intact metanephric rudiments can grow more extensively,

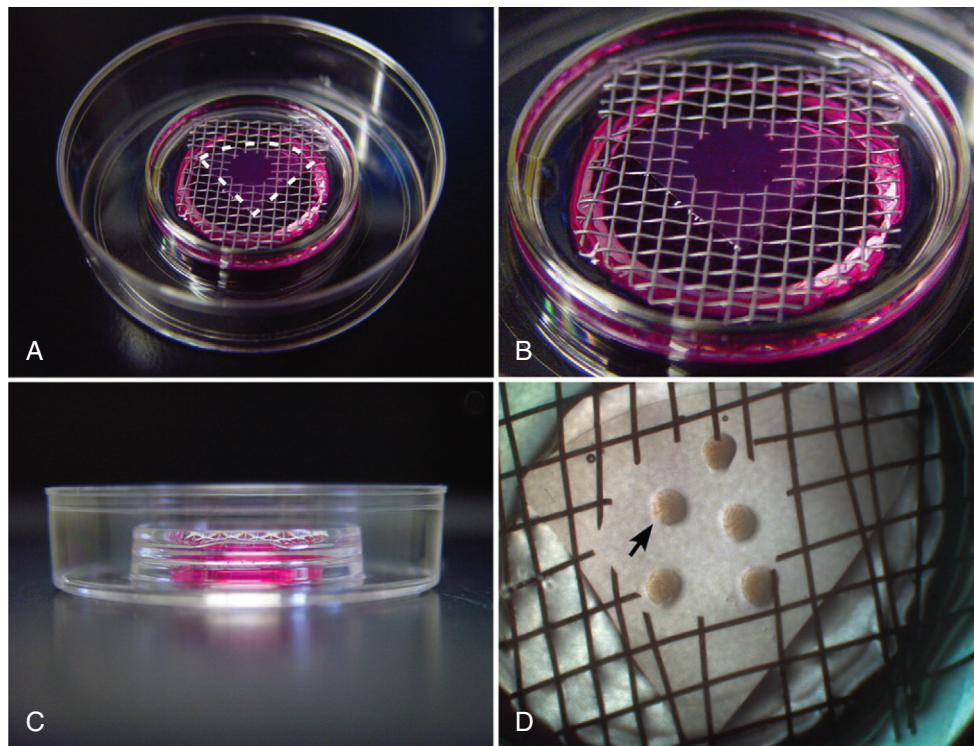


Fig. 1.9 Metanephric organ explants. (A, B) Top and (C) lateral view of a kidney organ culture. Embryonic kidney explants are grown at the air–growth medium interface on top of a floating porous polycarbonate filter (dashed lines in A) supported on a metal mesh. (D) Kidneys grown after 4 days of culture. Reproduced with permission from Cold Spring Harbor Protocols.⁷⁴¹

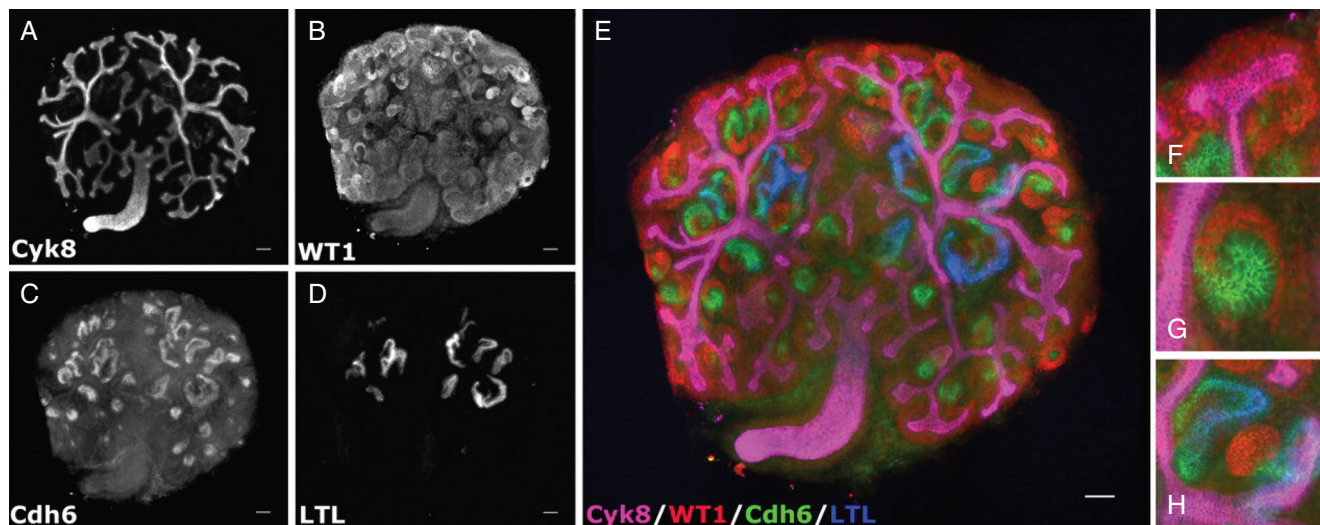


Fig. 1.10 Recapitulation of branching and nephrogenesis in renal explant cultures. (A) Ureteric tree stained for cytokeratin 8 (*Cyk8*). (B) Condensed metanephric mesenchyme stained for WT1. (C) Epithelial derivatives of the metanephric mesenchyme stained for E-cadherin (*Cdh6*). (D) Proximal tubules stained with *Lotus tetraglobulus* lectin (LTL). (E) Merged image of A–D. (F) WT1-expressing cells represent the nephron progenitor cells that surround the UB. (G) *Cdh6*-expression marks the mesenchyme-to-epithelial transformation of nephron progenitor cells. (H) Early patterning of nascent nephrons along a proximodistal axis. Reproduced with permission from Cold Spring Harbor Protocols.⁷⁴¹

displaying both sustained UB branching and early induction of nephrons even when cultured for a week. The isolated mesenchyme experiment has proven useful in the analysis of renal agenesis phenotypes, where there is no outgrowth of the UB. In these cases, the mesenchyme can be placed in contact with the neural tube to determine whether it has the intrinsic ability to differentiate. Most often, when renal agenesis is due to the mutation of a transcription factor,

tubular induction is not rescued by the neural tube, as could be predicted for transcription factors, which would be expected to act in a cell-autonomous fashion.³⁹ In the converse situation, in which renal agenesis is caused by loss of a gene function in the UB (e.g., *Emx2*), it is usually possible for the embryonic neural tube to induce tubule formation in isolated mesenchymes.⁴⁰ Therefore the organ culture induction assay can be used to test hypotheses concerning whether a particular

gene is required in the UB or the MM. Recently, as chemical inhibitors specific for various signal transduction pathways have been synthesized and become available, it has been possible to add these to organ cultures and observe effects that are informative about the roles of specific pathways in development of the kidney. Examples are the use of drugs to block the ERK/MAP kinase, PI3K/Akt, Notch signaling, and Wnt signaling pathways in renal explant cultures.^{41–45} Synthetic antisense oligonucleotides and small interfering or silencing RNA (siRNA) have also been successfully used to inhibit gene expression in kidney organ cultures.^{45–55}

GENETIC MOUSE MODELS

Many of the genetic pathways regulating many aspects of development are strongly conserved between mice and humans. Due to the relative ease of manipulating the genome of the mouse, its small size, and shorter gestation period, the mouse has become the primary model organism for the study of mechanism of human development and diseases. Anatomic and functional features of the human kidney are for the most part highly similar in the mouse, albeit at a smaller scale, while a number of genes identified as essential for normal mouse kidney development are also known to be associated with congenital anomalies of the kidney and urinary tract (CAKUT) and other kidney diseases.^{56–58}

The wealth of our understanding of metanephric kidney development over the past two decades is owed to the harnessing of homologous gene recombination to introduce targeted mutations and novel alleles into specific genes in cultured mouse embryonic stem cells. This paved the way for the creation of numerous genetically engineered mice that have become highly valuable tools for the study of renal developmental biology and the etiology of certain genetic diseases of the kidneys and the urinary tract (Table e1.2). The technology is applied in many ways. In its simplest is the creation of null mutations within the germline that generate gene knockout mouse models. The limitation of this method is that certain genes are essential for early development and their inactivation in the germline can cause premature lethality, thus precluding the analysis of the function of those genes in organogenesis. A novel improvement to this is the use of a conditional gene-targeting strategy, allowing for the creation of conditional alleles. This involves introducing small recognition sites for recombinase enzymes of which the Cre recombinase is the most routine now (Fig. 1.11). A conditional “floxed” allele of the target gene locus is created by incorporating two *loxP* sites within two separate introns, flanking the exons that can be excised or recombined. In principle, normal transcription from the locus is expected prior to recombination of the floxed allele and should effectively promote a wild type phenotype. The Cre recombinase itself is engineered under the control of a tissue-specific promoter. Breeding between tissue-specific transgenic Cre animals and those harboring a conditional allele for a particular gene ultimately results in a cell- or tissue-type specific inactivation of the gene of interest. This strategy has also been refined in some cases, so that the Cre expression is also temporally regulated by the use of a drug such as doxycycline or tamoxifen. A number of Cre lines are now available to target genes specifically within different subpopulations of renal cells and progenitors. The transgenic Cre lines driven by the *Hoxb7*,

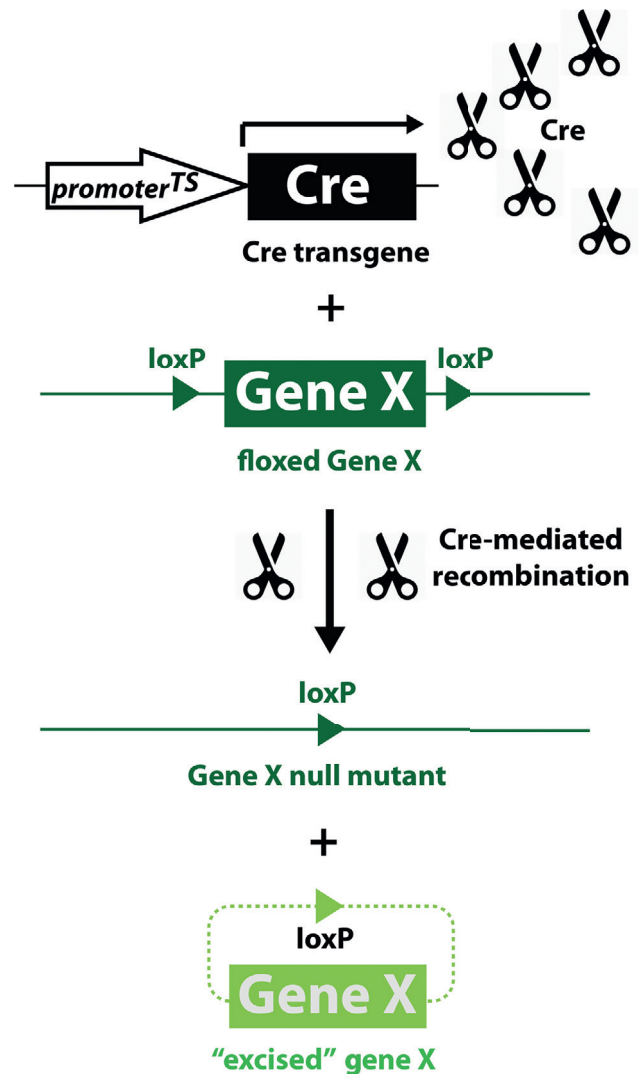


Fig. 1.11 Cre-loxP homologous gene recombination system. Simplified overview of Cre recombinase-mediated homologous recombination for generation of tissue-specific conditional null mutations of a target gene. Cre recombinase expression is engineered to be driven under the control of tissue specific promoter ($\text{promoter}^{\text{TS}}$). The target gene (or typically certain exons within the target chromosomal locus) are flanked with *loxP* sites (recognition sites for Cre recombinase). Cre activity in specific cell types mediates the excision of the *loxP*-flanked (floxed) target gene, creating a null allele.

Six2, *Foxd1*, and *Nphs2* promoters are among the most highly cited gene excision drivers for targeted genetic inactivation in the ureteric, nephrogenic, stromal, and podocyte cell lineages.^{59–64} Lately, the introduction of the more precise CRISPR-Cas9 gene editing tool, an ingenious application of the adaptive immune response of prokaryotes against bacteriophages, is facilitating the way researchers create customized mutant cells and animal models to study various aspects of renal development (Fig. 1.12).^{65–68}

The discovery of new kidney-relevant genes or signaling pathways has also been achieved by random chemical mutagenesis in mice. The alkylating agent N-ethyl-N-nitrosourea (ENU) is routinely used to introduce random point mutations in mouse spermatogonia. The introduced mutations can result in loss or gain of function and are either recessive or

Table e1.2 Summary of Knockout and Transgenic Models for Kidney Development

Gene Mutation or Knockout	Phenotype	Associated Human Diseases	References
Renal Agenesis			
<i>Celsr1</i>	Renal agenesis, hydroureter, hydronephrosis	Spina bifida, unilateral renal agenesis, hydronephrosis	331
<i>Ctnnb1</i> (β -catenin)	Renal agenesis or severe renal hypoplasia, premature differentiation of UB epithelia (UB-selective)	Mental retardation, multiple cancers, eye defects	181, 185
<i>Emx2</i>	Renal agenesis	Schizencephaly (cerebral cleft abnormalities)	40
<i>Emx2, Pax2</i>	Duplicated kidneys and ureter, ureteral obstruction	CAKUT, VUR	531
<i>Esrp1</i>	Renal agenesis, renal hypoplasia		532
<i>Etv4, Etv5</i>	Renal agenesis or severe renal hypodysplasia		210, 212
<i>Eya1</i>	Renal agenesis	Branchiootorenal syndrome (brachial fistulae, deafness)	136, 150
<i>Fgf9, Fgf20</i>	Renal agenesis		305
<i>Fgf10, Gdnf, Gfra1</i>	Renal agenesis		222
<i>Fgfr1, Fgfr2</i>	Renal agenesis (MM-selective)		304
<i>Fras1, Frem1, Frem2</i>	UB failure, defect of GDNF expression	Fraser syndrome (cryptophthalmos, syndactyly, CAKUT); Manitoba-oculo-tricho-anal (MOTA) syndrome	165, 166, 225, 227, 228
<i>Gata3</i>	Renal agenesis, gonad dysgenesis (null mutation)	Hypoparathyroidism, sensorineural deafness, and renal dysplasia (HDRS) syndrome; autoimmune disease	179, 180, 182, 183
<i>Gdf11</i>	UB failure, skeletal defects		145, 533
<i>Gdnf, Gfra1, Ret</i>	Renal agenesis or rudimentary kidneys, aganglionic megacolon	Hirschsprung disease, Multiple endocrine neoplasm type IIA/B (MEN2A/MEN2B), and familial medullary thyroid carcinoma (FMTC)	142–144, 148, 161, 164, 534–537
<i>Gen1</i>	Renal agenesis, duplex kidneys, hydronephrosis, ureteral obstruction		538
<i>Gli3</i>	Renal agenesis, severe renal agenesis, absence of renal medulla and papilla	Pallister-Hall (PH) syndrome (polydactyly, imperforate anus, abnormal kidneys, defects in the gastrointestinal tract, larynx, and epiglottis)	265, 266
<i>Greb11</i>	Renal agenesis	CAKUT	539–541
<i>Grem1</i>	Renal agenesis; apoptosis of the MM		146
<i>Grlh2</i>	Occasional unilateral renal agenesis, CD barrier dysfunction, diabetes insipidus	Autosomal dominant deafness, ectodermal dysplasia	542, 543
<i>Grip1</i>	Renal agenesis	Fraser syndrome (cryptophthalmos, syndactyly, CAKUT)	229–231
<i>Hnf1b</i>	Renal agenesis, renal hypoplasia, hydroureter, duplex kidneys	CAKUT, diabetes mellitus, renal cysts, renal carcinoma	544
<i>Hoxa11, Hoxd11</i>	Distal limbs, vas deferens	Radioulnar synostosis with amegakaryocytic thrombocytopenia	545
<i>Hs2st1</i>	Lack of UB branching and mesenchymal condensation		546
<i>Isl1</i>	Renal agenesis, renal hypoplasia, hydroureter (MM-selective)		547
<i>Itga8</i> ($\alpha 8$ Integrin)	Renal agenesis, renal hypodysplasia	Fraser syndrome (cryptophthalmos, syndactyly, CAKUT)	169

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Table e1.2 Summary of Knockout and Transgenic Models for Kidney Development (Cont'd)

Gene Mutation or Knockout	Phenotype	Associated Human Diseases	References
<i>Itgb1</i> ($\beta 1$ Integrin)	Renal agenesis, disrupted UB branching, hypoplastic collecting duct system (collecting duct-selective); podocyte dedifferentiation (podocyte-selective)	Fraser syndrome (cryptophthalmos, syndactyly, CAKUT)	177, 548, 549
<i>Kif26b</i>	Renal agenesis, failed UB attraction to the MM		167
<i>Lamc1</i>	UB failure, delayed nephrogenesis, water transport defects		232
<i>Lhx1</i> (<i>Lim1</i>)	Renal agenesis (null mutant); renal hypoplasia, UB branching defect, hydronephrosis, distal ureter obstruction (UB-selective); arrested nephrogenesis, nephron patterning defects (MM-selective)	Mayer-Rokitansky-Küster-Hauser (MRKH) syndrome (müllerian duct agenesis)	141, 550, 551
<i>Lrp4</i>	Delayed UB induction, failed MM induction, syndactyly, oligodactyly	Cenani-Lenz syndrome	552–555
<i>Mark2</i> (<i>Par1b</i>), <i>Mark3</i> (<i>Par1a</i>)	Renal hypoplasia, proximal tubule dilation, immature glomeruli		556
<i>Npnt</i>	Delayed UB association with the MM		168
<i>Osr1</i>	Lack of MM, adrenal gland, gonads, defects in formation of pericardium and atrial septum		137, 149, 281, 282
<i>Pax2</i>	Renal hypoplasia, VUR	CAKUT, VUR, optic nerve colobomas	46, 47
<i>Pax2</i> , <i>Pax8</i>	Defect in intermediate mesoderm transition, failure of pronephric duct formation	CAKUT, VUR, optic nerve colobomas	557
<i>Pbx1</i>	Unilateral renal agenesis, expansion of nephrogenic precursors	CAKUT, hearing loss, abnormal ears	357, 558–560
<i>Ptf1a</i>	Failure of UB induction, anal atresia, persistent cloaca, skeletal malformation	Pancreatic and cerebellar agenesis; diabetes mellitus	561–563
<i>Rara</i> , <i>Rarb</i>	Renal hypoplasia, dysplasia, hydronephrosis, skeletal and multiple visceral abnormalities		14, 17, 88
<i>Sall1</i>	Renal agenesis, severe renal hypodysplasia	Townes-Brock syndrome (anal, renal, limb, ear anomalies)	140, 564
<i>Shh</i>	Bilateral or unilateral renal agenesis, unilateral ectopic dysplastic kidney, defective ureteral stromal differentiation	Vertebral defects, anal atresia, cardiac defects, tracheoesophageal fistula, renal anomalies, and limb abnormalities (VACTERL) syndrome	59
<i>Six1</i>	Lack of UB branching and mesenchymal condensation	Branchiootorenal syndrome	136, 150
<i>Sox8</i> , <i>Sox9</i>	Renal genesis, renal hypoplasia	Campomelic dysplasia (limb and skeletal defects, abnormal gonad development)	565
<i>Tln1</i> , <i>Tln2</i>	Renal agenesis		566
<i>Wnt5a</i>	Renal agenesis, renal dysplasia, duplex kidneys, hydronephrosis	CAKUT	567–569
<i>Wt1</i>	Renal and gonadal agenesis, severe lung, heart, spleen, adrenal, and mesothelial abnormalities	Wilms tumor, aniridia, genitourinary abnormalities, and retardation (WAGR) syndrome; Denys-Drash syndrome	39, 48, 437, 438
Hypoplasia/Dysplasia/Low Nephron Mass			
<i>Adamts1</i>	Hypoplasia of the renal medulla, hydronephrosis		319, 321
<i>Adamts1</i> , <i>Adamts4</i>	Hypoplasia of the renal medulla, hydronephrosis		570

Table e1.2 Summary of Knockout and Transgenic Models for Kidney Development (Cont'd)

Gene Mutation or Knockout	Phenotype	Associated Human Diseases	References
<i>Agtr2</i> <i>Ald1a2 (Raldh2)</i>	Various collecting duct defects Renal hypoplasia, hydronephrosis, ectopic ureter	CAKUT	257, 422, 423 182
<i>Bmp1ra (Alk3)</i>	Hypoplasia of renal medulla, fewer UB branches (UB-selective)	Juvenile polyposis syndrome	254
<i>Bmp4</i> <i>Bmp7</i>	Severe renal hypodysplasia Renal hypoplasia, reduced MM survival	Microphthalmia, orofacial cleft	242 294
<i>Cask</i>	Renal hypoplasia and dysplasia, premature depletion of nephrogenic precursor cells	Microcephaly, mental retardation	308, 571
<i>Cdc42</i>	Renal hypoplasia, oligonephronia, defects in mesenchyme to epithelial transition (CM-selective)		355, 572
<i>Cfl1</i>	Renal hypodysplasia, ureter duplication		573
<i>Chd4 (Mi-2β)</i>	Renal hypodysplasia	Sifrim-Hitz-Weiss syndrome (autosomal dominant intellectual disability disorder with variable congenital defects in cardiac, skeletal, and urogenital systems)	318, 574
<i>Ctnnb1 (β-catenin)</i>	Severely hypoplastic kidney, lack of nephrogenic zone and S-shaped body (CM-selective)	Mental retardation, multiple cancers, eye defects	289
<i>Dchs1, Dchs2</i>	Renal hypoplasia, impaired ureteric branching, expansion of nephrogenic precursor zone	Van Maldergem syndrome 1—periventricular nodular heterotopia, intellectual disability, deafness, renal hypoplasia, tracheal anomalies, and skeletal dysplasia	353, 356, 575
<i>Dicer1</i>	Renal hypoplasia, dysplasia, cysts (UB-selective); renal hypoplasia characterized by premature termination of nephrogenesis (MM-selective)		576, 577
<i>Dkk1</i>	Overgrown renal papilla (renal tubule and collecting duct restricted)		250
<i>Dlg1</i>	Renal hypoplasia and dysplasia, premature depletion of nephrogenic precursor cells		308, 571
<i>Dstn (Destrin)</i>	Renal hypodysplasia, ureter duplication		573
<i>Egfr</i>	Hypoplasia of the renal papilla, moderate polyuria and urine concentration defects		248
<i>Egln1 (Phd2), Egln3 (Phd3)</i>	Renal hypoplasia, oligonephronia, abnormal postnatal nephron formation, glomerulosclerosis (stroma-specific)	Familial erythrocytosis; abnormally high EPO levels; high altitude adaptation hemoglobin (HALAH)	578
<i>Esrrg</i> <i>Fat4</i>	Agensis of renal papilla Renal hypoplasia, impaired ureteric branching, failed nephrogenesis (mesenchyme-to-epithelial transition), expansion of nephrogenic precursor zone (stroma-selective)	Van Maldergem syndrome 2 (VMLDS2): dysmorphic faces, tracheomalacia, microtia, intellectual disability, and skeletal dysplasia	53 352, 353, 356, 575
<i>Fgf7</i>	Small kidneys, reduction in nephron number		253
<i>Fgf8</i>	Renal dysplasia, arrested nephrogenesis at pretubular aggregate stage (MM-selective)	Kallmann syndrome, hypogonadism	286, 287

Continued on following page

Table e1.2 Summary of Knockout and Transgenic Models for Kidney Development (Cont'd)

Gene Mutation or Knockout	Phenotype	Associated Human Diseases	References
<i>Fgf10</i>	Renal hypoplasia, multiorgan developmental defects, including the lungs, limb, thyroid, pituitary and salivary glands		252
<i>Fgfr1, Fgfr2</i> <i>Fgfr2</i>	Renal agenesis (MM-selective) Renal hypoplasia, hydronephrosis (UB-selective)		304 60
<i>Foxc2</i>	Renal hypoplasia	Lymphedema-distichiasis syndrome	237, 579
<i>Foxd1</i>	Accumulation of undifferentiated CM, attenuated UB branching, stromal patterning defects		28, 341, 343
<i>Frs2</i> <i>Fzd4, Fzd8</i>	Mild renal hypoplasia (UB-selective) Impaired UB branching, renal hypoplasia		580 581
<i>Hdac1, Hdac2</i>	Renal hypoplasia, renal dysplasia, arrest of nephrogenesis at the renal vesicle stage		582, 583
<i>Lats1, Lats2</i>	Renal hypoplasia, impaired UB-branching and UB tip specification, impaired nephrogenesis and renal interstitial differentiation		354, 584
<i>Lgr4</i>	Severe renal hypoplasia and oligonephronia; renal cysts	Aniridia-genitourinary anomalies; mental retardation	585, 586
<i>Lmx1b</i> <i>Map2k1 (Mek2), Map2k (Mek1)</i> <i>Mdm2</i>	Renal dysplasia, skeletal abnormalities Renal hypodysplasia, megaureter Renal hypoplasia and dysplasia, severely impaired UB branching and nephrogenesis (UB-selective); depletion of nephrogenic precursors (MM-selective)	Nail-patella syndrome Cardiofaciocutaneous syndrome	435, 442 208 587, 588
<i>Mf2</i> <i>Mitf</i>	Renal hypoplasia, oligonephronia Oligonephronia	Microphthalmia, Waardenburg syndrome type 2A	589 590
<i>Nf2</i> <i>Notch1, Notch2</i>	Renal hypoplasia, renal dysplasia Loss of nephron derivatives, nephron segmentation defects	Alagille syndrome (cholestatic liver disease, cardiac disease, kidney dysplasia, renal cysts, renal tubular acidosis)	584 293
<i>Pbx1</i>	Reduced UB branching, expansion of nephrogenic precursors, delayed mesenchyme-to-epithelial transformation, dysgenesis of adrenal gland and gonads		357, 558
<i>Plxnb2</i>	Renal hypoplasia and ureter duplication		591
<i>Pou3f3 (Brn1)</i>	Impaired development of distal tubules, loop of Henle, and macula densa; distal nephron-patterning defect		320
<i>Prr</i>	Renal hypoplasia, renal dysplasia, oligonephronia		592, 593
<i>Psen1, Psen2</i>	Severe renal hypoplasia, severe defects in nephrogenesis		594
<i>Ptgs2</i> <i>Rbpj</i>	Oligonephronia Severe renal hypoplasia, oligonephronia, loss of proximal nephron segments, tubular cysts (MM-selective)		595 596, 597
<i>Sall1</i>	Severe renal hypoplasia, cystic dysplasia of nephrogenic derivatives (tubules and glomeruli)	Townes-Brocks branchiootorenal-like syndrome	358

Table e1.2 Summary of Knockout and Transgenic Models for Kidney Development (Cont'd)

Gene Mutation or Knockout	Phenotype	Associated Human Diseases	References
<i>Shp2</i>	Severe impairment of UB branching, renal hypoplasia		207
<i>Six1</i>	Hydronephrosis, hydroureter, abnormal development of ureteral smooth muscle		361
<i>Six2</i>	Renal hypoplasia and premature depletion of nephrogenic precursors (homozygous loss); increased UB branching and augmentation of nephron endowment (haploinsufficiency)		278, 337
<i>Tbx18</i>	Hydronephrosis, hydroureter, abnormal development of ureteral smooth muscle		359, 361
<i>Tfap2b</i>	MM failure, craniofacial and skeletal defects		598
<i>Trp53 (p53)</i>	Oligonephronia, precocious depletion of nephrogenic precursors	Multiple cancers	599
<i>Trps1</i>	Impaired UB branching, renal hypoplasia	Trichorhinophalangeal syndrome (skeletal defects)	600
<i>Vangl2</i>	Impaired UB branching and renal hypoplasia	Neural tube defects	332
<i>Wnt4</i>	Failure of MM induction		289
<i>Wnt7b</i>	Complete absence of medulla and renal papilla (UB-selective)		245
<i>Wnt9b</i>	Vestigial kidney, failure of MM induction; cystic kidney (CD-selective)		135, 325
<i>Wnt11</i>	Impaired ureteric branching, renal hypoplasia		211
<i>Yap</i>	Renal hypoplasia renal dysplasia, hydronephrosis, severe disruption of UB branching (UB-selective), oligonephronia, defects in mesenchyme to epithelial transition (CM-selective)	Coloboma, hearing impairment, cleft palate, cognitive deficit, hematuria	355, 572, 584
Mislocalized or Ectopic UB/Increased UB Branching			
<i>Bmp4</i>	Duplex ureter, hydroureter, renal hypodysplasia	Microphthalmia, orofacial cleft	242
<i>Cer1</i>	Increased ureteric branching, altered spatial organization of ureteric branches		601
<i>Cfl1</i>	Renal hypodysplasia, ureter duplication		573
<i>Foxc1</i>	Duplex kidneys, ectopic ureters, hydronephrosis, hydroureter		237
<i>Gata3</i>	Ectopic ureteric budding, duplex kidneys, hydroureter (UB-selective)	Hypoparathyroidism, sensorineural deafness, and renal dysplasia (HDRS) syndrome; autoimmune disease (rheumatoid arthritis)	179, 180, 182, 183
<i>Hnf1b, Pax2</i>	Renal hypoplasia, duplex kidneys, ectopic ureters, megaureter, hydronephrosis	CAKUT	602
<i>Hspb11 (lft25)</i>	Duplex kidneys		603
<i>lft27</i>	Duplex kidneys		603
<i>Lzts2</i>	Duplex kidneys/ureters, hydronephrosis, hydroureter		604
<i>Plxnb1</i>	Increased ureteric branching		605
<i>Plxnb2</i>	Renal hypoplasia and ureter duplication		591
<i>Robo2</i>	Increased UB branching	CAKUT, VUR	238, 239

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Table e1.2 Summary of Knockout and Transgenic Models for Kidney Development (Cont'd)

Gene Mutation or Knockout	Phenotype	Associated Human Diseases	References
<i>Ror2</i>	Duplex ureter, hydronephrosis	Skeletal dysplasia, shortened limbs, brachydactyly, facial dysmorphism (brachydactyly, type B1, Robinow syndrome)	569, 606
<i>Sema3a</i>	Increased ureteric branching (UB-selective)		50
<i>Slit2</i>	Increased UB branching	CAKUT, VUR	238, 239
<i>Spry1</i>	Supernumerary UBs, multiple ureters		87, 218
<i>Wnt5a</i>	Duplex ureter, hydronephrosis	Robinow syndrome (skeletal dysplasia, shortened limbs, brachydactyly, facial dysmorphism)	569, 606
Renal Cysts			
<i>Angpt1, Angpt2</i>	Interstitial medullary cysts, urinary concentration defects		407
<i>Aqp11</i>	Abnormal vacuolization of proximal tubules; polycystic kidneys		607
<i>Arhgap35 (GRLF1)</i>	Glomerular cysts		73
<i>Bcl2</i>	Renal hypoplasia and cysts		608
<i>Bicc1</i>	Polycystic kidneys		609
<i>Bpck</i>	Polycystic kidneys, hydrocephalus	Meckel syndrome 3	610
<i>Erbb4</i>	Renal cysts (overexpression in renal tubules)		611
	Dilated and mispolarized tubules, increased renal fibrosis (renal tubule deletion)		
<i>Fat4</i>	Renal cysts, disrupted hair cell organization in inner ear	Van Maldergem syndrome (mental retardation, abnormal craniofacial features, deafness, limb malformations, renal hypoplasia)	330, 612
<i>Glis3</i>	Polycystic kidney, neonatal diabetes	Congenital hypothyroidism, diabetes mellitus, hepatic fibrosis, congenital glaucoma	613, 614
<i>Gpc3</i>	Disorganized tubules and medullary cysts	Simpson-Golabi-Beckwith syndrome	615–617
<i>Hnf1b</i>	Polycystic kidney disease (tubule-selective)	CAKUT, diabetes mellitus, renal cysts, renal carcinoma	326, 329, 544
<i>lft88 (Orpk)</i>	Polycystic kidneys; defective left-right asymmetric patterning	ARPKD	618, 619
<i>Ilk</i>	Medullary cysts (stroma-specific)		620
<i>Invs</i>	Polycystic kidneys, inverted viscera	Nephronophthisis	621, 622
<i>Kif3A</i>	Polycystic kidney disease (tubule-selective)		623
<i>Maflb (Kreisler)</i>	Decreased glomeruli, cysts, and tubular dysgenesis	Musculoaponeurotic fibrosarcoma	624, 625
<i>Mks1</i>	Renal hypoplasia and cysts	Meckel syndrome (multicystic dysplasia, neural tube defect)	626
<i>Pkd1, Pkd2</i>	Renal cysts	ADPKD, ARPKD	627
<i>Pten</i>	Abnormal ureteric bud branching, cysts (UB-selective)	Cowden disease, Bannayan-Riley-Ruvalcaba syndrome, various tumors	209
<i>Sall1</i>	Nephrogenic tubule and glomerular cysts (stroma specific)	Townes-Brocks branchiootorenal-like syndrome	358
<i>Taz</i>	Polycystic kidneys, emphysema		628, 629
<i>Tek (Tie2)</i>	Interstitial medullary cysts, urinary concentration defects	Cutaneous and mucosal venous malformation, congenital glaucoma	407
<i>Vhl</i>	Renal cysts (tubule-selective)	Von Hippel-Lindau syndrome	630
<i>Xylt2</i>	Polycystic kidneys and liver		631
<i>Zeb2</i>	Glomerular cysts	Mowat-Wilson syndrome (Hirschsprung disease with associated mental retardation)	632

Table e1.2 Summary of Knockout and Transgenic Models for Kidney Development (Cont'd)

Gene Mutation or Knockout	Phenotype	Associated Human Diseases	References
Later Phenotypes (Tubular, Vascular, and Glomerular Defects)			
<i>Ace</i>	Atrophy of renal papillae, vascular thickening and hypertrophy, perivascular inflammation	Chronic systemic hypotension	258, 259
<i>Actn4</i>	Glomerular developmental defects, FSGS	SRNS	451, 452
<i>Adam10</i>	Loss of principal cells of the CD, hydronephrosis, polyuria	Alzheimer disease; reticulate acropigmentation of Kitamura	271
<i>Agt</i>	Atrophy of renal papillae, vascular thickening and hypertrophy, perivascular inflammation	Chronic systemic hypotension	256, 427
<i>Agtr1a</i> (AT1A)	Hypertrophy of juxtaglomerular apparatus and expansion of renin cell progenitors, mesangial cell hypertrophy	Chronic systemic hypotension	633
<i>Agtr1a/Agtr1b</i> (AT1A/AT1B)	Atrophy of renal papillae, vascular thickening and hypertrophy, perivascular inflammation	Chronic systemic hypotension	261
<i>Ampd</i>	Podocyte foot process effacement, proteinuria	Minimal change nephropathy	634
<i>Angpt1</i>	Simplification and <i>dilation</i> of glomerular capillaries; detachment of glomerular endothelium from the GBM; loss of mesangial cells; loss of ascending vasa recta (compound deletion with <i>Angpt2</i>)		382, 407
<i>Angpt2</i>	Cortical peritubular capillary abnormalities (null allele); apoptosis of glomerular capillaries, proteinuria (transgenic overexpression); loss of ascending vasa recta (compound deletion with <i>Angpt1</i>)		390, 391, 407
<i>Arhgdia</i> (<i>RhoGDIα</i>)	Podocyte effacement and proteinuria	SRNS, FSGS	77, 80, 635
<i>Bmp7</i>	Hypoplastic kidney, impaired maturation of nephron, reduced proximal tubules (podocyte-selective)		398
<i>Cd151</i>	Podocyte foot process effacement, disorganized GBM, tubular cystic dilation	Nephropathy (FSGS) associated with pretibial epidermolysis bullosa and deafness	636, 637
<i>Cd2ap</i>	Podocyte effacement, proteinuria	FSGS	484
<i>Cdc42</i>	Congenital nephrosis; impaired formation of podocyte foot processes (podocyte-selective)		474
<i>Cmas</i>	Congenital nephrosis; impaired formation of podocyte foot processes, defective sialylation		638
<i>Col4a1</i> , <i>Col4a3</i> , <i>Col4a4</i> , <i>Col4a5</i>	Disorganized GBM, proteinuria	Alport syndrome	639–642
<i>Coq6</i>	Nephrotic syndrome and deafness	SRNS, FSGS, sensorineural deafness	462, 643
<i>Crb2</i>	Podocyte effacement and proteinuria	SRNS, FSGS	644
<i>Crk1</i> , <i>Crk2</i> , <i>CrkL</i>	Albuminuria, altered podocyte cytoarchitecture (podocyte-selective)		645
<i>Cxcl12</i> (<i>SDF1</i>), <i>Cxcr4</i> , <i>Cxcr7</i>	Petechial hemorrhage in the kidneys, glomerular aneurysm, fewer glomerular fenestrations, reduced mesangial cells, podocyte foot process effacement, mild renal hypoplasia	WHIM (warts, hypogammaglobulinemia, infections, and myelokathexis) syndrome	401, 402, 646

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Table e1.2 Summary of Knockout and Transgenic Models for Kidney Development (Cont'd)

Gene Mutation or Knockout	Phenotype	Associated Human Diseases	References
<i>Dicer1</i>	Podocyte damage, albuminuria, end-stage renal failure (podocyte-selective); reduced renin production, renal vascular abnormalities, striped fibrosis (renin cell-selective)	Pleuropulmonary blastoma	430, 492–494
<i>Dnm1, Dnm2 (Dynamin 1/2)</i>	Podocyte foot process effacement and proteinuria (podocyte-selective)		647
<i>Dot1l</i>	Increased intercalated at the expense of principal CD cells; polyuria		275, 277
<i>Efnb1 (Ephrin B1)</i>	Podocyte foot process effacement and proteinuria (podocyte-selective)		648
<i>Efnb2 (Ephrin B2)</i>	Dilation of glomerular capillaries		396
<i>Egln1 (Phd2), Egln3 (Phd3)</i>	Renal hypoplasia, oligonephronia, abnormal postnatal nephron formation, abnormally elevated erythropoietin production, dilation of renal blood vessels, glomerulosclerosis (stroma-specific)	Familial erythrocytosis; abnormally high EPO levels; high altitude adaptation hemoglobin (HALAH)	578
<i>Elf5</i>	Paucity in principal CD cells		274
<i>Fat1</i>	Foot process fusion, failure of foot process formation, proteinuria	SRNS, FSGS, hematuria with neurologic defects; glioblastoma, colorectal cancer, head and neck cancer	459, 649
<i>Fermt2 (Kindlin-2)</i>	Rac1 hyperactivation, podocyte effacement and proteinuria		650
<i>Flt1 (Vegfr1)</i>	Nephrotic syndrome		491
<i>Foxc1 and Foxc2</i>	Impaired podocyte differentiation, dilated glomerular capillary loop, poor mesangial migration; proteinuria and glomerulosclerosis	Anterior segment dysgenesis/Axenfeld-Rieger syndrome (iris hypoplasia and defective cornea); lymphedema-distichiasis syndrome (lower limb swelling and extra eyelashes)	83, 651, 652
<i>Foxi1</i>	Tubular acidosis; absence of CD intercalated cells	Tubular acidosis and deafness	268, 269
<i>Fyn</i>	Podocyte foot process effacement, abnormal slit diaphragms, proteinuria		478, 653
<i>Gata3</i>	Impaired maintenance of mesangial cells, dilation of glomerular capillaries, glomerulosclerosis and mesangial matrix expansion, proteinuria		654
<i>Gnas (Gαs)</i>	FSGS, mesangial expansion, proteinuria, urinary concentration defect (renin cell-specific)	Pseudohypoparathyroidism, McCune-Albright syndrome, endocrine tumors	655, 656
<i>Gne (Mnk)</i>	Hyposialylation defect, foot process effacement, GBM splitting, proteinuria and hematuria		657
<i>Grlh2</i>	Occasional unilateral renal agenesis, CD barrier dysfunction, diabetes insipidus	Autosomal dominant deafness, ectodermal dysplasia	542, 543
<i>Ilk</i>	Nephrotic syndrome (podocyte-selective); collecting duct obstruction (UB-selective)		236, 482
<i>Insr</i>	Podocyte effacement, GBM alteration, proteinuria (podocyte-selective)	Diabetic nephropathy	658
<i>Itga3 (Integrin α3)</i>	Reduced UB branching, glomerular defects, poor foot process formation		249, 251
<i>Itga6 (Integrin α6)</i>	Collecting duct dilation and dysplasia	Epidermolysis bullosa, collecting duct dysplasia	659

Table e1.2 Summary of Knockout and Transgenic Models for Kidney Development (Cont'd)

Gene Mutation or Knockout	Phenotype	Associated Human Diseases	References
<i>Itgb1</i> (<i>Integrin</i> $\beta 1$)	Podocyte loss, capillary and mesangial degeneration, glomerulosclerosis (podocyte-selective)		548, 549
<i>Itgb4</i> (<i>Integrin</i> $\beta 4$)	CD dilation and dysplasia	Epidermolysis bullosa	659
<i>Kdr</i> (<i>Flk1/Vegfr2</i>)	Thrombotic microangiopathy, ascites; renal papilla dysplasia, urine concentration defects, loss of peritubular papillary capillaries (stroma specific)	Capillary infantile hemangioma	345, 377
<i>Kirrel</i> (<i>Neph1</i>)	Abnormal slit diaphragm function, FSGS		82
<i>Lama5</i>	Defective glomerulogenesis, abnormal GBM, poor podocyte adhesion, loss of mesangial cells		233
<i>Lamb2</i>	Proteinuria prior to the onset of foot process effacement	Pierson syndrome	234, 660
<i>Lmx1b</i>	Impaired differentiation of podocytes, cytoskeletal disruption in podocytes	Nail-patella syndrome	661–663
<i>Mafb</i> (<i>Kreisler</i>)	Abnormal podocyte differentiation		434
<i>Magi2</i>	Podocyte effacement and proteinuria	SRNS, FSGS	664–666
<i>Mib1</i>	Loss of principal CD cells, polyuria, urine concentration defects, sodium wasting, hydronephrosis	Left ventricular noncompaction	272
<i>Mpp5</i> (<i>Pals1</i>)	Tubular cysts, podocyte effacement, and proteinuria		667
<i>Mpv17</i>	Nephrotic syndrome		668
<i>Mtor</i>	Proteinuria, podocyte autophagy defects (podocyte-selective)		669
<i>Myo1E</i>	Podocyte foot process effacement and proteinuria	SRNS	458, 670, 671
<i>Nck1</i> , <i>Nck2</i>	Failure of foot process formation (podocyte-selective)		64
<i>Nid1</i>	Abnormal GBM		672
<i>Notch1</i> , <i>Notch2</i>	Lack of glomerular endothelial and mesangial cells (standard knockout) lack of podocytes and proximal tubular cells (MM-selective); impaired nephrogenesis (cap mesenchyme-selective)	Alagille syndrome (cholestatic liver disease, cardiac disease, kidney dysplasia, renal cysts, renal tubular acidosis)	362, 363, 597, 673
<i>Nphs1</i> (<i>Nephrin</i>)	Absent slit diaphragms, podocyte effacement, proteinuria	Congenital nephrosis of the Finnish type, childhood-onset steroid-resistant nephritic syndrome, childhood- and adult-onset FSGS	444
<i>Nphs2</i> (<i>Podocin</i>)	Congenital nephrosis, FSGS, vascular defects	SRNS, FSGS	443, 674
<i>Npnt</i>	Mislocalization of $\alpha 8 \beta 1$ integrin, mesangial hyperproliferation and sclerosis (nephron and podocyte specific deletion)		675
<i>Nrp1</i>	Glomerular aneurysm, impaired mesangial development, glomerulosclerosis		510
<i>Par1a</i> , <i>Par1b</i>	Proximal tubule dilation and immature glomeruli		556
<i>Pdgfb</i> , <i>Pdgfrb</i>	Lack of mesangial cells, ballooned glomerular capillary loop		508, 509
<i>Pik3c3</i> (<i>Vps34</i>)	FSGS, defects in vesicular trafficking (podocyte-selective)		676, 677
<i>Podxl</i>	Podocyte effacement, reduced glomerular capillary fenestrae, abnormal GBM, anuria	Congenital FSGS, omphalocele, and microcoria	678, 679

Continued on following page

Table e1.2 Summary of Knockout and Transgenic Models for Kidney Development (Cont'd)

Gene Mutation or Knockout	Phenotype	Associated Human Diseases	References
<i>Prkci</i> (<i>aPKCλ/i</i>)	Defect of podocyte foot processes, nephrotic syndrome (podocyte-selective)		472, 473
<i>Ptpro</i> (<i>GLEPP1</i>)	Broadened podocyte foot processes with altered interdigitation patterns	SRNS	657, 680
<i>Rab3A</i>	Albuminuria, disorganization of podocyte foot process structure		681
<i>Rbpj</i>	Decreased renal arterioles, absence of mesangial cells, and depletion of renin cells (stromal cell-selective) Reduction in juxtaglomerular cells, impaired renin synthesis (renin cell-selective) Loss of principal CD cells (UB-specific)		274, 364, 432
<i>Rhpn1</i>	FSGS, podocyte foot process effacement, GBM thickening		464
<i>Robo2</i>	Abnormal pattern of podocyte foot process interdigitation, focal effacement of foot processes, proteinuria	CAKUT, VUR	682
<i>Slc5a2</i> (<i>SGLT2</i>)	Elevated urinary excretion of glucose, calcium, and magnesium	Glucosuria	683
<i>Sh3gl1</i> , <i>Sh3gl2</i> , <i>Sh3gl3</i> (<i>Endophilin 1/2/3</i>)	Podocyte foot process effacement and proteinuria, neuronal defects		647
<i>Sirpa</i>	Irregular podocyte foot process interdigitation, mild proteinuria		684
<i>Sox4</i>	Oligonephronia, podocyte effacement, GBM defects (MM-selective)		685
<i>Sox17</i> , <i>Sox18</i>	Vascular insufficiency in kidneys and liver; ischemic atrophy of renal and hepatic parenchyma; defective postnatal a	HLT (hypotrichosis-lymphedema-telangiectasia) syndrome (hair, vascular, and lymphatic disorder)	403, 406
<i>Sv2b</i>	Podocyte foot process effacement and proteinuria		686
<i>Synj1</i>	Podocyte foot process effacement and proteinuria; neuronal defects		647
<i>Tcf21</i> (<i>Pod1</i>)	Lung and cardiac defects, sex reversal and gonadal dysgenesis, vascular defects, disruption in UB branching, impaired podocyte differentiation, dilated glomerular capillary, poor mesangial migration		13, 339, 440
<i>Tek</i> (<i>Tie2</i>)	Loss of ascending vasa recta and medullary capillary plexus, urinary concentration defects	Cutaneous and mucosal venous malformations, congenital glaucoma	407
<i>Tfcp2l1</i>	Loss of CD intercalated cells		273
<i>Tjp1</i> (<i>ZO-1</i>)	Podocyte effacement and proteinuria		687
<i>Trp63</i> (<i>TP63</i>)	Loss of CD intercalated cells	ADULT (acro-dermato-ungual-lacrima-tooth) syndrome; limb-mammary syndrome	270
<i>Trpc6</i>	Protected from angiotensin-mediated or proteinuria or complement-dependent glomerular injury (null mutation); podocyte foot process effacement and proteinuria (transgenic overexpression in the podocyte lineage)	SRNS, FSGS	460, 489, 490, 688–691
<i>Vangl2</i>	Immature and poorly branched glomerular tuft	Neural tube defects	332

Table e1.2 Summary of Knockout and Transgenic Models for Kidney Development (Cont'd)

Gene Mutation or Knockout	Phenotype	Associated Human Diseases	References
<i>Vegfa</i>	Endotheliosis, disruption of glomerular filtration barrier formation, nephrotic syndrome (podocyte-selective); peritubular capillary rarefaction and polycythemia (tubule-specific)		375, 376, 378, 692
<i>Vhl</i>	Glomerulonephritis (podocyte-selective)	Von Hippel-Lindau syndrome	400
<i>Wasl</i> (<i>N-wasp</i>)	Podocyte effacement, proteinuria		693
<i>Wnt7b</i>	Impaired development medullary microvasculature		694
<i>Wnt11</i>	Glomerular cysts		695

ADPKD, Autosomal dominant polycystic kidney disease; *ARPKD*, autosomal recessive polycystic kidney disease; *CAKUT*, congenital anomalies of the kidney and urinary tract; *CD*, collecting duct; *CM*, cap mesenchyme; *FSGS*, focal segmental glomerulosclerosis; *GBM*, glomerular basement membrane; *GDNF*, glial cell-derived neurotrophic factor; *MM*, metanephric mesenchyme; *SRNS*, steroid-resistant nephrotic syndrome; *UB*, ureteric bud; *VUR*, vesicoureteral reflux.

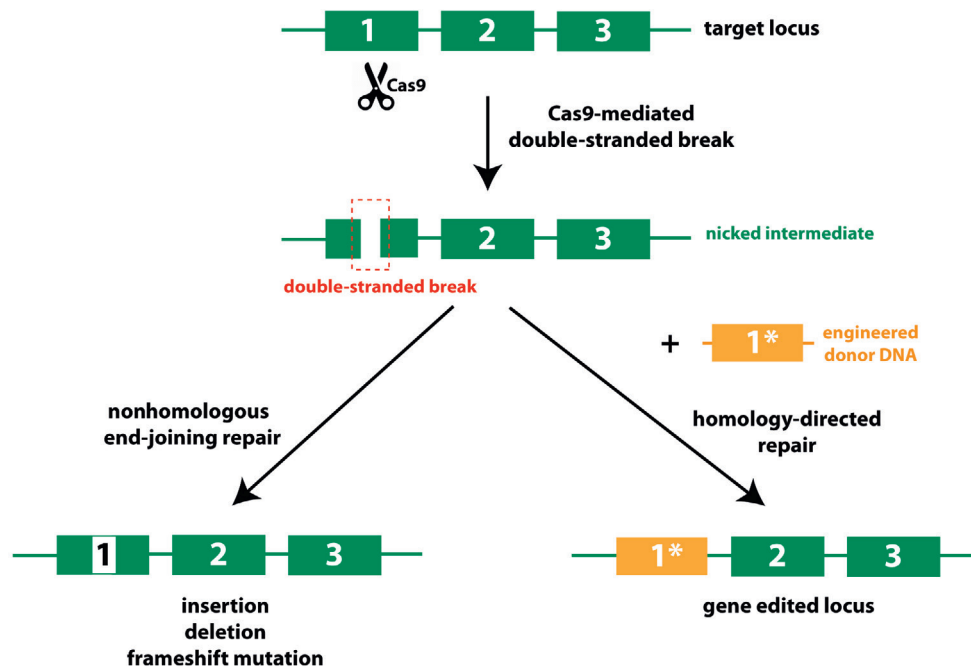


Fig. 1.12 Gene editing using the CRISPR-Cas9 system. Simplified overview for targeted introduction of mutations using the CRISPR-Cas9 system. Site-specific cleavage of a target gene locus with the Cas9 endonuclease creates a double-strand DNA break. The nicked target gene can be repaired by an error-prone, nonhomologous repair mechanism resulting in nucleotide insertion, deletion, or frameshift mutations. Introduction of a homologous engineered donor DNA results in a homologous substitution into the target locus which then results in a precisely edited gene.

dominant. Breeding of ENU-injected male mice results in offspring that can then be screened for various renal phenotypes (e.g., dysplasia, cysts, proteinuria) and heritability.^{69–71} Some examples of genes whose relevance to kidney development functions was revealed from ENU mutagenesis include *Arhgap23* (ciliogenesis and glomerular cyst defects), *Six1* (renal hypoplasia), *Scl5a2* (*SGLT2*; glucosuria), *Pou3f3* (*Bmn1*; oligonephronia and smaller loop of Henle), and *Aqp11* (proximal tubule vacuolation and cysts).^{72–74}

Significant innovations in high-throughput gene sequencing are also facilitating the identification of monogenic mutations associated with heritable kidney diseases.^{75,76} Putative disease-causing genes identified from genome-wide association studies can then be validated in corresponding gene-targeted mouse models. Similarly, a gene whose mutation has been identified to cause kidney maldevelopment and dysfunction in mice can be investigated for incidence of mutations across the human genome (e.g., *Arhgdia* in nephrotic syndrome).^{77–80} Other genome-wide approaches that have led to the discovery of novel genes in kidney development and disease include gene trap consortia^{81,82} and genome-wide transcriptome and proteome projects.^{83–85}

IMAGING AND LINEAGE TRACING STUDIES

Reporter mouse lines are animal models engineered to express transgenes encoding an enzyme (e.g., bacterial β -galactosidase or firefly luciferase) or fluorescent protein tag in a cell lineage-specific manner (Fig. 1.13). In the simplest model, a single transgene comprising the reporter coding region is placed downstream of a cell-specific promoter. Such a model is useful to identify and follow distinct cell lineages where the chosen promoter is active. *Hoxb7-EGFP* is the first

fluorescent transgene developed to visualize renal development.⁸⁶ Enhanced green fluorescent protein (EGFP), placed under the control of the *Hoxb7* promoter, specifically labels the wolffian duct and the ureteric epithelial lineage. *Hoxb7-EGFP* has therefore proven to be invaluable in studying the rates and pattern of ureteric branching morphogenesis and ureteral development, including disruption of these events in the context of particular mutant backgrounds.^{60,87–89} An alternative model uses two separate transgenes, namely a cell-specific Cre driver and an independent reporter transgene (e.g., R26R). With the R26R reporter model, β -galactosidase transgene expression is switched on by a Cre-mediated removal of an upstream stop codon.⁹⁰ Inducible reporter systems incorporate genetic regulatory elements that allow for both cell-specific promoter- and drug-dependent activation of the reporter gene. The most commonly used strategies for controlled transgene expression in mice are the Tet-operon/repressor bi-transgenic and the estrogen-receptor ligand binding domain (Cre-ERT2) systems. These inducible reporter systems allow for timed pulse labeling of cell lineages, resulting in permanent tagging of progenitor cells and their direct derivatives. Inducible reporter systems have become valuable tools for the mapping of cell fates. Inducible reporter systems have been used to establish that the *Six2*⁺ cells are the progenitors of all epithelial structures of the mature nephron.⁶¹ In some cases, the Cre transgene is engineered with an associated fluorophore expression cassette under the control of a shared promoter, thus allowing easy identification of cells with targeted gene mutations. A wide variety of reporter and Cre transgenes (see Table e1.2) are now available to characterize the development and organization of multiple compartments of the kidney.

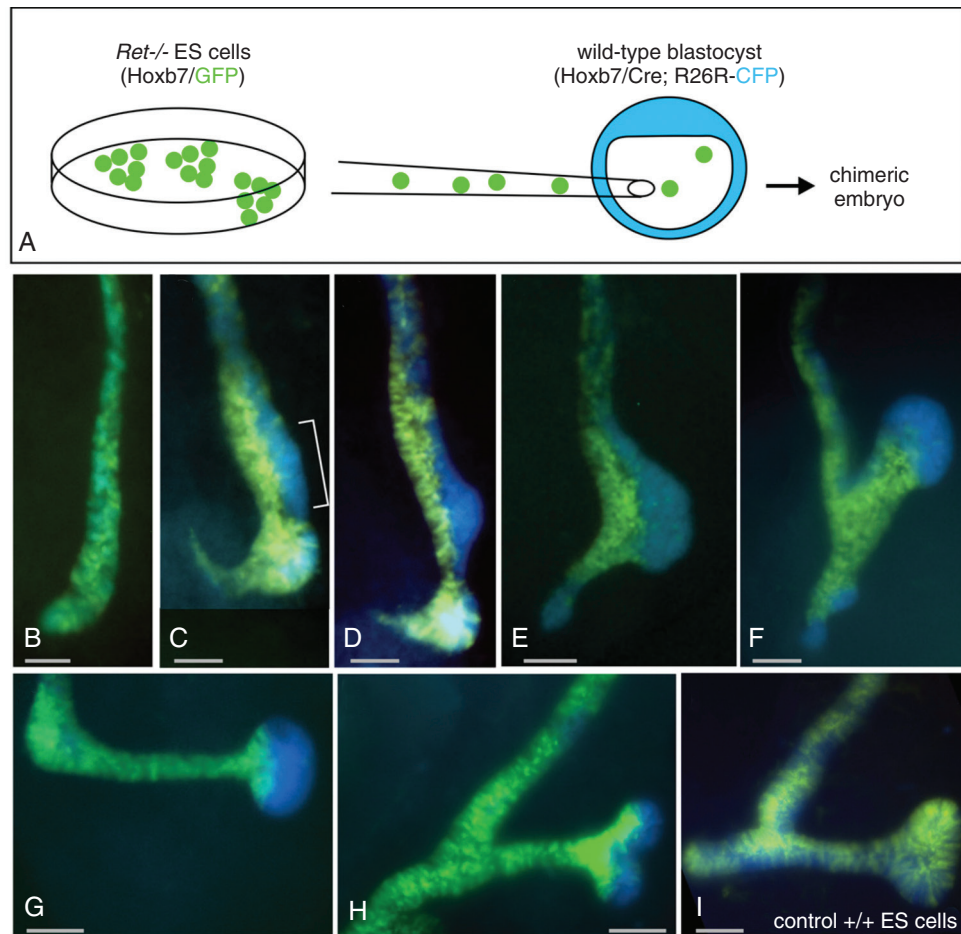


Fig. 1.13 Cell fate tracing through genetic expression of fluorophores. Segregation of *Ret*-deficient cells in the outgrowth and branching of the ureteric bud (UB). (A) *Ret*-null embryonic stem cells (ES) expressing *Hoxb7-GFP* were mixed with a wild-type transgenic blastocyst (*Hoxb7-Cre; R26R-CFP*). This generates chimeric animals in which *Ret*-null cells exhibit green fluorescent protein (GFP) fluorescence, while wild-type UB cells express cyan fluorescent protein (CFP). (B) At E9.5, *Ret*-null epithelial cells are intermingled with wild-type cells in the wolffian duct (WD). (C) At E10, when the dorsal side of the WD begins to swell, the region where UB will emerge becomes enriched with CFP-expressing but not *Ret*-null cells. (D, E) At around E10.5 dpc, the UB is exclusively formed by wild-type cells. (F) Upon elongation of the UB at E11, the bulbous distal tip of the UB is formed by wild-type cells but the *Ret*-null cells begin to contribute to the trailing trunk structure. (G, H) During the initial branching of the UB at around E11.5, *Ret*-null cells are excluded from the distal ampullary UB tips. (I) In contrast, control cells expressing *Ret* and GFP contribute to the whole branching UB structure. dpc, Days post coitum. Reproduced with permission from *Developmental Cell*.¹⁸⁶

Reporter mice expressing fluorescent tags also come in handy for segregation and isolation of particular cell types on the basis of fluorescence, facilitating global gene expression analysis, elucidation of transcriptional regulatory networks, and epigenetic interactions that orchestrate particular morphogenetic events (Table 1.3). Purified single or bulk populations of fluorescence-tagged cells can be used for high-throughput discovery of transcriptomes, genome-wide DNA-binding protein interactions, histone modifications, and nucleosome complexes through revolutionary next-generation DNA sequencing methods such as RNA-seq (RNA sequencing) and ChIP-seq (chromatin immunoprecipitation sequencing).^{91–93}

NONMAMMALIAN MODEL SYSTEMS FOR KIDNEY DEVELOPMENT

Organisms separated by millions of years of evolution from humans still provide useful models to study the genetic basis and function of mammalian kidney development. This stems

from the fact that all of these organisms possess excretory organs designed to remove metabolic wastes from the body, and that genetic pathways involved in other aspects of invertebrate development may serve as templates to dissect pathways in mammalian kidney development. In support of the latter argument, elucidation of the genetic interactions and molecular mechanism of the Kirrel (Neph1) orthologue and nephrin-like molecules SYG1 and SYG2 in synapse formation in the soil nematode *Caenorhabditis elegans* is providing major clues to the function of these genes in glomerular and slit diaphragm formation and function in mammals.⁹⁴

The excretory organs of invertebrates differ greatly in their structure and complexity and range in size from a few cells in *C. elegans*, to several hundred cells in *Drosophila*, to the more recognizable kidneys in amphibians, birds, and mammals. In *C. elegans*, the excretory system consists of a single large H-shaped excretory cell, a pore cell, a duct cell, and a gland cell.^{95,96} *C. elegans* provides many benefits as a model system: the availability of powerful genetic tools

Table 1.3 Mouse Strains for Conditional Gene Targeting and Lineage Marking of Cells

Gene Promoter	Cre	rtTA	Cre-ERT2	Fluorescent Reporter	Renal Expression	Extrarenal Expression	Reference
<i>11Hsd2</i>	✓			✓	Principal cells of collecting duct, connecting tubules	Amygdala, cerebellum, colon, ovary, uterus, epididymis salivary glands	696
<i>Aqp2</i>	✓		✓		Principal cells of collecting duct	Testis, vas deferens	697
<i>Atp6v1b1</i>	✓			✓	Collecting ducts (intercalated cells), connecting tubule		698, 699
<i>Bmp7</i>	✓		✓	✓	Cap mesenchyme		700
<i>Cdh16 (Ksp-Cadherin)</i>	✓		✓		Renal tubules, collecting ducts, ureteric bud, wolffian duct, mesonephros	Müllerian duct	701, 702
<i>Cited1</i>			✓	✓	Cap mesenchyme		703
<i>Emx1</i>	✓		✓	✓	Renal tubules (proximal and distal tubules)	Cerebral cortex, thymus	704
<i>Foxd1</i>	✓		✓	✓	Stromal cells		705
<i>Gdnf</i>			✓	✓	Cap mesenchyme		706
<i>Ggt1</i>	✓				Cortical tubules		707
<i>Hoxb6</i>	✓		✓		Metanephric mesenchyme	Lateral mesoderm, limb buds	547, 708
<i>Hoxb7</i>	✓	✓		✓	Ureteric bud, wolffian duct, collecting ducts, distal ureter	Spinal cord, dorsal root ganglia	59
<i>Kap</i>	✓				Proximal tubules	Brain	709, 710
<i>Klf3</i>			✓		Collecting ducts	Gonads	711, 712
<i>Nphs1</i>	✓	✓		✓	Podocytes	Brain	713
<i>Nphs2</i>	✓	✓	✓	✓	Podocytes		714, 715
<i>Osr1</i>	✓		✓	✓	Metanephric mesenchyme	Intermediate mesoderm	716
<i>Osr2</i>	✓				Condensing metanephric mesenchyme; glomeruli	Palatal mesenchyme	
<i>Pax2</i>	✓			✓	Pronephric duct, wolffian duct, ureteric bud, cap mesenchyme	Inner ear, midbrain, cerebellum, olfactory bulb	717
<i>Pax3</i>	✓				Metanephric mesenchyme	Neural tube, neural crest	713, 718, 719
<i>Pax8</i>	✓	✓	✓	✓	Renal tubules (proximal and distal tubules) and collecting ducts (Tet-On inducible system)		720, 721
<i>Pck1</i>	✓		✓	✓	Proximal tubules	Liver	630
<i>Pdgfb</i>			✓		Endothelium	Systemic vascular endothelium	722
<i>Pdgfrb</i>	✓	✓	✓	✓	Mesangial cells, vascular smooth muscles	Pericytes, vascular smooth muscles	396, 510, 723
<i>Prox1</i>	✓		✓	✓	Ascending vasa rectae, lymphatic vessels	Systemic lymphatic vasculature	724–726
<i>Rarb</i>	✓			✓	Metanephric mesenchyme		550
<i>Ren1</i>	✓		✓	✓	Juxtaglomerular cells, afferent arterioles, mesangial cells	Adrenal gland, testis, sympathetic ganglia	420
<i>Ret</i>			✓		Ureteric bud, collecting ducts	Dorsal root ganglion, neural crest	727
<i>Sall1</i>			✓		Metanephric mesenchyme (tamoxifen-inducible system)	Limb buds, central nervous system, heart	728
<i>Six2</i>	✓	✓	✓	✓	Cap mesenchyme		61
<i>Slc22a6</i>			✓		Proximal tubules		721
<i>Slc5a2</i>	✓				Proximal tubules		729
<i>Sox18</i>			✓		Cortical and medullary vasculature	Blood vessel and precursor of lymphatic endothelial cells	730–732

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Table 1.3 Mouse Strains for Conditional Gene Targeting and Lineage Marking of Cells (Cont'd)

Gene Promoter	Cre	rtTA	Cre-ERT2	Fluorescent Reporter	Renal Expression	Extrarenal Expression	Reference
<i>Spink3</i>	✓				Medullary tubules (distal or connecting tubules?)	Mesonephric tubules, pancreas, lung, liver, gastrointestinal tract	718, 719, 733
<i>T (Brachyury)</i>	✓		✓		Whole kidney (both ureteric bud and metanephric mesenchyme)	Pan-mesodermal	286
<i>Tbx18</i>	✓				Ureteral mesenchyme	Heart, limb buds	734
<i>Tcf21</i>	✓				Metanephric mesenchyme, cap mesenchyme, podocytes, stromal cells	Epicardium, lung mesenchyme, gonad, spleen, adrenal gland	246
<i>Tek</i>	✓	✓	✓	✓	Endothelium	Systemic vasculature endothelium	735–737
<i>Tie1</i>	✓	✓			Endothelium	Systemic vasculature endothelium	738
<i>Umod</i>	✓		✓		Thick ascending limbs of loops of Henle	Testis, brain	739
<i>Wnt4</i>	✓		✓	✓	Renal vesicles, nascent nephrons (comma- and S-shaped bodies)	Lungs, developing gonads	61, 740

^aGenitourinary Development and Molecular Anatomy Project (GUDMAP); <http://www.gudmap.org>.

Cre (noninducible Cre recombinase transgene); rtTA (reverse tetracycline transactivator, tetracycline inducible expression system); Cre-ERT2 (Cre-estrogen receptor ligand binding domain fusion transgene, tamoxifen-inducible expression system); fluorescent reporter (promoter-driven expression of a fluorescent protein such as green fluorescent protein and its variants.)

including “mutants by mail,” a short life and reproductive cycle, a publicly available genome sequence and resource database (<http://www.wormbase.org>), the ease of performing genetic enhancer-suppressor screens in worms, and the fact that they share many genetic pathways with mammals. Major contributions in our understanding of the function of polycystic and cilia-related genes have been made from studying *C. elegans*. The *Pkd1* and *Pkd2* homologues, *LOV1* and *LOV2* of *C. elegans*, are involved in cilia development and mating behavior.^{97,98} Strides in understanding the function of the slit diaphragm have also been made from *C. elegans* as described earlier.

Similar to *C. elegans*, the relative ease of large-scale genetic screens and phenotypic characterization in *Drosophila* makes it another valuable complimentary model for understanding the genetic basis of developmental processes. The excretory system of *Drosophila* consists of two parts, the nephrocytes and the malpighian tubules, which are functionally analogous to podocytes and renal tubules, respectively. A fundamental difference from vertebrate kidneys is that the nephrocytes and malpighian tubules are not physically connected. Nephrocytes either surround the heart (epicardial nephrocytes) or the esophagus (garland cells) and have elaborate membrane invaginations that closely resemble the glomerular filtration barrier. Remarkably, mutations of *Drosophila* homologs of genes known to be essential to form slit diaphragms and maintain podocyte functions also impair nephrocyte morphology and filtration functions.^{99–103} Similarly, conserved genes have been identified that regulate normal patterning and function of Malpighian tubules and vertebrate renal tubules.^{104–109} Functional readouts such as impaired

nephrocyte filtration or uptake of tracers, ultrastructural analysis, and mortality screens can be executed efficiently in *Drosophila*, which can facilitate the characterization of novel gene functions that are vital for renal filtration.^{110–112}

The pronephros is the functional kidney of the larva of some fishes (with the exception of jawless fishes which only develop the pronephros) and amphibians, while the mesonephros serve as the kidney in adults of these aquatic animals. The pronephros of the zebrafish (*Danio rerio*) larva consists of two tubules connected to a fused, single, midline glomerulus. The zebrafish pronephric glomerulus expresses many of the same genes found in mammalian glomeruli (e.g., *Vegfa*, *Nphs1*, *Nphs2*, and *Wt1*) and contain podocytes and fenestrated endothelial cells.¹¹³ Advantages to the zebrafish as a model system include its short reproductive cycle, transparency of the larvae with easy visualization of defects in pronephric development without sacrificing the organism, the availability of the genome sequence, the ability to rapidly knockdown gene function using morpholino oligonucleotides, and the ability to perform functional studies of filtration using fluorescently tagged labels of varying sizes.¹¹⁴ These features make zebrafish amenable to both forward and reverse genetic screens. Currently, multiple labs perform knockdown screens of mammalian homologs and genome-wide mutagenesis screens in zebrafish in order to study renal function. The pronephros of the clawed frog *Xenopus laevis* has also been used as a simple model to study early events in nephrogenesis.^{115,116} Similar to the fish, the pronephros of the clawed frog consists of a single glomus, paired tubules, and a duct. The fact that *X. laevis* embryos develop rapidly outside the body (all major organ systems are formed by 6 days of age),

the ease of injecting DNA, mRNA, and protein, ability to perform grafting, and in vitro culture experiments establish the frog as a valuable model system to dissect early inductive and patterning cues.^{117,118} The availability of transgenic zebrafish and *Xenopus* lines that express the fluorescent protein EGFP in pronephric and mesonephric kidneys provide an opportunity to visualize real-time kidney development and function.^{119–121}

STEM CELL DERIVED KIDNEY ORGANIDS

The discovery that a combination of small molecules and cytokines can effectively induce human pluripotent stem cells (hPSCs) to develop into intermediate mesoderm laid the groundwork for ingenious in vitro methods to generate kidney organoid cultures in which most fundamental processes in early renal development are recapitulated.^{65,122–128} These hPSC-derived kidney organoids can form many renal structures, including the glomerulus, proximal tubule, loop of Henle, distal tubule, collecting ducts, interstitium, and a primitive endothelial network. Gene editing of hPSCs through CRISPR-Cas9 technology (see Fig. 1.12) provides a unique opportunity for a time- and cost-effective strategy to interrogate novel gene functions relevant to embryonic kidney development.^{66,129–131} One particular example is the study of epithelial cyst pathogenesis in kidney organoids from polycystin-deficient hPSCs.⁶⁵ Although their use in renal replacement therapy remains to be realized, kidney organoids have immediate practical application in high-throughput screening of drugs for nephrotoxicity.¹³²

GENETIC ANALYSIS OF MAMMALIAN KIDNEY DEVELOPMENT

INTERACTION OF THE URETERIC BUD AND THE METANEPHRIC MESENCHYME

The classic studies with the organ culture system that started in the 1950s provided an extensive framework that formed the basis for further studies of organ development.^{38,133,134} These elegant studies demonstrated that the epithelialization of the MM requires a UB-derived factor which we now know as Wnt9b.¹³⁵ However, the modern era of studies on the early development of the kidney began with the observation of renal agenesis phenotypes in gene-targeted or knockout mice, the earliest among these being the knockout of several transcription factors, including the *Wt1*, *Pax2*, *Eya1*, *Osr1* (*Odd1*), *Six1*, *Sall1*, *Lhx1* (*Lim1*), and *Emx2*.^{39,40,47,136–141} The knockout of genes for several secreted signaling molecules such as GDNF (glial cell line–derived neurotrophic factor), GDF11 (growth differentiation factor 11), gremlin, and the receptors Ret and GFR α 1 also resulted in renal agenesis, at least in the majority of embryos.^{142–148}

Clinical Relevance

Perturbation of cell–cell communication during embryonic kidney development can have wide-ranging detrimental consequences, including renal agenesis, CAKUT, proteinuria, kidney cysts, defective urine osmoregulation, acidosis, and predisposition to hypertension and chronic kidney diseases.

EARLY LINEAGE DETERMINATION OF THE METANEPHRIC MESENCHYME

In most embryos exhibiting renal agenesis, an appropriately localized putative MM is often uninvaded by a UB outgrowth. Two exceptions are the *Osr1* (*Odd1*)- and *Eya1*-mutant embryos, where this distinct patch of MM is absent, suggesting that *Osr1* and *Eya1* represent early determinants of the MM (Fig. 1.14). Together, the phenotypes of these knockout mice have provided an initial molecular hierarchy of early kidney development.^{136,149} *Osr1* marks the intermediate mesoderm from which the mesenchymal cells within the mesonephric and metanephric kidney are derived and is subsequently downregulated upon epithelial differentiation. Mice lacking *Osr1* do not form the MM and do not express several other factors required for metanephric kidney formation, including *Eya1*, *Six2*, *Pax2*, *Sall1*, or *Gdnf*.¹⁴⁹

Eya1-mediated specification of the MM cell fate is thought to occur via interaction with another transcription factor, *Six1*. *EYA1* and *SIX1* mutations are found in humans with branchiootorenal (BOR) syndrome.¹⁵⁰ It is now known through in vitro experiments that *Eya1* and *Six1* form a regulatory complex that appears to be involved in transcriptional regulation.^{151,152} Interestingly, *Eya1* was shown to have an intrinsic phosphatase activity that regulates the activation of the *Eya1/Six1* complex.^{152,153} Moreover, *Eya* and *Six* family genes are coexpressed in several tissues in mammals, *Xenopus* and *Drosophila*, further supporting a functional interaction between these genes.^{136,138,139,154,155} Direct transcriptional targets of this complex appear to include the pro-proliferative factor *Myc*.¹⁵² In the *Eya1*-deficient urogenital ridge the putative MM is completely absent.¹⁵⁶ Consistent with this finding, *Six1* is either absent or poorly expressed in the presumptive location of the MM of *Eya1*-null embryos.^{152,154–156} *Eya1* is expressed in the *Six1*-null mesenchyme, suggesting that *Eya1* is upstream of *Six1*.^{138,139}

The transcription factor *Wt1* is another essential regulator of early MM development. *Wt1* expression is weak in the uninduced MM but increases in the condensed cap mesenchyme surrounding the branching UB tips. *Wt1* expression remains throughout nephrogenesis but eventually becomes restricted to the presumptive podocytes at the proximal end of the S-shaped nephron. Mature podocytes continue to express *Wt1* at high levels. Genetic loss of *Wt1* in mice prevents UB outgrowth and causes apoptosis of the MM, whereas human mutations of *WT1* have been linked to renal tumors.^{39,157} Among the numerous identified transcriptional targets of *Wt1* known to be required for kidney development are *Bmp7*, *Pax2*, and *Sall1*.⁴⁸ More recently, it has been shown that *Wt1* regulates antagonistic fibroblast growth factor (FGF) and bone morphogenetic protein (BMP)/SMAD (contraction of homologous Sma and MAD genes of worms and fruitflies) signaling pathways, effectively promoting the proliferation and survival of the MM.^{158,159} The absence of *Wt1* significantly downregulates the expression of the genes for several FGF ligands, including *Fgf8*, *Fgf10*, *Fgf16*, and *Fgf20*, which support mesenchymal proliferation.^{48,159} Furthermore, *Wt1* deficiency in nephron progenitors leads to specific loss of *Gas1*, a gene encoding for an extracellular glycosphingolipid-tethered protein that modulates PI3K-Akt signaling downstream of FGFs.¹⁵⁸ The impairment of FGF signaling upon loss of *Wt1* is exacerbated by the upregulation of BMP/SMAD signaling,

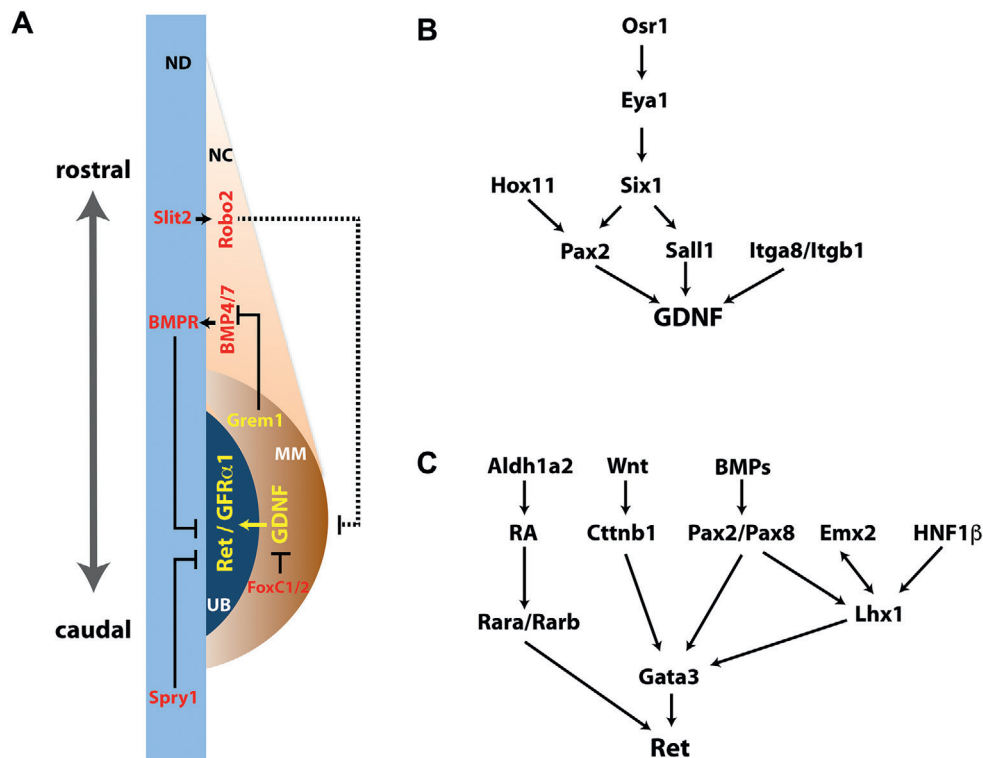


Fig. 1.14 Genetic interactions during early metanephric kidney development. (A) Regulatory interactions that control the strategically localized expression of glial cell-derived neurotrophic factor (GDNF) and Ret and the subsequent induction of the ureteric bud. The anterior part of GDNF expression is restricted by Foxc1/2 and Slit2/Robo2 signaling. Spry1 suppresses the postreceptor activity of Ret. BMP4/7-BMPR signaling inhibits the response to GDNF, an effect counteracted by Grem1. Genetic regulatory network that controls the expression of (B) GDNF and (C) Ret. *BMPR*, Bone morphogenetic protein receptor; *MM*, metanephric mesenchyme; *NC*, nephrogenic cord; *ND*, nephric duct; *UB*, ureteric bud.

which promotes apoptosis.¹⁵⁹ This is thought to occur through the loss of *Bmper* expression, a direct target of *Wt1*, which inhibits BMP4 signaling.

URETERIC BUD INDUCTION

In many cases of renal agenesis, a failure of the GDNF-Ret signaling axis has been identified.¹⁶⁰ GDNF, a member of the TGF- β superfamily and secreted by the MM, activates the Ret-GFR α 1 receptor complex that is expressed by cells of the nephric duct and the UB. Activation of the Ret tyrosine kinase is of central importance to UB induction. Most mutant embryos lacking *Gdnf*, *Ret*, or *Gfra1* exhibit partial or complete renal agenesis due to severe impairment of UB induction while exogenous GDNF is sufficient to induce sprouting of ectopic buds from the nephric duct.^{142–144,148,161–164} Consistently, other genes linked to renal agenesis are known to regulate the normal expression of GDNF. These include transcription factors (e.g., *Eya1*, *Pax2*, *Six1*, *Hox11* paralogues, and *Sall1*) and proteins required to stimulate or maintain GDNF expression (e.g., GDF11, Kif26b, nephronectin, α 8 β 1-integrin, and Frs1) (see Fig. 1.14).^{136,139,140,145,165–173}

As described earlier, *Eya1* mutants fail to form the MM. *Pax2* is a transcriptional regulator of the paired box family and is expressed widely during the development of both UB and mesenchymal components of the urogenital system.¹⁷⁰ In *Pax2*-null embryos, *Eya1*, *Six1* and *Sall1* are expressed,¹⁵⁶ suggesting that *Eya1* and *Six1* are likely upstream of *Pax2*. Through a combination of molecular and in vivo studies, it

has been demonstrated that *Pax2* appears to act as a transcriptional activator of *Gdnf* and regulates the expression of *Ret*.^{171,174} *Pax2* also appears to regulate kidney formation through epigenetic control as it is involved in the assembly of a histone H3–lysine 4 methyltransferase complex through the ubiquitously expressed nuclear factor PTIP, which regulates histone methylation.¹⁷⁵ The *Hox* genes are conserved in all metazoans and specify positional information along the body axis. *Hox11* paralogues include *Hoxa11*, *Hoxc11*, and *Hoxd11*. Mice carrying mutations in any one of these genes do not have kidney abnormalities; however, triple-mutant mice for these genes demonstrate a complete absence of metanephric kidney induction.¹⁷² Interestingly, in this mutant, the formation of condensing MM and the expression of *Eya1*, *Pax2*, and *Wt1* remain unperturbed, suggesting that *Hox11* is not upstream of these factors. Although there seems to be some hierarchy, *Eya1*, *Pax2*, and *Hox11* appear to form a complex to coordinately regulate the expression of *Gdnf*.¹⁷⁶

Sall1 indirectly controls the expression of GDNF. *Sall1* is necessary for the expression of the kinesin *Kif26b* by the MM cells.¹⁶⁷ In the absence of either *Sall1* or *Kif26b*, the nephronectin receptor α 8 β 1-integrin expressed by the MM mesenchyme is downregulated. The loss of *Sall1*, *Kif26b*, *Itga8* (α 8 integrin), *Itgb1* (β 1-integrin), and *Npnt* (nephronectin) compromises the adhesion of the MM cells to the UB tips, ultimately causing loss of *Gdnf* expression and failure of UB outgrowth.^{168,169,177} The absence of Frs1, an extracellular matrix protein linked to Fraser syndrome, which is expressed

selectively in the UB epithelium and nascent epithelialized nephrons but not the MM, causes loss of *Gdnf* expression.¹⁶⁵ *Fras1* likely regulates MM induction and GDNF expression via multiple signaling pathways. *Fras1* deficiency results in downregulation of *Gdf11*, *Hox11*, *Six2*, and *Itga8*, and an increase in *Bmp4*, which altogether cooperatively controls *Gdnf* expression.¹⁶⁵

GENES REQUIRED BY THE URETERIC BUD

Several components of the genetic network supporting the development of the nephric duct and the UB have been identified (see Fig. 1.14). *Pax2* and *Pax8* are required to maintain the expression of *Lhx1*.¹⁷⁸ *Pax2*, *Pax8*, and *Lhx1* altogether likely coordinate the expression of *Gata3*, which is necessary for elongation of the nephric duct.¹⁷⁹ *Gata3* and *Emx2*, which are required for the expression of *Ret* in the nephric duct, are both regulated by β -catenin (*Ctnnb1*), an effector of the canonical Wnt signaling pathway.^{40,180,181} Acting likely in parallel with *Gata3* to maintain *Ret* expression in the UB is *Aldh1a2* (*Raldh2*), a gene in the retinoic acid synthesis pathway.¹⁸² Surprisingly, this genetic regulatory hierarchy cannot fully account for the distinctive phenotypes arising from the mutation of each individual gene, suggesting that additional important components of the nephric duct genetic network have yet to be identified. Nephric duct specification fails in *Pax2/Pax8* mutants but not in the case of *Lhx1* deficiency, where only the caudal portion of the nephric duct degenerates.¹⁷⁸ The absence of *Gata3* or *Aldh1a2* causes misguided elongation of the nephric duct, terminating into either blind-ended ureters or abnormal connections between the bladder and urethra.¹⁷⁹ The curtailed caudal growth of the nephric duct when either *Lhx1* or *Gata3* is lost prevents the formation of the first UB and consequently causes renal agenesis.^{179,183,184} The absence of *Aldh1a2* leads to the formation of ectopic ureters and hydronephrotic kidneys.¹⁸² *Emx2* deficiency does not prevent caudal extension of the nephric duct toward the presumptive MM but the evagination of the UB is aborted, thereby resulting in renal agenesis.⁴⁰ Without β -catenin, nephric duct cells undergo precocious differentiation into collecting duct epithelia.¹⁸⁵ *Ret* does not affect the nephric duct fate but has importance in later UB development and insertion of the nephric duct to the cloaca.^{162,182,186} Identification of additional targets of *Pax2*, *Pax8*, *Lhx1*, *Gata3*, and β -catenin are necessary in order to fully understand these seemingly disparate mutant phenotypes.

UB induction and subsequent branching requires a unique spatial organization of Ret signaling. The bulbous UB tip is a region enriched with proliferative ureteric epithelial cells, in contrast to the emerging stalk regions of the developing ureteric tree.^{41,187} It is now well appreciated that receptor tyrosine kinase signaling primarily through Ret is key to the proliferation of UB tip epithelia. Exogenous GDNF supplemented in explanted embryonic kidneys can cause expansion of the UB tip region toward the source of the ligand.^{187–189} ERK kinase activation is prominent within the ampullary UB terminals where Ret expression is elevated.⁴¹ Consistently, chimera analysis in mice reveals that Ret-deficient cells do not contribute to the formation of the UB tips.¹⁶² Altogether, these studies underscore the importance of strategic levels of Ret expression and activation of proliferative signaling pathways in the stereotypical sculpting of the nascent collecting duct network.

A ligand-receptor complex formed by GDNF, GFR α 1, and Ret is necessary for autophosphorylation of Ret on its intracellular tyrosines (see Fig. 1.12). A number of downstream adaptor molecules and effectors have been identified to interact with active phosphorylated Ret, including Grb2, Grb7, Grb10, ShcA, Frs2, PLC γ 1, Shp2, Src, and Dok adaptor family members (Dok4/5/6).^{190–201} These downstream Ret effectors altogether are likely contributors to the activation of the Ras/SOS/ERK and PI3K/Akt pathways supporting the proliferation, survival, and migratory behavior of the UB epithelium.^{41,43,202} Knock-in mutations of the interaction site for Shc/Frs2/Dok adaptors on the short isoform of Ret lead to the formation of rudimentary kidneys.^{203–206} Specific mutation of the PLC γ 1 docking site on Ret leads to renal dysplasia and ureter duplications.²⁰³ The loss of *Shp2* and the upstream ERK regulators *Map2k1* (*Mek2*) and *Map2k2* (*Mek1*) in the UB lineage also cause severe renal hypoplasia phenocopying that is observed in occasional Ret-deficient kidneys.^{207,208} UB-specific inactivation of *Pten*, a target of the PI3K/Akt pathway, disrupts UB branching.²⁰⁹ Taken together, these findings underscore the significance of Ret signaling in normal UB branching.

A number of transcriptional targets of Ret activation in microdissected UB stimulated with GDNF have been elucidated (Fig. 1.15).²¹⁰ Among these are *Ret* itself and *Wnt11*, which stimulates *Gdnf* expression in the MM,²¹¹ suggesting that a positive feedback loop exists for the GDNF-Ret signaling pathway. Ret activation also positively regulates the ETS transcription factors *Etv4* and *Etv5*, which are also necessary for normal UB branching morphogenesis. *Etv4*-null homozygous mutants and compound heterozygous mutants for *Etv4* and *Etv5* manifest severe renal hypoplasia or renal agenesis, suggesting that these transcription factors are indispensable targets of Ret for proper UB development.²¹⁰ In chimeric animals, *Etv4/Etv5*-deficient cells, like Ret-deficient cells, fail to integrate within the UB tip domain.^{162,212}

The gene *Sprouty* was identified as a general antagonist of receptor tyrosine kinases and was discovered for inhibiting the FGF and EGF signaling pathways that pattern the *Drosophila* airways, wings, and ovarian follicles.^{213–215} Of the four mammalian *Sprouty* homologues, *Spry1*, *Spry2*, and *Spry4* are expressed in developing kidneys.²¹⁶ *Spry1* is expressed strongly at the UB tips, whereas *Spry2* and *Spry4* are found in both the UB and the MM.²¹⁷ *Sprouty* molecules are thought to uncouple receptor tyrosine kinases with the activation of ERK pathways either through competitive binding with the Grb2/SOS complex or the kinase Raf, effectively repressing ERK activation. Interestingly, *Spry1* expression is distinctively upregulated upon GDNF activation of Ret.²¹⁰ This suggests that Ret activates a negative feedback mechanism via *Spry1* in order to control activated ERK levels and modulate cell proliferation in the UB. Studies on *Spry1* knockout mice reveal some intriguing facets about Ret dependence of UB induction and branching.^{87,218–222} *Spry1* deficiency leads to ectopic UB induction and it can rescue renal development in the absence of either GDNF or Ret.^{222,223} Germline inactivation of *Spry2* does not overtly affect renal development but can rescue renal hypoplasia in mice engineered to express Ret mutants impaired in activating the Ras/ERK pathway.²¹⁷ The transcriptional targets of Ret, such as *Etv4*, *Etv5*, and *Wnt11*, are retained in *Gdnf/Spry1* or *Ret/Spry1* compound null mutants.^{222,223} These findings indicate that Ret signaling is not absolutely required for UB development. In fact,

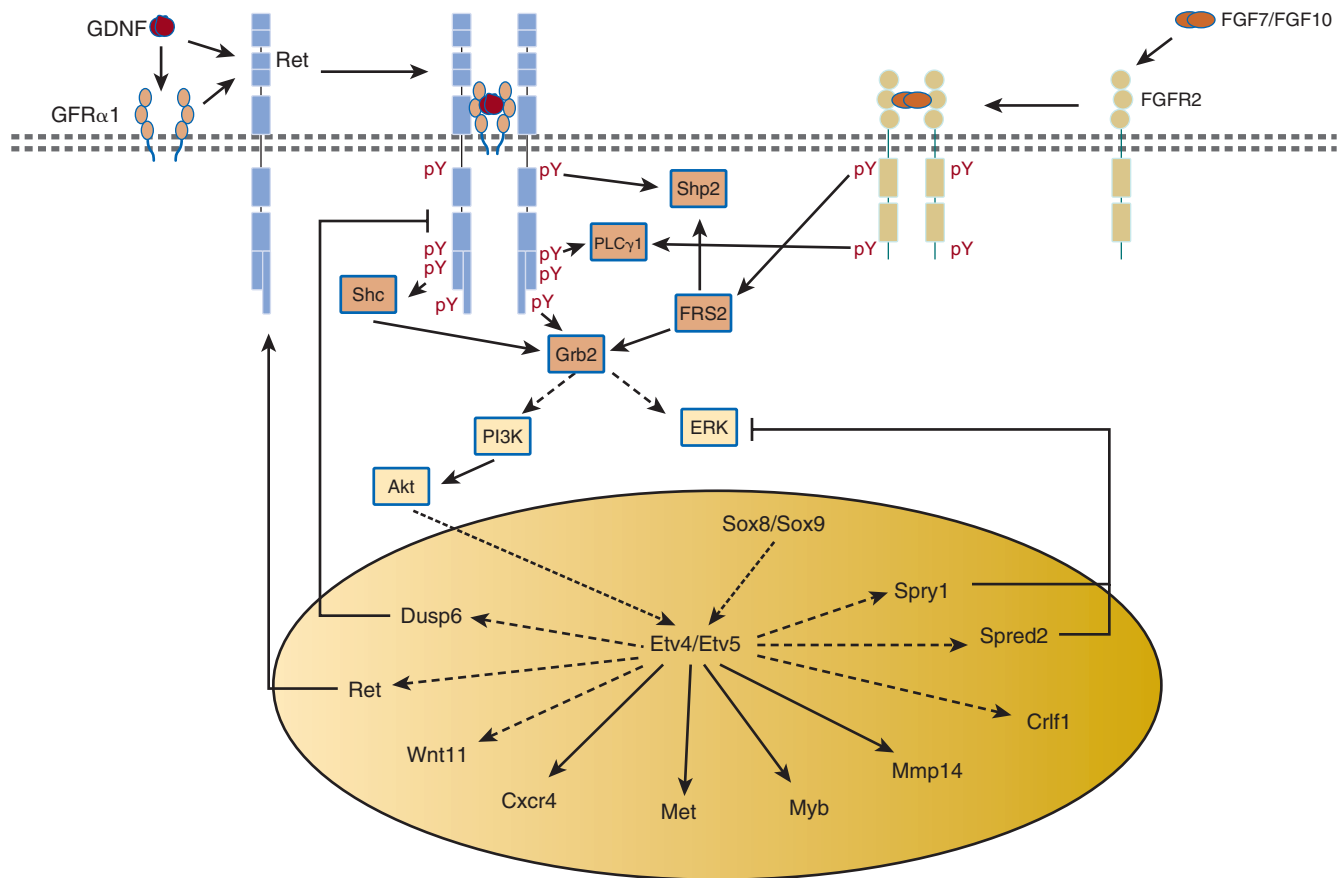


Fig. 1.15 Ret signaling pathway. Ret is activated and becomes autophosphorylated on intracellular tyrosine residues upon association with glial cell-derived neurotrophic factor (GDNF) and GFR α 1. Signaling molecules such as Grb2, Shc, FRS2, PLC γ 1, and Shp2 bind directly to the phosphorylated tyrosine residues within the intracellular domain of Ret. Recruitment of Shc, FRS2, and Grb2 leads to activation of ERK and PI3K/Akt pathways. GDNF-Ret signaling leads to the specific activation of a host of genes, some of which are strongly dependent on the upregulation of the transcription factors Etv4 and Etv5 (solid arrows). Etv4/Etv5 activation requires activation of the PI3K/Akt but not the ERK pathway. Sox8 and Sox9 are believed to act in parallel to reinforce transcriptional responses to GDNF-Ret engagement. Some of these pathways are shared with the FGF7/10-FGFR2 receptor signaling system. Spry1 and Spred2 negatively regulate ERK signaling, whereas Dusp6 likely mitigates the dephosphorylation of the Ret receptor, thus acting as part of a negative feedback regulatory loop.

signaling via FGF10 and FGFR2 receptors is sufficient for renal development despite the absence of GDNF or Ret, provided *Spry1* is inactivated. Nevertheless, patterns of renal branching are distinctively altered in *Gdnf/Spry1* and *Gdnf/Ret* compound mutants with UB tips often displaying more heterogeneous shapes and orientation. These indicate that there remain some distinctive roles of GDNF-Ret signaling that cannot be fully compensated by FGF10/FGFR2 during UB development.

ADHESION PROTEINS IN EARLY KIDNEY DEVELOPMENT

A current theme in cell biology is that growth factor signaling often occurs coordinately with signals from the extracellular matrix transduced by adhesion receptors such as members of the integrin family. The α 8 β 1 integrin complex is expressed by cells of the MM interacting with the novel ligand nephronectin (*Npnt*) expressed specifically by UB cells.^{169,224} In most *Itga8* (α 8 integrin) mutant embryos, UB outgrowth is arrested upon contact with the MM.¹⁶⁹ In a small portion of embryos, this block is overcome, and a single, usually hypoplastic, kidney develops. Knockout mice for *Npnt* exhibit

renal agenesis or severe hypoplasia.¹⁶⁸ Thus the interaction of α 8 β 1 integrin with nephronectin must have an important role in the continued growth of the UB toward the MM. Both *Itga8* and *Npnt* knockout phenotypes appear to result from a reduction in *Gdnf* expression.¹⁶⁸ The attraction of the UB to the mesenchyme is also governed by the maintenance of proper cell-cell adhesion within mesenchymal cells. Kif26b, a kinesin specifically expressed in the MM, is important for tight condensation of mesenchymal cells.¹⁶⁷ Genetic inactivation of *Kif26b* results in renal agenesis resulting from impaired UB induction. In *Kif26b*-mutant mice, the compact aggregation of mesenchymal cells is compromised, resulting in distinctive loss of polarized expression of integrin α 8 and severe downregulation of *Gdnf* expression. Hence, dysregulation of mesenchymal cell adhesion causes the failure to attract and induce the ureteric epithelia. Genetic evidence further shows that nephronectin localization at the basement membrane of the UB is critical for *Gdnf* expression by the MM. Genetic inactivation of basement membrane proteins associated with Fraser syndrome (*Fras1*, *Frem1/Qbrick*, and *Frem2*) lead to renal agenesis characterized by severe downregulation of *Gdnf* expression.^{165,166,225–228} On the basis of interaction of

nephronectin with *Fras1*, *Frem1*, and *Frem2*, it has been proposed that the Fras1/Frem1/Frem2 ternary complex anchor nephronectin to the UB basement membrane, thus stabilizing engagement with $\alpha 8\beta 1$ integrin expressed by the MM (Fig. 1.16).²²⁵ Grip1, a PDZ-domain protein known to interact with Fras1, is required to localize the Fras1/Frem1/Frem2 complex on the basal aspect of the UB epithelium.²²⁹ *Grip1* mutations phenocopy Fraser syndrome, including renal agenesis, thus further highlighting the importance of the strategic localization of nephronectin on the UB surface towards the opposing MM.^{229–231}

The establishment of epithelial basement membranes during metanephric kidney development involves the stage-specific assembly of different laminin α and β subunits with

a common laminin $\gamma 1$ subunit. The UB-specific inactivation of the gene *Lamc1*, which encodes for laminin $\gamma 1$, leads to impaired UB induction and branching, ultimately causing either renal agenesis or hypomorphic kidneys with water transport deficits.²³² *Lamc1* deficiency prevents the formation of basement membranes, causing downregulation of both growth factor (GDNF, Wnt11, and FGF2) and integrin-based signaling. This highlights another example of how signaling through the extracellular matrix intersects with growth factor signaling to influence morphogenesis. The importance of basement membrane assembly in the development of other renal structures is emphasized by genetic studies on the genes *Lama5* and *Lamb2*, which encode for laminins $\alpha 5$ and $\beta 2$, respectively. Loss of *Lama5* causes either renal agenesis or

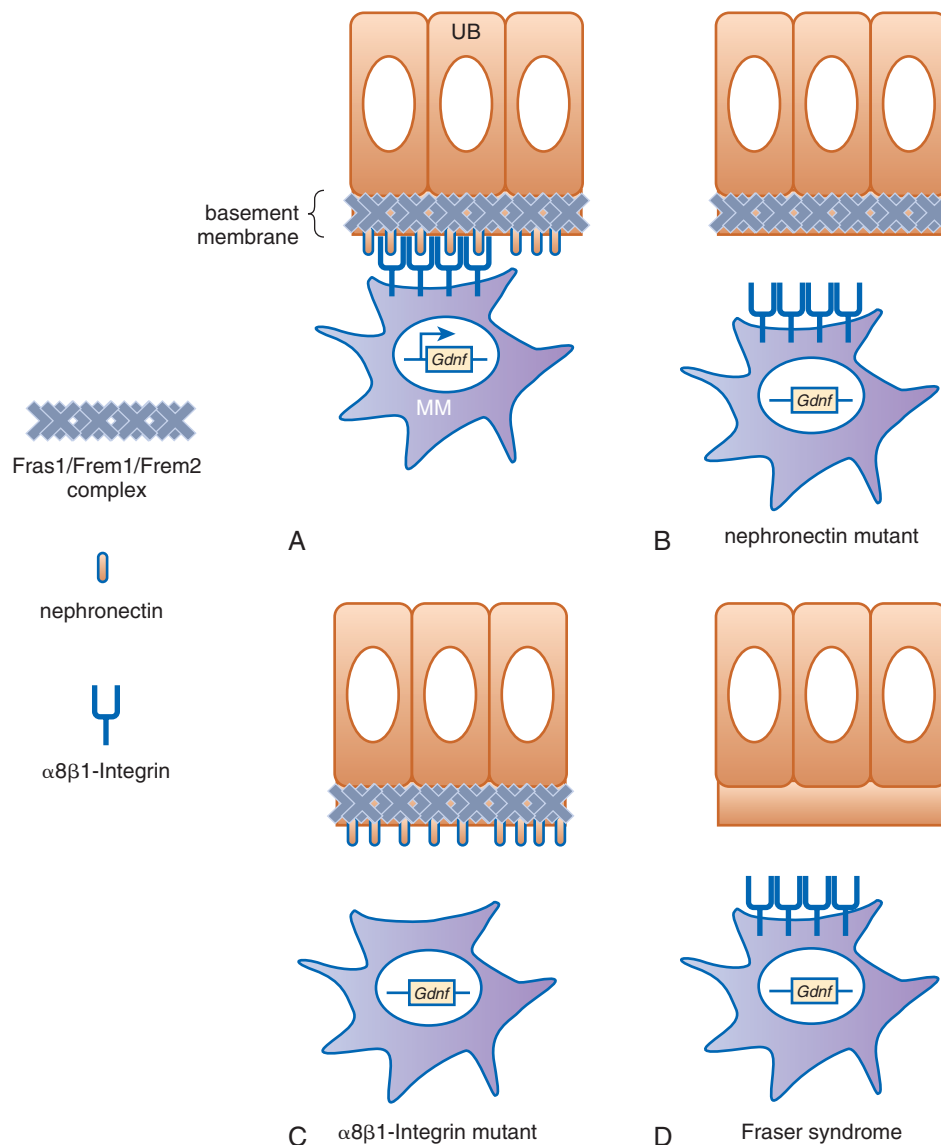


Fig. 1.16 Molecular model of renal defect in Fraser syndrome. (A) Adhesion to the ureteric bud (UB) epithelium positively regulates the expression of glial cell-derived neurotrophic factor (GDNF) by the metanephric mesenchyme (MM). Adhesion and GDNF expression are impaired in the absence of (B) nephronectin (expressed by the UB), (C) $\alpha 8\beta 1$ integrin (expressed by the MM), (D) or the Fras1/Frem1/Frem2 complex. Fras1, Frem1, and Frem2 are implicated in Fraser syndrome, and are believed to coordinately anchor nephronectin to the UB basement membrane and stabilize the conjugation with $\alpha 8\beta 1$ integrin. Modified from Kiyosumi, Takeichi M, Nakano I, et al.: Basement membrane assembly of the integrin $\alpha 8\beta 1$ ligand nephronectin requires Fraser syndrome-associated proteins. *J Cell Biol.* 2012;197:677–689.

disruption of glomerulogenesis, whereas deficiency for *Lamb2* leads to a defective glomerular filtration barrier.^{233,234}

The UB branching program is stereotypically organized such that the proliferative UB epithelial cells are largely confined to the bulbous UB tips, whereas cell division is dampened within the elongated nonbranching UB stalks of the growing ureteric tree. TROP2/Tacstd2, an adhesion molecule related to EpCAM, is expressed prominently in the UB stalks where it colocalizes with collagen-1.²³⁵ TROP2, unlike EpCAM, which is expressed throughout the UB tree, is not expressed at the UB ampullary tips. Consistently, dissociated and sorted UB cells expressing high levels of TROP2 are nonproliferative and express low levels of *Ret*, *Gfra1*, and *Wnt11*, which are notable UB tip markers. Elevated expression of TROP2 is also associated with poor attachment of epithelial cells to collagen matrix and suppression of cell spreading and motility, thus emphasizing the importance of this adhesion molecule in negative regulation of UB branching and the sculpting of the nascent collecting duct network. The formation of patent lumens within epithelial tubules of the kidney is also dependent on coordinated cell adhesion. β 1 Integrin is tethered to the actin cytoskeleton via a ternary complex formed between integrin-like kinase (ILK) and parvin. ILK has been shown to be important in mediating cell cycle arrest and cell contact inhibition in the collecting duct epithelia.²³⁶ The targeted ablation of the *Ilk* gene in the UB did not cause remarkable defects in UB branching but eventually caused postnatal lethality due to obstruction of collecting ducts arising from dysregulated intraluminal cell proliferation. Thus cell adhesion molecules may suppress cell division to regulate distinctive aspects of renal branching and tubulogenesis.

POSITIONING OF THE URETERIC BUD

A crucial aspect of kidney development that is of great relevance to renal and urological congenital defects in humans relates to the positioning of the UB (see Fig. 1.14). Incorrect positioning of the bud, or duplication of the bud, results in abnormally shaped kidneys and incorrect insertion of the ureter into the bladder, with a resultant ureteral reflux that can predispose to infection and scarring of the kidneys and urological tract.

Foxc1 is a transcription factor of the Forkhead family, expressed in the intermediate mesoderm and the MM adjacent to the wolffian duct. In the absence of *Foxc1*, the expression of GDNF adjacent to the wolffian duct is less restricted than in wild type embryos. *Foxc1* deficiency results in ectopic UBs, hypoplastic kidneys, and duplicated ureters.²³⁷ Additional molecules that regulate the location of UB outgrowth are Slit2 and Robo2, signaling molecules best known for their role in axon guidance in the developing nervous system. Slit2 is a secreted factor, and Robo2 is its cognate receptor. Slit2 is mainly expressed in the Wolffian duct, whereas Robo2 is expressed in the mesenchyme.²³⁸ UBs form ectopically in embryos deficient in either *Slit2* or *Robo2*, similar to the *Foxc1* mutant. However, in contrast to the *Foxc1* phenotype, none of the ureters in *Slit2/Robo2* mutants failed to undergo the normal remodeling that results in insertion in the bladder.²³⁸ Instead, the ureters remained connected to the nephric duct in *Slit2* or *Robo2* mutants. The domain of *Gdnf* expression is expanded anteriorly in the absence of either *Slit2* or *Robo2*. Indeed, mutations in *Robo2* have been identified in patients

with vesicoureteral junction defects and vesicoureteral reflux.²³⁹ The expression of *Pax2*, *Eya1*, *Foxc1*, and *Six2*, all thought to regulate *Gdnf* expression, was not dramatically different in the absence of *Slit2* or *Robo2*, suggesting that Slit2-Robo2 signaling was not upstream of these genes. Instead, recent findings strongly suggest that Slit2-Robo2 signaling, as in other systems, acts as a repulsive guidance cue.²⁴⁰ Consistent with this are findings that *Slit2* is most strongly expressed at an increasing gradient, at regions more anterior from the normal site of UB induction. Thus the absence of *Robo2* most likely compromises normal separation of the wolffian duct from the nephrogenic mesenchyme, ultimately broadening the nephrogenic zone and provoking ectopic UB induction.

Spry1, as described earlier in this chapter, negatively regulates the Ras/Erk signaling pathway and is expressed strongly in the posterior wolffian duct and the UB tips.²⁴¹ Embryos lacking *Spry1* develop supernumerary UBs, but unlike mutants of *Foxc1*, *Slit2*, or *Robo2* they do not display changes in *Gdnf* expression.²¹⁸ The phenotype of *Spry1* mutants can be rescued by reducing the *Gdnf* expression dosage.²¹⁸ *Spry1* deletion also rescues the renal agenesis defect in mice lacking either *Ret* or *Gdnf*.²²² Consistently, renal agenesis and severe renal hypoplasia in mice expressing *Ret* specifically mutated on a tyrosine phosphorylation site known to couple with the Ras/ERK pathway can be reversed in the absence of *Spry1*.²²³ Thus *Spry1* appears to regulate UB induction sites by dampening receptor tyrosine kinase-dependent proliferative signaling.

Another negative regulator of branching is BMP4, which is expressed in the mesenchyme surrounding the wolffian duct. *Bmp4* heterozygous mutants have duplicated ureters, and, in organ culture, BMP4 blocks the induction of ectopic UBs by GDNF-soaked beads.²⁴² Furthermore, knockout of *Grem1*, which encodes for the secreted BMP inhibitor Gremlin, causes renal agenesis, supporting a role for BMP in the suppression of UB formation.²⁴³

FORMATION OF THE COLLECTING DUCT SYSTEM

The overall shape, structure, and size of the kidneys are largely guided by the stereotypical branching of the UB and the subsequent patterning of the collecting duct system. During late gestation, past embryonic stage E15.5 in the mouse, the trunks of the UB tree undergo extensive elongation to establish the array of collecting ducts found in the renal medulla and papilla. The radial arrangement of elongated collecting ducts together with the loops of Henle (derived from the nephrogenic mesenchyme) establishes the corticomedullary axis by which nephron distributions are patterned. After birth, further elongation of the newly formed collecting duct network is partly responsible for the postnatal growth of the kidney.

Elongation of the collecting duct involves oriented cell division characterized by the parallel alignment of the mitotic spindle of proliferating ductal epithelia with the longitudinal axis of the duct.²⁴⁴ Oriented cytokinesis, therefore, guarantees that the daughter cells contribute to lengthening of the duct with minimal effect on tubular lumen diameter. The renal medulla and pelvis are nonexistent in mice lacking *Wnt7b*.²⁴⁵ Notably, the collecting ducts and loops of Henle are stiffer due to reorientation of cell division toward a radial instead of a longitudinal axis. *Wnt7b* expression is restricted within

the nonbranching stalk of the ureteric tree and is absent in the ampullary UB tips. Failed development of the renal medullary and papillary regions is also recapitulated in mice where *Ctnnb1* is ablated in the renal stroma, suggesting that Wnt7b activates the canonical β -catenin-dependent Wnt signaling pathway involving the ureteric epithelia and the surrounding stroma.^{245–247} However, the relevant reverse signal from the interstitial stroma to the collecting duct that drives oriented cell division in the duct epithelia remains unknown.

The normal development of the collecting ducts is also dependent on cell survival cues provided by diverse ligands, such as Wnt7b, EGF, HGF, and interactions with the extracellular matrix.^{245,248,249} Papillary collecting ducts display higher incidences of apoptosis in mice lacking Wnt7b or EGFR.^{245,248} Conversely, loss of *Dkk1* (Dickkopf1), a secreted antagonist of Wnt7b, results in overgrowth of the renal papilla.²⁵⁰ Conditional inactivation of *Dkk1* using the *Pax8-Cre* transgene (expressed in renal tubules and the collecting ducts) causes increased proliferation of papillary epithelial cells. The HGF-receptor Met, $\alpha\beta1$ integrin (*Itga3/Itgb1*), and laminin $\alpha5$ (*Lama5*) are all required to maintain the expression of Wnt7b, and thus likely support the viability of collecting duct cells.^{177,249,251}

Poor development of the renal medulla and papilla are also observed in mutant mice lacking *Fgf7*, *Fgf10*, *Fgf2*, *Bmpr1a* (*Alk3*), the components of the renin-angiotensin system, *Shh* (Sonic hedgehog), or the orphan nuclear steroid hormone receptor gene *Esrsg*. FGF7 and FGF10 are the cognate ligands of FGFR2. Renal hypoplasia is observed when *Fgf2* is conditionally removed from the ureteric lineage and is more severe than in mutants lacking *Fgf7* or *Fgf10*, suggesting that these related ligands may have some functional redundancy in the development of the UB and collecting ducts.^{60,252,253} Kidneys lacking *Bmpr1a* show an attenuated phosphorylation of SMAD1, an effector of the BMP and TGF β ligands, and a concomitant increase in Myc and β -catenin levels.²⁵⁴ Although the significance of these results is not clear, the elevated expression of β -catenin indicates a novel crosstalk between BMP and Wnt signaling pathways in collecting ducts. Signaling through angiotensin is relevant to both early UB branching and the morphogenesis of medullary collecting ducts.²⁵⁵ Genetic inactivation of angiotensinogen, its processing enzyme ACE, and its target angiotensin-II AT1R receptors (*Agtr1a* and *Agtr1b*) results in similar phenotypes characterized by hypoplastic kidneys with modestly sized renal papillae.^{256–261} Furthermore, the postnatal growth and survival of renal papilla grown ex vivo are dependent on the presence of AT1R.²⁶² Interestingly, in cultures of renal papilla explants, angiotensin appears to regulate the Wnt7b, FGF7, and $\alpha\beta1$ integrin signaling pathways such that the loss of endogenous angiotensin or pharmacologic inhibition of AT1R causes significant dampening of the expression of *Wnt7b*, *Fgf7*, *Ctnnb1*, *Itga3*, and *Itgb1*.²⁶² *Shh* is expressed in the more distal derivatives of the UB, the medullary collecting ducts, and the ureter.⁵⁹ The germline deletion of *Shh* results in either bilateral renal agenesis or a single ectopic dysplastic kidney.^{263,264} It has been shown that *Shh* controls the expression of early inductive and patterning genes (*Pax2* and *Sall1*), cell cycle regulators (*Mycn* and *Cnd1*), and signaling effectors of the Hedgehog pathway (*Gli1* and *Gli2*). Interestingly, genetic removal of *Gli3* in an *Shh*-null background restores the expression of *Pax2*, *Sall1*, *Cnd1*, *Mycn*, *Gli1*, and *Gli2*,

providing physiologic proof for the role of *Gli3* as a repressor of the *Shh* pathway in renal development.²⁶⁴ Frameshift mutations resulting in truncation of the expressed *Gli3* protein are linked to Pallister-Hall syndrome and the presence of hydronephrosis and hydroureter in both humans and mice.^{265,266} *Esrsg* has a strong and localized expression within collecting duct epithelia later in gestation and its inactivation in mice causes complete aplasia of the renal medulla and papillae. However, the ligand of *Esrsg* remains to be identified and little is known regarding its downstream targets.

The mature collecting duct epithelia consist of two major cell subtypes: the abundant principal cells which strongly express the aquaporins, ion channels, and pumps mediating Na⁺ and K⁺ transport; and the fewer intercalated cells which are responsible for secretion of protons and bicarbonate ions (see Fig. 1.14). A third collecting duct cell subtype, now called the transition cell, has been recently identified and coexpresses principal and intercalated cell-specific genes.²⁶⁷ Fate mapping analyses in an *Aqp2-Cre* and *Atp6v1b1-Cre* fluorescent reporter mouse line estimate that principal cells constitute ~60% of collecting duct epithelia, while intercalated cells and transition cells constitute ~30% and ~10%, respectively. Genetic studies suggest a cell type plasticity among collect duct epithelia with gene mutations identified as leading to disproportionate increase in one cell at the expense of the other with detrimental consequences in the maintenance of fluid, electrolyte, and acid balance.^{267–272} Lineage tracing analyses indicate that both principal and intercalated cells can dynamically switch fates, transforming into an intermediate cell type, the transition cell.²⁶⁷ Gene expression analyses implicate the Notch signaling pathway as an important regulator of collecting duct cell composition and patterning. Intercalated cells strongly express the Notch ligand *Jag1*, whereas principal cells express the Notch2 receptor.^{267,273} Genetic induction of Notch signaling within collecting ducts in adult mice notably increased the amount of *Aqp2*⁺ principal cells while the number of *Atp6v1b1*⁺ intercalated cells decreased in parallel, without altering the number of transition cells.²⁶⁷ Similarly, loss of the transcription factors *Tfcp2l1*, *Foxi1*, or the p53-related *Trp63* results in the paucity of intercalated cells.^{268,269} *Tfcp2l1* induces the expression of intercalated cell-specific genes including *Jag1* and *Atp6v1b1*.²⁷³ It has been proposed that Notch activation in the UB suppresses *Foxi1*, while the identification of putative *Trp63* binding sites in the promoters of several Notch ligands suggests that *Trp63* may repress Notch signaling.^{270,271} Conversely, inhibition of Notch signaling in ureteric epithelia leads to a disproportionate predominance of intercalated cells at the expense of principal cells, and the onset of polyuria, urinary concentration defects, and hydronephrosis.^{271,272,274} A specific target of Notch signaling in principal cells is the transcription factor *Elf5*, which positively regulates expression of principal cell-specific genes *Aqp2* and *Avpr2*.²⁷⁴ One other identified epigenetic regulator of principal cell fate is the histone methyltransferase encoded by *Dot1l*, which normally represses the aquaporin gene *Aqp5* and the intercalated cell-specific gene *Atp6v1b1*.^{275–277} *Dot1l* deficiency elevates the expression of both *Aqp5*, whose product interferes with cell surface localization of the principal cell-specific aquaporin *Aqp2* while concomitantly promoting the acquisition of intercalated cell traits with the upregulation of *Atp6v1b1*.

MOLECULAR GENETICS OF NEPHROGENESIS

EPITHELIALIZATION OF THE METANEPHRIC MESENCHYME

The generation of a sufficient number of nephrons requires a highly regulated balance between the expansion of progenitor compartments and the commitment toward epithelial fate to become renal vesicles. This has important clinical implications as the impaired renewal of nephron precursors or their perturbed differentiation can ultimately cause a wide range of renal pathologies due to significant paucity of functional nephrons. Signaling through Wnt, FGF, the BMP family of ligands, and Fat4 have been identified as important regulators of the delicate balance between progenitor self-renewal and differentiation.

All nephrogenic structures (podocytes, parietal epithelial cells, proximal tubules, loop of Henle, distal tubules, and the connecting tubule directly conjoined with the collecting duct) descend from a common progenitor pool that expresses the transcription factor *Six2*.⁶¹ *Six2* expression is notably elevated in the cap mesenchyme that condenses adjacent to the UB and is downregulated once this cap mesenchyme organizes into pretubular aggregates. It is now recognized that *Six2* activity is required to keep these nephron progenitor cells in a naïve, proliferative precursor state.^{61,278} *Six2* has been demonstrated to function as both a transcriptional activator that promotes cell cycling and proliferation, and as part of a repressor complex silencing differentiation-related genes.^{279,280} *Six2* can synergize with *Sall1*, to promote transcription of genes relevant to progenitor status (e.g., *Wt1*, *Eya1*, and *Gdnf*). *Six2* and *Sall1* also co-occupy promoters of their own genes, thus acting as positive feedback regulators of progenitor fate. *Six2* also interacts with *Osr1*, *Tcf* (*Lef*), and *Aes* (*Groucho*/*TLE*) to form a repressor complex that antagonizes expression of genes related to epithelialization (e.g., *Fgf8* and *Wnt4*).^{281,282} Complete loss of *Six2* causes premature ectopic formation of renal vesicles at E12.5 and the untimely depletion of nephron precursors.^{61,278} In contrast, overexpression of *Six2* prevented epithelialization of the cap mesenchyme.²⁷⁸

Six2⁺ nephron progenitors respond to stimulation with the UB-secreted factor Wnt9b and transition into epithelial renal vesicles.²⁸³ During the transition from cap mesenchyme to renal vesicles, expressions of a second Wnt family member *Wnt4* and an FGF family member *Fgf8* are activated. Canonical Wnt signaling involving β -catenin-dependent gene transcription is necessary and sufficient for the early inductive actions of Wnt9b and *Wnt4*, although it is also known that *Wnt4* can activate a noncanonical alternative pathway during the final phase of nephrogenic epithelialization.^{284–289} The interaction between *Six2*- and Wnt-signaling pathways appears to be intricately dosage- and context-dependent.^{280,282,283} When *Six2* expression remains high, Wnt signaling promotes progenitor renewal.²⁹⁰ In this situation, the stabilization of β -catenin promotes the association of *Six2* downstream target *Myc* with β -catenin, favoring precursor proliferation.^{283,291} However, with a sustained canonical Wnt-signaling pathway, β -catenin accumulates and displaces *Aes* converting the *Tcf* complex into a differentiation driver.²⁸⁰ The stabilization of

β -catenin eventually interferes with *Six2* expression, thus attenuating *Six2* expression. Notch signaling is required to prime nephron progenitors for differentiation and contributes to the silencing of *Six2* expression.²⁹² All nephron segments fail to form when Notch signaling is lost within the *Six2*⁺ precursor lineage.²⁹³

The growth factor BMP7 is also required for the formation of the nephrogenic compartment.^{294–296} Activation of the Jnk pathway mediates the proliferative effect of BMP7 in uncommitted nephron precursors.^{297,298} BMP7, through activation of the p38-MAPK, causes upregulation of the transcriptional repressor *Trps1*.²⁹⁹ Loss of *Trps1* severely impairs the formation of renal vesicles. It has been speculated that *Trps1* may indirectly relieve repression of *Cdh1* expression. Additionally, BMP7-dependent phosphorylation and nuclear translocation of SMAD1/5/8 is required for Wnt9b-induced epithelialization.³⁰⁰ Thus BMP7 has dual essential roles in promoting both progenitor replenishment and priming for epithelialization. How these pathways integrate with *Six2*-dependent signaling complexes remains poorly understood. BMP signaling could either positively or negatively modulate β -catenin signaling. In other systems, SMAD proteins can associate and synergize with β -catenin and *Tcf*.³⁰¹ BMP signaling can also activate the PTEN pathway, which indirectly suppresses β -catenin activity.^{302,303}

The FGFs FGF2, FGF8, FGF9, and FGF20, and their cognate receptors FGFR1 and FGFR2, are essential to form nephrons. Compound loss of FGFR1 and FGFR2 in the MM leads to renal agenesis.³⁰⁴ FGF9 and FGF20 are required to maintain the multipotency and proliferative state of nephron precursors.³⁰⁵ FGF2 is required for the condensation of the cap mesenchyme.³⁰⁶ FGF8 is not needed for the formation of renal vesicles but is required for the survival of the newly formed nephrogenic epithelia. Renal vesicles lacking *Fgf8* fail to express *Wnt4* and *Lhx1*, and do not progress into S-shaped intermediate nephrons.^{286,287} Potential downstream targets of FGFR1/FGFR2 relevant to nephrogenesis are the closely related MAGUK family proteins encoded by the genes *Cask* and *Dlg1*.³⁰⁷ The absence of *Cask* and *Dlg1* causes impaired proliferation and cell death in nephron precursors, with a distinctive dampening of the Ras/ERK signaling pathway.³⁰⁸ *Fgf8* expression is severely attenuated when both *Cask* and *Dlg1* are absent. Moreover, *Cask/Dlg1* deficiency causes the formation of a loose cluster of cap mesenchyme around the UB. As *Dlg1* has been implicated in the directed migration of Schwann cells,^{309,310} it is tempting to speculate that *Cask* and *Dlg1* may play a supportive role in the UB-directed condensation of the cap mesenchyme.

NEPHRON SEGMENTATION AND TUBULOGENESIS

The mature nephron is a highly compartmentalized structure with individual segments having distinguishable molecular, cellular, and anatomic attributes. Nephron segments are organized along a proximal-distal axis, from the most proximal renal corpuscle or glomerulus, followed by the proximal tubule, the loop of Henle, the distal tubule, and the most distal connecting tubule that links directly to the UB-derived collecting duct. The segmental patterning of nephrons involves a complex series of events instructed by inductive cues between neighboring cells and controlled by epigenetic

signaling mechanisms. A variety of human diseases result from the mispatterning of nephrons.³¹¹

Elegant imaging studies combined with high-throughput single-cell gene expression analysis demonstrate that nephron patterning is determined as early as the recruitment of mesenchymal nephron progenitors from the cap mesenchyme.³¹² Recent evidence indicates that renal vesicle arises not from a single event in time. Instead, nephron progenitors progressively incorporate into the nascent nephrons, with the timing of their recruitment predicting their acquisition of proximal–distal fates (i.e., initial recruits commit to distal fates while the last recruits contribute to the formation of more proximal fates) (Fig. 1.17).

By the renal vesicle stage, gene expression asymmetry highlights an early establishment of proximal and distal domains. Genes such as *Fgf8*, *Lhx1*, *Dll1*, *Dkk1*, *Hnf1b*, *Sox9*, and *Pou3f3* are markedly elevated in the distal portion of the renal vesicles, whereas *Wt1*, *Foxc2*, and *Mafb* are largely restricted in the proximal end.^{57,313} Some genes are expressed in both regions, such as *Wnt4*, *Jag1*, *Cdh6*, and *Ccnd1*, albeit nonuniformly and more elevated in the distal domain.³¹³ By the S-shaped body stage, nephron segmentation has become more evident, with several more marker genes having distinctively regionalized expression patterns.

As the earliest nephron precursors that interact with the UB become destined to acquire distal fates, it can be speculated that localized Wnt9b signaling orients the proximal–distal axis. Interestingly, in chick mesonephros, overexpression of Wnt3 in the coelomic lining redirects the glomerular development farthest away from the coelomic

lining while the distal nephron occasionally become fused with the coelomic lining.³¹⁴ During nephrogenesis, a gradient of β -catenin activity is established along the nephron's longitudinal axis with the highest β -catenin activity found in the distal end, progressively decreasing towards the proximal end.⁴⁵ Although β -catenin is absolutely required to initiate nephron induction, its activity must be attenuated in order to complete an epithelial differentiation program.²⁸⁹ Indeed, a β -catenin gradient is already established within renal vesicles and a persistent constitutive activation of β -catenin prevents epithelialization.^{45,289} In organ cultures, pharmacologic manipulation of β -catenin activity can alter proximal and distal fate acquisition. Attenuation of β -catenin activity accelerates glomerular development. Conversely, augmentation of β -catenin activity favors the expression of *Lgr5*, a marker of distal fate, while repressing proximal identity. Additionally, it has been identified that modulation of β -catenin activity involves an integration of Wnt, Notch, and BMP signaling pathways.

Although many genes are now known as marking nephron segments, only a few other genes and pathways have been characterized that strongly influence proximal versus distal fate determination. Loss of *Hnf1b* in the nephrogenic precursors causes marked loss of proximal and median domain markers at the S-shaped body stage, causing the formation of immature and cystic glomeruli connected to the collecting duct by a severely truncated renal tubule.^{315–317} A mutation in *Sall1* that prevents Sall1–NuRD interaction downregulates the distal marker *Lgr5*, and specifically impairs the development of the loop of Henle and distal tubules.³¹⁸ Other genes

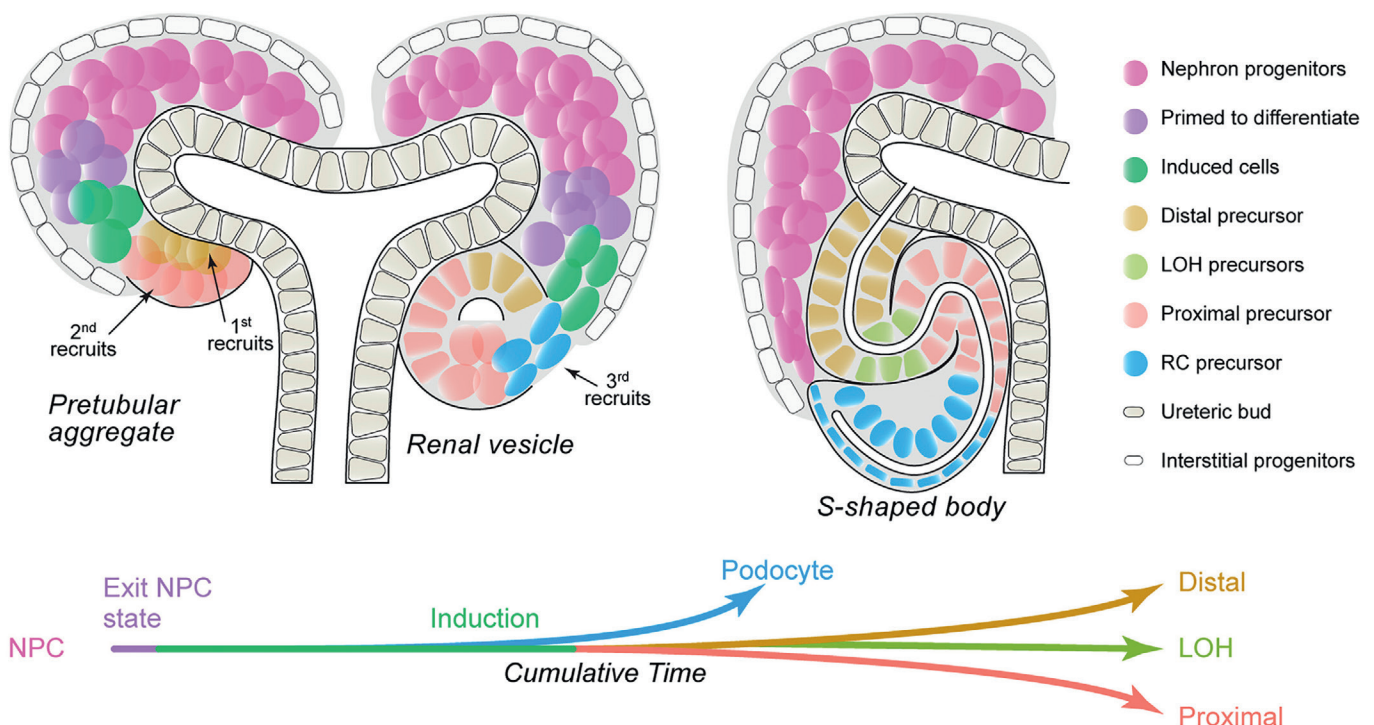


Fig. 1.17 Gradual recruitment of progenitor cells during nephrogenesis. Early recruited nephron progenitors clustering around the ureteric bud give rise to pretubular aggregates. New progenitors continue to be recruited and incorporate to the proximal end of the renal vesicle. Distal, medial, and proximal domains are established by the S-shaped body stage. NPC, Nephron progenitor cell; LOH, loop of Henle; RC, renal corpuscle. Adapted from Lindström NO, De Sena Brandine G, Tran T, et al. Progressive recruitment of mesenchymal progenitors reveals a time-dependent process of cell fate acquisition in mouse and human nephrogenesis. *Dev Cell* 2018;45:651–660.

required to generate distal tubules and the loop of Henle are *Pou3f3* and *Adamts1*.^{319–321}

In addition to cell differentiation, the spatial orientation of cells is essential for tubule elongation and morphogenesis. In epithelia, cells are uniformly organized along an apical–basal plane of polarity. However, in addition, cells in most tissues require positional information in the plane perpendicular to the apical–basal axis. This type of polarization is referred to as planar cell polarity and is critical for the morphogenesis of metazoans.^{322,323} Using cell lineage analysis and close examination of the mitotic axis of dividing cells, it has been shown that the lengthening of renal tubules is associated with mitotic orientation of cells along the tubule axis, demonstrating intrinsic planar cell polarity.²⁴⁴ Dysregulation of oriented cell division can give rise to cysts due to abnormal widening of tubule diameters.³²⁴ To date, molecules implicated in planar cell polarity and nephrogenic tubule elongation include *Wnt9b*, *Hnf1b*, *Pkhd1*, *Fat4*, *Celsr1*, and *Vangl2*.^{244,246,325–332}

CESSATION OF NEPHROGENESIS

Nephrogenesis is a time-limited event that is not reactivated postinjury to the kidneys of humans and mice at adulthood. The last wave of nephrogenesis is observed around the 36th wave of gestation in humans and shortly after birth in mice.^{333,334} Cessation of nephron generation is characterized by the exhaustion of nephron progenitor cells and the completion of epithelial differentiation of the remaining nephrogenic precursor. Although various morphologic and molecular changes occurring at this time have been characterized in mouse kidneys, the exact trigger is not fully understood.^{333,335,336}

In mice, the nephrogenic zone progressively shrivels after birth and is replaced by mature tubules by P6. The precursor markers *Six2* and *Cited1* are significantly downregulated at P2 and are undetectable by P3, concomitant with the disappearance of the cap mesenchyme. Multiple newly induced nephrons are found associated with each UB tip at P3 that are no longer found at P7. Ureteric branching also ends between birth and P3 accompanied by a significant reduction of *Ret* and *Wnt11* and loss of the UB ampullary shape.^{333,335} In contrast, *Wnt9b* expression in the UB remains high even at P4, consistent with findings that the UB tips isolated from P3 kidneys remain competent in promoting survival and inducing epithelialization of recombined mesenchyme taken from embryonic kidneys.³³³ This strongly argues against the possibility that weakened trophic support from the UB contributes to the progressive decline of the nephrogenic precursor population and is supported by the rarity of apoptosis within the postnatal nephrogenic zone. *Foxd1* expression is also downregulated by P3, but given the expansion of the cap mesenchyme in *Foxd1*- or stroma-deficient kidneys, it is unlikely that postnatal loss of the *Foxd1*⁺ stromal compartment provokes the abrupt halting of nephrogenesis.³³³ There is also no evidence that nephrogenic progenitors have switched to stromal fates. Instead, what is now apparent is that the remaining cap mesenchyme becomes globally committed to epithelialize. Interestingly, when nephron progenitor proliferation is enhanced and kidney size is increased as seen in *Six2* haploinsufficient mice, the timing of nephrogenesis cessation is unaltered.³³⁷

Two opposing factors appear more likely to contribute to the termination of nephrogenesis: (1) a rapid decline in the proliferation rate of the *Six2*⁺ cell compartment, which was observed between E15.5 and P0; and (2) an acceleration of nephrogenic differentiation postnatally.^{333,335} One model proposed posits that the novel spatial relationship between cell types in the kidney around birth (as compared to early embryonic stages) dramatically alters the molecular context of the nephrogenic niche, thus shifting the balance between progenitor self-renewal and commitment to differentiation.^{335,336,338} Another proposed model argues that increased oxygen tension after birth, which turns on the expression of glycolysis-related genes, could act as an active trigger that ends nephrogenesis.³³⁶ However, this latter model, although applicable in mice, may not be a shared mechanism in humans where nephrogenesis ceases before parturition. A thorough understanding of the mechanism of nephrogenesis cessation is necessary to better assess the potential of regenerative therapies for the kidney.

MOLECULAR GENETICS OF THE STROMAL CELL LINEAGE

The maintenance of reiterative ureteric branching and nephron induction largely accounts for the growth and enlargement of embryonic kidneys. Genetic studies reveal that interstitial stroma provide additional inductive cues that regulate UB branching and nephrogenesis (Fig. 1.18). These studies also underscore the pivotal role played by the stroma in establishing the stereotypical radial patterning of the kidney. In embryonic kidneys, the stroma is organized into two distinct zones: an outer stromal region within the nephrogenic zone expressing the winged helix transcription factor *Foxd1*, and a deeper region expressing the basic-helix–loop–helix (bHLH) transcription factor *Tcf21* (*Pod1*).^{28,29,339,340} Without either *Foxd1* or *Tcf21*, UB branching and nephrogenesis are notably impaired, resulting in a distinctive perturbation of the corticomedullary renal histoarchitecture.^{28,29,339}

The most prominent features of the genetic loss of *Foxd1* include the thickening of the renal capsule and the formation of large metanephric mesenchymal condensates.^{28,341} The morphologically altered renal capsule in *Foxd1*-mutant kidneys has notably lost expression of *Aldh1a2* (*Raldh2*) and *Sfrp1* (a regulator of Wnt signaling), and is abnormally interspersed with endothelial cells and *Bmp4*-positive cells.³⁴¹ The identity of these *Bmp4*-expressing cells populating the renal capsule in *Foxd1*-deficient kidneys is unknown, although they are clearly distinct from the presumptive medullary stroma based on lineage tracing for *Foxd1*-promoter expression. As BMP4 is a known chemotactic agent for endothelial cells,³⁴² it is very likely that the ectopic *Bmp4*-positive cells account for the presence of endothelial cells within the broadened renal capsule of *Foxd1*-mutant kidneys. The accumulation of the cap mesenchyme is also likely contributed in part by ectopic *Bmp4* signaling in the absence of *Foxd1* because *Bmp4* has been shown to antagonize epithelialization of the cap mesenchyme.³⁴² Transcriptome analysis reveals that the gene *Dcn*, which encodes for the collagen-binding proteoglycan decorin, is a specific target that is repressed by *Foxd1* in the cortical interstitium.³⁴³ *Dcn* expression is normally localized within the medullary stroma but is normally absent in the cortical stroma of wild type kidneys. In the absence of *Foxd1*, *Dcn*

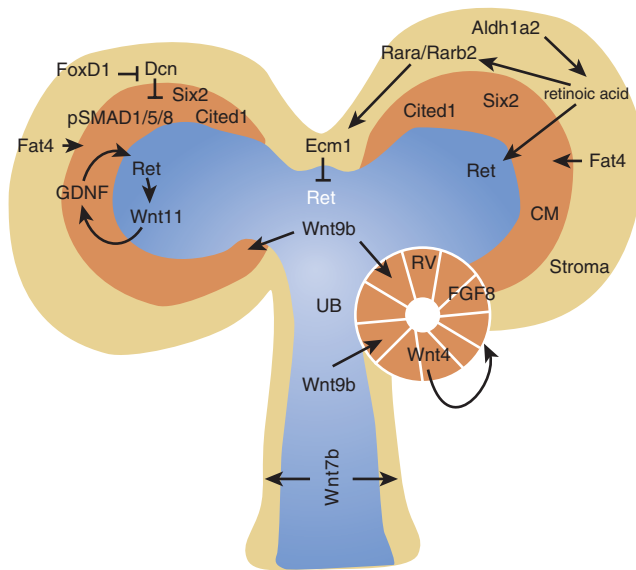


Fig. 1.18 Tripartite inductive interactions regulating ureteric branching and nephrogenesis. Six2 and Cited1 are expressed in the self-renewing nephron progenitors within the cap mesenchyme (CM) surrounding the ureteric bud (UB). The UB tip domains express high levels of Ret, which is activated by glial cell-derived neurotrophic factor (GDNF) from the surrounding CM. Wnt11 is upregulated in response to Ret activation and stimulates GDNF synthesis in the CM. Wnt9b expressed by the UB, and Fat4 by the Foxd1-positive stroma are required to initiate nephrogenesis from a subset of the CM. This results in the formation of a transient renal vesicle (RV) expressing FGF8 and Wnt4, factors that sustain epithelialization. The stroma expresses *Aldh1a2*, a gene required for retinoic acid synthesis, and genes for the retinoic acid receptors (*Rara* and *Rarb*). Retinoic acid signaling stimulates elevated expression of Ret in the UB tip domain while at the same time suppressing Ret expression via *Rara/Rarb2* and *Ecm1* in the stroma to initiate bifurcation of the UB tip to generate new branches. Foxd1 in the cortical stroma also represses *Dcn*, thus relieving the *Dcn*-mediated suppression of BMP7-dependent signaling, which results in phosphorylation of SMAD1/5/8 (pSMAD1/5/8) and epithelialization of the cap mesenchyme.

becomes abundantly expressed in the presumptive cortical stromal region. Functional cell-culture-based assays and epithelialization assays of mesenchymal aggregates reveals that *Dcn* inhibits Bmp7 signaling and mesenchyme-to-epithelial transformation. The antagonistic effect of *Dcn* on epithelial differentiation is further enhanced in vitro when the mesenchymal aggregates are grown in collagen IV, thus recapitulating the persistence of the cap mesenchyme as seen in *Foxd1*-mutant kidneys where both *Dcn* and collagen IV are upregulated in the cortical interstitium. These findings are corroborated by the partial rescue of the *Foxd1*-null phenotype through genetic inactivation of *Dcn*. Fate mapping studies also reveal that the *Foxd1*⁺ stromal mesenchyme are the precursors of renal mural cells (renin cells, smooth muscle cells, pericytes, perivascular fibroblasts, and glomerular mesangial cells) and endothelial cells comprising peritubular capillaries.^{32,33,344,345}

Tcf21 is expressed in the medullary stroma as well as in the condensing MM.^{339,340} *Tcf21* is also expressed in a number of differentiated renal cell types that derive from these mesenchymal cells and include developing and mature podocytes of the renal glomerulus, cortical and medullary peritubular

interstitial cells, pericytes surrounding small renal vessels, and adventitial cells surrounding larger blood vessels (see Fig. 1.6).²⁴⁶ The defect in nephrogenesis observed in *Tcf21*-null mice is similar to the defect seen in *Foxd1* knockout mice, with a disruption of branching morphogenesis with an associated arrest and delay in nephrogenesis.^{339,341,346} Interestingly, as with *Foxd1*, *Tcf21* also represses *Dcn*, although the significance of *Dcn* upregulation in *Tcf21*-null mutant mice has not been formally addressed.³⁴⁷ The analysis of chimeric mice that are derived from *Tcf21*-mutant embryonic stem cells and GFP-expressing embryos demonstrated both cell autonomous and noncell-autonomous roles for *Tcf21* in nephrogenesis.³⁴⁶ Most strikingly, the glomerulogenesis defect is rescued by the presence of wild type stromal cells (i.e., mutant cells will epithelialize and form nephrons normally as long as they are surrounded by wild-type stromal cells). In addition, there is a cell-autonomous requirement for *Tcf21* in stromal mesenchymal cells to allow differentiation into interstitial and pericyte cell lineages of the cortex and medulla as *Tcf21*-null cells were unable to contribute to these populations.

Although many of the defects in the *Tcf21*-mutant kidneys phenocopy those seen in the *Foxd1*-mutant kidneys, there are important differences. Kidneys from *Tcf21*-null mice have vascular anomalies and defective pericyte differentiation that were not reported in *Foxd1*-mutant mice.^{339,346} These differences might result from the broader domain of *Tcf21* expression, which also includes the condensing mesenchyme, podocytes, and medullary stromal cells, in addition to the stromal cells that surround the condensates. In contrast to *Foxd1*, *Tcf21* is not highly expressed in the thin rim of stromal cells found immediately beneath the capsule, suggesting that *Foxd1* and *Tcf21* might mark early and late stromal cell lineages, respectively, with overlap in the stroma that surrounds the condensates.²⁹ However, definitive co-labeling studies to address this issue have not been performed. As both *Tcf21* and *Foxd1* are transcription factors, it is interesting to speculate that they might interact or regulate the expression of a common stromal “inducing factor.”

Retinoids secreted by the renal stroma are also recognized as important for the maintenance of a high level of Ret receptor expression in the UB tip, promoting the proliferation of UB epithelial cells and the growth of the ureteric tree.^{17,348–350} The defective UB branching seen in *Foxd1*-null mutants is most likely a direct consequence of the loss of cortical expression of *Aldh1a2*, a gene involved in retinol synthesis.³⁴¹ More recently, it has been shown that renal stroma immediately around the UB tips is also important in regulating the bifurcation of the tips and the creation of new UB branches.³⁵¹ Autocrine retinoid signaling in the stromal cells juxtaposed to the UB tips stimulates the expression of extracellular matrix 1 (*Ecm1*). *Ecm1* is specifically expressed at the UB cleft, where it suppresses and restricts Ret expression domains within the UB tips. In the absence of *Ecm1*, Ret expression in the UB tips broadens, effectively attenuating UB branching due to impaired formation of UB bifurcation clefts. Thus stromal retinoids promote and confine Ret expression domains and more likely cell proliferation patterns within the UB tips.

Recent studies provide valuable insight on how stroma-based signaling intersects with UB-derived inductive cues to promote proper differentiation of the nephrogenic mesenchyme.^{352–356} When the stromal lineage is selectively annihilated by

Foxd1-Cre-driven expression of diphtheria toxin, the zone of condensing mesenchymal cells capping the UBs is abnormally broadened while the development of pretubular aggregates is strongly hindered. This reiterates findings from the *Foxd1*-null mice suggesting that regulation of nephrogenesis involves a crosstalk between stroma and UB-derived inductive signals. In particular, it was shown that Fat4-dependent Hippo signaling initiated by the stroma integrates with canonical Wnt signaling derived from the ureteric lineage in order to balance the nephron precursor propagation and differentiation. The absence of Fat4 in the stromal compartment or its ligands Dchs1 and Dchs2 from the cap mesenchyme phenocopies the expansion of the nephrogenic precursor domain and failed epithelial differentiation of nephron progenitors seen in stroma-deficient kidneys.^{353,356} It was postulated that Fat4, acting through the Hippo pathway, promotes the differentiation of the epithelial transition of nephrogenic precursors.^{353–356} This was further reiterated by the rescue of the depletion of nephrogenic precursors by *Fat4* deficiency in *Wnt9b* knockouts.³⁵² Interestingly, the loss of Vangl2, a signaling partner of Fat4 known to regulate renal tubular diameter, fails to rescue the loss of nephron progenitors in *Wnt9b* knockouts, suggesting that Fat4-mediated signaling during early differentiation of nephrogenic precursors is likely independent of the planar cell polarity pathway.^{330,352} Transcriptional regulation of stroma-dependent restriction of nephron progenitor expansion involves the transcription factors *Sall1*, *Foxd1*, and *Pbx1*.^{28,341,357} More importantly, *Sall1* is likely a major upstream regulator of many of the stroma functions as *Sall1* binds directly to several stroma-related gene loci, including *Fat4*, *Dcn*, *Pbx1*, *Tcf21*, *Meis1*, and *Hoxd10*.³⁵⁸ Loss of *Sall1* in the stroma downregulates *Fat4* expression and results in an excess of *Six2*⁺ nephron precursors.

The T-box transcription factor *Tbx18* is strongly expressed during early urogenital tract development in the ureteral mesenchyme and a subset of kidney stromal mesenchyme originating from *Foxd1* positive precursor cells.^{359,360} Later in renal development, *Tbx18* expression is also found in the renal capsule, vascular smooth muscle cells, pericytes, mesangial cells, and the mesenchyme surrounding the renal papillae and calyces.³⁶⁰ The most overt phenotype of *Tbx18*-inactivation is the onset of hydronephrosis and hydroureter as a result of impaired development of the ureteral smooth muscle cells.^{359,361} This underscores the importance of *Tbx18* in the normal differentiation of the ureteral mesenchyme. A more recent detailed phenotypic characterization of *Tbx18*-null mutant kidneys reveals an additional significance of this transcription factor in the overall development of the renal vasculature.³⁶⁰ The branching and overall density of the renal vasculature are notably reduced in the absence of *Tbx18*. *Tbx18* is also specifically required in the normal development of the glomerular microvasculature. Loss of *Tbx18* causes significant oligonephronia and dilation of glomerular capillaries. These vascular phenotypes likely result from the degeneration of the stromal mesenchyme adjacent to the developing vasculature and the failure to sustain the proliferation of mesangial precursors.

Mice carrying a hypomorphic allele of *Notch2* that is missing two epidermal growth factor (EGF) motifs are born with a reduced number of glomeruli that lack both endothelial and mesangial cells, as discussed in the section on nephron segmentation.^{362,363} The downstream Notch signaling target,

Rbpj, has also been described as crucial for the proper development of the renal vasculature and the glomerular mesangium. Conditional inactivation of *Rbpj* in the *Foxd1*-expressing stromal lineage leads to profound renal maldevelopment and early postnatal death.³⁶⁴ *Rbpj* deficiency in the renal stroma results in poor branching and simplification of the renal vascular network. *Rbpj* conditional mutant kidneys have a greater proportion of larger vessels and a concomitant reduction in microvascular density. Glomeruli are dilated and lack mesangial cells in *Rbpj* conditional knockouts. Furthermore, loss of *Rbpj* results in loss of renin cells, abnormal thickening of blood vessels, and renal fibrosis. Altogether, these studies highlight the distinctive significance of Notch signaling within the stromal mesenchyme in the establishment and organization of the renal vasculature.

MOLECULAR GENETICS OF VASCULAR FORMATION

Although we now have a comprehensive if not complete understanding of the mechanisms underlying ureteric branching and nephrogenesis, we still know little how the complex vascularization of the kidney is coordinated with epithelial and stromal development. During kidney development, the formation and elaborate patterning of an arterial, venous, and capillary blood vascular network involves a combination of both vasculogenesis and angiogenesis (see Fig. 1.8). In addition, lymphangiogenesis underlies the development of the lymphatic vasculature from veins. The major renal vessels align close to the branching UB and likely elaborate through angiogenesis into large-caliber afferent and efferent distributaries.³³ Vasculogenesis in the kidney likely arises from sporadic endothelial cells within the MM that organize to form a primitive vascular network and then give rise to most of the peritubular capillaries.^{32,365,33} Additionally, vasculogenesis within the S-shaped nephron intermediates establishes the glomerular capillaries.^{34,35,366,367}

Grafting studies demonstrate that transplanted embryonic kidneys can become vascularized from the invasion of host-derived extrinsic blood vessels.^{368,369} However, more recent fate mapping studies provide compelling evidence for the existence of endogenous endothelial precursors within the kidney.^{31–35} Cultured mouse embryonic kidneys contain a heterogeneous intrinsic pool of endothelial cells that express the endothelial-specific markers Kdr, Cd31, and Cd146.³³ As early as E11.5 embryonic stage in the mouse, *Kdr*⁺ cells are readily identifiable as either single-cell clusters or primitive capillaries. The primitive capillaries consist of *Cd31*⁺/*Cd146*⁺ cells, whereas the single cells predominantly express *Cd146*⁺ but not *Cd31*. At E12.5, *Cd31*⁺ cells have formed an elaborate chain-like network, including being found adjacent to *Pax2*⁺ cap mesenchyme while singular *Cd146*⁺ cells have become scarce. Lineage tracing analysis demonstrates that a subset of endogenous renal endothelial cells that give rise to peritubular capillaries but not glomerular endothelial cells are derived from *Foxd1*⁺ stromal cells.^{32,33}

Isolated E11.5, E12.5, and E13.5 kidneys, and also cultured E11.5 kidneys, produce VEGF-A, a potent factor known to promote vascularization.³³ Indeed, the pharmacologic inhibition of VEGFR signaling completely abolishes the establishment of endothelial cell networks in cultured embryonic kidney explants, suggesting that VEGF-A signaling is essential

for renal vascular development. More recent imaging studies reveal that kidney vascularization is initiated at E11 in the mouse, when systemic vessels from embryonic circulation circumscribe around the UB.^{370,371} From E13.5, the endothelial network surrounds the cap mesenchyme and UB in a cyclical manner by which endothelia form across and in close contact with the bifurcating UB, which begs the question as to whether endothelial development coregulate ureteric branching. It is, however, easily conceivable that given the importance of oxygen levels in nephrogenesis,^{372–374} the intimate integration of the renal vascular plexus, which carries oxygen-delivering erythrocytes, is essential for nephron maturation.

Conditional gene targeting experiments and cell-selective deletion of *Vegfa* from podocytes demonstrates that VEGF-A signaling is required for formation and maintenance of the glomerular filtration barrier.^{375,376} Glomerular endothelial cells express VEGFR2 as they migrate into the vascular cleft. Although a few endothelia migrate into the developing glomeruli of *Vegfa* podocyte conditional knockout mutants (likely due to a small amount of VEGF-A produced by presumptive podocytes at the S-shaped stage of glomerular development prior to Cre-mediated genetic deletion), the endothelia fail to develop fenestrations and rapidly disappear, leaving capillary “ghosts” (Fig. 1.19). Deletion of a single *Vegfa* allele from podocytes leads to glomerular endothelial

defects known as endotheliosis, a phenomenon involving hypertrophy of glomerular endothelial cells and loss of fenestrations (endotheliosis), progressing to capillary occlusion and thrombosis (thrombotic microangiopathy).³⁷⁵ Inactivation of *Vegfr2* phenocopies endotheliosis and thrombotic microangiopathy, indicating that VEGF-A primarily signals through VEGFR2 in supporting the development and maintenance of the glomerular endothelium.³⁷⁷ As the dose of *Vegfa* decreased, the associated endothelial phenotypes became more severe. Upregulation of the major 164 angiogenic VEGF-A isoform in developing podocytes of transgenic mice led to massive proteinuria and collapse of the glomerular tuft by 5 days of age. Taken together, these results show a requirement for VEGF-A for development and maintenance of the specialized glomerular endothelia and demonstrate a major paracrine signaling function for VEGF-A in the glomerulus. Furthermore, tight regulation of the dose of VEGF-A is essential for proper formation of the glomerular capillary system. The molecular basis and mechanism of dosage sensitivity is unclear at present and is particularly intriguing given the documented inducible regulation of VEGF-A by hypoxia-inducible factors (HIFs) at a transcriptional level. Despite this, it is clear that in vivo, a single *Vegfa* allele is unable to compensate for loss of the other. Similarly, VEGF-A signaling is essential for the development of the

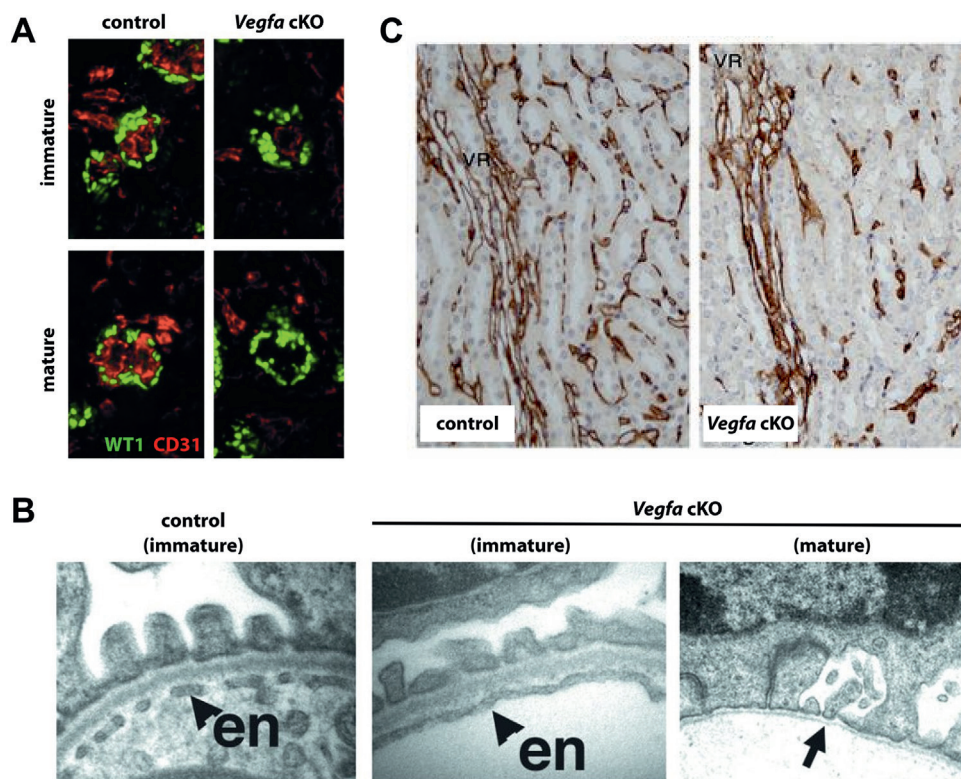


Fig. 1.19 VEGF-A is essential for the development of the glomerular and peritubular capillaries. (A) *Vegfa* inactivation in podocytes leads to recruitment of fewer endothelial cells in the glomerulus which are subsequently lost. Podocytes and endothelial cells are stained with WT1 (green) and CD31 (red), respectively. (B) Transmission electron micrograph showing that in podocyte-specific *Vegfa* knockouts, the glomerular endothelium (en) lack fenestrae (arrowheads) and eventually detach leaving the glomerular basement membrane bare (arrow). (C) *Vegfa* ablation in renal tubules leads to a significant reduction in peritubular capillaries (stained brown for CD34). VR, Vasa recta. Panels (A) and (B) were adapted from Eremina V, et al. Glomerular-specific alterations of VEGF-A expression lead to distinct congenital and acquired renal diseases. *J Clin Invest.* 2003;111:707–716. Panel (C) adapted from Dimke H. Tubulovascular cross-talk by vascular endothelial growth factor A maintains peritubular microvasculature in kidney. *J Am Soc Nephrol.* 2014;26:1027–1038.

peritubular capillaries. Ablation of *Vegfa* in renal tubules did not cause gross histologic disturbance in the kidney but lead to a dramatic reduction of peritubular capillaries and an abnormal elevation in renal erythropoietin production (see Fig. 1.19).³⁷⁸ This uncovered an important tubulo vascular crosstalk involved in the promotion of erythrocyte development by the kidney.

A second major receptor tyrosine kinase (RTK) signaling pathway required for maturation of developing blood vessels is the angiopoietin–Tie signaling system. Angiopoietin 1 (*Angpt1*) stabilizes newly formed blood vessels and is associated with loss of vessel plasticity and concurrent recruitment of pericytes or vascular support cells to the vascular wall.³⁷⁹ The molecular switch or pathway leading to vessel maturation through the activation of Tie2 (encoded by the gene *Tek*, the major receptor for Angpt1) is not known and appears to be independent of the platelet-derived growth factor (PDGF) signaling system that is required for pericyte recruitment. The importance of Angpt1 in promoting the development of the renal microvasculature was first suggested based on observations that exogenous Angpt1 enhances the growth of interstitial capillaries in mouse metanephric organ cultures.³⁸⁰ Because *Angpt1*-null mice perish embryonically at around E12.5, the *in vivo* role of *Angpt1* during renal development was gleaned using an inducible knockout strategy.^{381,382} Ablation of *Angpt1* deletion at around E10.5 results in general dilation of renal blood vessels, including the glomerular capillaries sometimes observed as simplified single enlarged loops.³⁸² A marked reduction of mesangial cells was also observed in *Angpt1*-deficient mutants. Without *Angpt1*, a few endothelial cells are seen detached from the glomerular basement membrane. In contrast, it is proposed that angiopoietin 2 (*Angpt2*) functions as a context-dependent antagonist of the Tie2 receptor.^{383,384} Consistent with this hypothesis is the fact that overexpression of *Angpt2* in transgenic mice results in a phenotype similar to the *Angpt1* or *Tek* knockout mice. Angpt1, Angpt2, Tie2, and the orphan receptor Tie1 are all expressed in the developing kidney.^{385–389} Whereas, *Angpt1* is quite broadly expressed in condensing mesenchyme, podocytes, and tubular epithelial cells, *Angpt2* is more restricted to pericytes and smooth muscle cells surrounding cortical and large vessels as well as in the mesangium. *Angpt2*-null mice are viable but exhibit defects in peritubular cortical capillary development.³⁹⁰ Podocyte-specific overexpression of *Angpt2* causes proteinuria and increased apoptosis in glomerular capillaries.³⁹¹ Both angiopoietin ligands function in concert with VEGF-A, although the precise degree of crosstalk between these pathways is still under investigation. VEGF-A and Angpt2, for example, have been shown to cooperate in promoting endothelial sprouting. Chimeric studies showed that Tie1 is required for the development of the glomerular capillary system because *Tie1*-null cells fail to incorporate in the glomerular endothelium.³⁹²

Renal vascular development also relies on a third tyrosine kinase-dependent signaling mediated by ephrins and Eph family receptors, which are better known for their involvement in axon guidance and specification of arterial and venous cell fates.^{393,394} Ephrins and their cognate receptors are expressed widely during renal development. Overexpression of *Ephb4* leads to defects in glomerular arteriolar formation, whereas conditional deletion of *Efnb2* (*EphrinB2*) from perivascular smooth muscle cells and mesangial cells leads to glomerular vascular abnormalities.^{395,396} How this occurs

is not entirely clear as *Efnb2* has a dynamic pattern of expression in the developing glomerulus, beginning in podocyte precursors and rapidly switching to glomerular endothelial cells and mesangial cells.³⁹⁷

Dysregulation of BMP within the podocyte compartment also results in glomerular vascular defects. Overexpression of BMP4 leads to defects in endothelial and mesangial recruitment, whereas overexpression of noggin, a natural BMP2 antagonist, leads to collapse of the glomerular tuft.^{398,399} *Bmp4* haploinsufficiency, on the other hand, leads to dysplastic kidneys and glomerular cysts with collapsed capillary tufts.³⁹⁹ Additional studies are required to fully understand the role of this family of growth factors in glomeruli.

An additional pathway that is likely to play a role in glomerular endothelial development and perhaps of the entire renal vasculature is the SDF1–CXCR4 axis. CXCR4, a G-protein-coupled chemokine receptor, is strongly expressed in endothelial cells. SDF1 (encoded by the gene *Cxcl12*), the only known ligand for CXCR4, is expressed in a dynamic segmental pattern in podocytes and later in the mesangial cells of the glomerulus.⁴⁰⁰ Embryonic deletion of either *Cxcl12* or *Cxcr4* does not preclude nephrogenesis but results in the defective formation of blood vessels, notably an abnormal patterning of the renal vasculature and the development of a simplified and dilated glomerular capillary tuft.⁴⁰¹ Genetic loss of CXCR7, which is thought to act as a decoy receptor for SDF1, interestingly phenocopies defective development seen in SDF1 and CXCR4-mutant mice. Unlike CXCR4, CXCR7 is specifically expressed by podocytes and not endothelial cells.⁴⁰² It has been proposed that CXCR7, acting as a scavenger receptor, establishes an SDF1 morphogen gradient preventing feedback inhibition of CXCR4 receptor expression in target cells such as the endothelium. Consistent with this, inactivation of *Cxcr7* distinctively causes downregulation of *Cxcr4* expression in the renal cap mesenchyme and the glomerular tuft. Thus the spatial regulation of SDF1–CXCR4 signaling appears to be important for the normal development of the glomerular vasculature.

Two transcription factors belonging to the large Sry-related HMG box (Sox) gene family, named *Sox17* and *Sox18*, have distinctive and overlapping expression in vascular endothelial cells.⁴⁰³ Complete loss of *Sox17* is embryonic lethal in mice due to endodermal dysmorphogenesis.⁴⁰⁴ In mice, *Sox18* ablation results in a mild coat defect but does not cause cardiovascular abnormalities.⁴⁰⁵ Nevertheless, a point mutation in *SOX18* in humans has been implicated in HLT (hypotrichosis-lymphedema-telangiectasia) syndrome, which affects hair, lymph, and blood vessel vasculature.⁴⁰⁶ The more severe consequence of the human *SOX18* mutation compared to the null mutation in mice was suggested to be due to a dominant-negative effect. *Sox17*, however, shows haploinsufficiency in a homozygous *Sox18* background, affecting neovascularization in kidneys, liver, and the reproductive system and causing early postnatal lethality.⁴⁰³ Kidneys from *Sox17/Sox18* double-null mutant mice have hypoplastic and atrophying medullary regions. In these compound *Sox17/Sox18* mutants, the radiating outer medullary vascular bundles of the vasa recta are missing without apparent abnormalities in the inner medullary or cortical regions. These defects within the outer medullary region result in variable degrees of hydronephrosis. Interestingly, midgestational loss of both Angpt1 and Angpt2 ligands or their cognate receptor Tie2 leads to rarefaction of the medullary capillary plexus and

the absence of outer medullary vascular bundles, particularly the fenestrated ascending vasa recta, leading to urinary concentration defects and interstitial fluid retention that culminates in the formation of interstitium-derived medullary cysts (Fig. 1.20).⁴⁰⁷ This raises an intriguing possibility that Sox17 and Sox18 pathways either converge or intersect with Angpt1/Angpt2-Tie2 signaling in coordinating late-stage angiogenesis in kidneys.

A least understood component of the renal vasculature are the lymphatic vessels. Similar to other organs, the renal lymphatics play important roles in interstitial fluid homeostasis and the regulation of the immune response. Lymphatic vessels have been identified surrounding the renal artery in the hilum, in the arcuate and interlobular arteries in the cortex, and are also found in the renal capsule.^{408–410} These lymphatic

vessels are known to express the hyaluronan receptor Lyve1.^{407,411} The medulla was previously thought to lack lymphatic vessels. However, more recently a study of angiopoietin-Tie2 signaling in the kidney reveals that the ascending vasa rectae represent hybrid vessels that express markers of both blood (*Cd34*, *Emcn*, *Pecam1*, and *Plvap*) and lymphatic endothelial cells (*Prox1* and *Vegfr3*) (see Fig. 1.19).⁴⁰⁷ This underscores a novel lymphatic circuit for the drainage of medullary interstitial fluid as part of the osmoregulation of urine. VEGF-C signaling via the receptor VEGFR3 is essential for lymphangiogenesis.^{412–415} *Vegfr3* deficiency in adult mice causes the extravasation of fibrinogen into the renal interstitium.⁴¹⁶ How the renal lymphatic vasculature is particularly remodeled upon inactivation of VEGF-C/VEGFR3 signaling remains unknown.

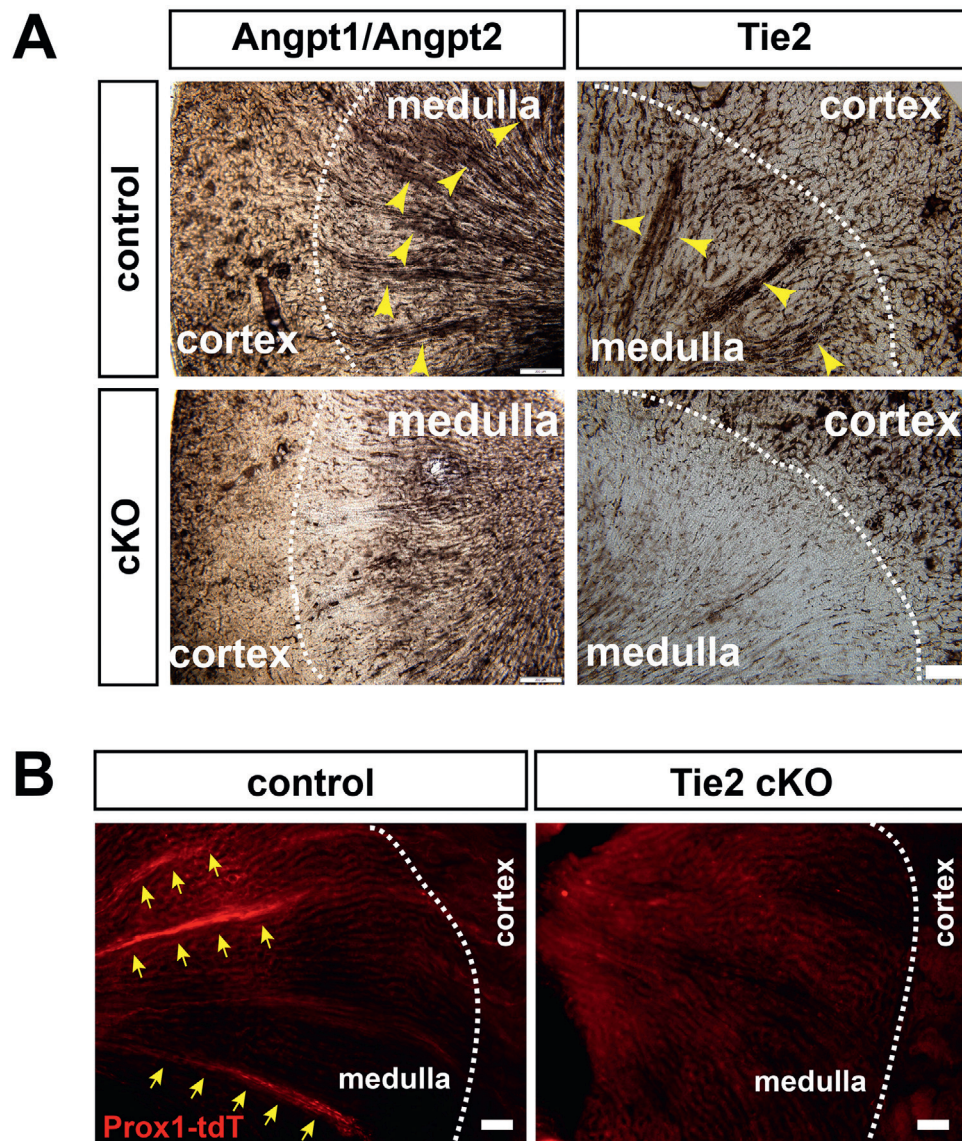


Fig. 1.20 Ascending vasa recta development depends on angiopoietin-Tie2 signaling. (A) Midgestational (E16.5) loss of both Angpt1 and Angpt2 or their cognate receptor Tie2 (Tek) leads to loss of vasa recta bundles (yellow arrowheads) in the outer renal medulla, particularly the ascending vasa rectae. (B) Ascending vasa rectae are novel lymphatic-like vessels expressing the lymphatic-specific Prox1 transcription factor as seen in transgenic reporter mice expressing the fluorophore tdTomato under the control of the *Prox1* promoter. *Prox1-tdT*⁺ vascular bundles representing the ascending vasa rectae are absent in kidneys of Tie2-deficient mouse mutants. *cKO*, Conditional knockout; *Prox1-tdT*, Prox1-tdTomato transgene. Adapted with permission from Kenig-Kozlovsky, Scott RP, Onay T, et al. Ascending vasa recta are angiopoietin/Tie2-dependent lymphatic-like vessels. *J Am Soc Nephrol*. 2018;29:1097–1107.

RENIN CELLS AND THE JUXTAGLOMERULAR APPARATUS

The juxtaglomerular apparatus consists of cells that line the afferent arteriole, the macula densa cells of the distal tubule, and the extraglomerular mesangial cells that are in contact with intraglomerular mesangium.⁴¹⁷ Renin-expressing cells may be seen in arterioles in early mesonephric kidneys in 5-week human fetuses and in metanephric kidneys by week 8, at a stage prior to hemodynamic flow changes within the kidney, and are derived from *Foxd1*-expressing stromal mesenchyme.⁴¹⁸ Renin-expressing cells reside within the MM and give rise not only to juxtaglomerular cells but also to mesangial cells.^{419,420}

The only known substrate for renin, angiotensinogen, is converted to angiotensin I and angiotensin II by angiotensin-converting enzyme (ACE).⁴²¹ The renin-angiotensin-aldosterone axis is required for normal renal development. In humans, the use of ACE inhibitors during pregnancy has been associated with congenital defects including renal anomalies.^{422,423} Two subtypes of angiotensin receptors exist: AT1 receptors are responsible for most of the classically recognized functions of the renin-angiotensin system (RAS), including pressor effects and aldosterone release mediated through angiotensin; functions of the type 2 receptors have been more difficult to characterize, but generally seem to oppose the actions of the AT1 receptors.⁴²⁴ Genetic deletion of angiotensinogen or ACE results in hypotension and defects in formation of the renal papilla and pelvis.^{256–259} Humans have one *AT1* gene whereas mice have two: *Agtr1a* and *Agtr1b*. Mice carrying a knockout for either AT1 receptor alone exhibit no major defects,^{425,426} whereas combined deficiency phenocopies the angiotensinogen and ACE phenotypes.^{260,261} Although AT2 receptor (*Agtr2*) expression is markedly upregulated in the embryonic kidney, genetic deletion of the AT2 receptor does not cause major impairment of renal development.^{427,428} However, an association between *Agtr2*-deficiency and malformations of the collecting duct system, including vesicoureteral reflux and ureteropelvic junction obstruction, has been reported.⁴²⁹

MicroRNAs (miRNAs) are regulatory RNAs that act as antisense posttranscriptional repressors by binding the 3' untranslated region of target mRNAs. Eukaryotes express hundreds of miRNAs that can regulate thousands of mRNAs, and they have been shown to play an important role in development and disease, including in differentiation, signaling pathways, proliferation, apoptosis, and tumorigenicity. Dicer1 is an endoribonuclease that processes precursor miRNAs. Deletion of *Dicer1* from renin-expressing cells results in severely reduced number of juxtaglomerular cells, reduced renin production, and lower blood pressure. The kidney develops severe vascular abnormalities and striped fibrosis along the affected blood vessels, suggesting that miRNAs are required for normal morphogenesis and function of the kidney.⁴³⁰

Gene promoter analysis indicates that renin expression is dependent on the Notch signaling pathway. The intracellular domain of Notch (NIC) and the transcription factor Rbpj bind and cooperatively stimulate reporter gene expression from the renin promoter.⁴³¹ Genetic studies, however, indicate that Notch signaling has a broader role in the juxtaglomerular apparatus.⁴³² The conditional ablation of *Rbpj* in renin cells results in severe paucity of juxtaglomerular cells with

consequential decrease in overall renin expression and the development of lower blood pressure. Lack of increase in apoptosis in *Rbpj* conditional mutant kidneys suggests that *Rbpj* may have altered the cell fate specification of renin cell precursors.

PODOCYTE DEVELOPMENT

Presumptive podocytes are located at the proximal end of the S-shaped body, lining the emerging vascular cleft (Fig. 1.21). Immature podocytes are simple columnar epithelia expressing E-cadherin (Fig. 1.22). Postmitotic mature podocytes, on the other hand, normally lose E-cadherin expression and atypically express vimentin, an intermediate filament protein more common among mesenchymal cells but absent in most epithelial cells. The most distinctive morphologic feature of a fully differentiated podocyte is its arborized and stellate appearance (Fig. 1.22). Podocytes ensheath the glomerular capillaries with their foot processes, effectively forming the final layer of the glomerular filtration barrier. Foot processes emanating from adjacent podocytes interdigitate in *trans* and form a unique and porous intercellular junction called the slit diaphragm through which primary urinary filtrate passes. Three-dimensional reconstruction of podocyte ultrastructure obtained by block-face scanning electron microscopy reveals the morphologic transformation of podocytes during development and the formation of interdigitating foot processes.^{22,433} Columnar-shaped immature podocytes are linked by tight and adherens junctions that progressively migrate from the apical to the basal side. Once the junctional complex has descended close to the basement membrane, podocytes begin to flatten, spread, and interdigitate with short primitive foot processes underneath the junctions. As the primitive processes grow, the tight and adherens junctions relocate from the cell body to between the processes forming the immature foot processes. Finally, the junctional complexes are gradually replaced with slit diaphragms, resulting in mature foot processes.

The transcription factors *Wt1*, *Tcf21*, *Maifb*, *Foxc2*, and *Lmx1b* are highly expressed by developing podocytes and are important for the elaboration of podocyte foot processes and the establishment of slit diaphragms.* Complete loss of *Wt1* leads to renal agenesis.³⁹ However, specific loss of a *Wt1* splice isoform results in poor development of podocyte foot processes.⁴³⁶ The *Wt1*-null phenotype in mice can also be rescued using a yeast artificial chromosome containing the human *WT1* gene and, depending on the level of expression of WT1, the mice developed a range of glomerular pathologies ranging from crescentic glomerulonephritis to mesangial sclerosis, clinical features observed in Denys-Drash syndrome arising from a mutant *WT1* allele in humans.⁴³⁷ Transgenic mice expressing a Denys-Drash-mutant *Wt1* allele under the regulation of a podocyte-specific promoter also developed glomerular disease with abnormalities observed in the adjacent endothelium.⁴³⁸ Genome-wide analysis of the *Wt1* targets in podocytes reveals that *Wt1* autoregulates its own transcription and acts as master regulator of a complex transcriptional network that regulates podocyte development, structure, and function, including transcription factors (*Lmx1b*, *Tcf21*, *Maifb*,

*References 39, 83, 339, 340, 434, and 435.

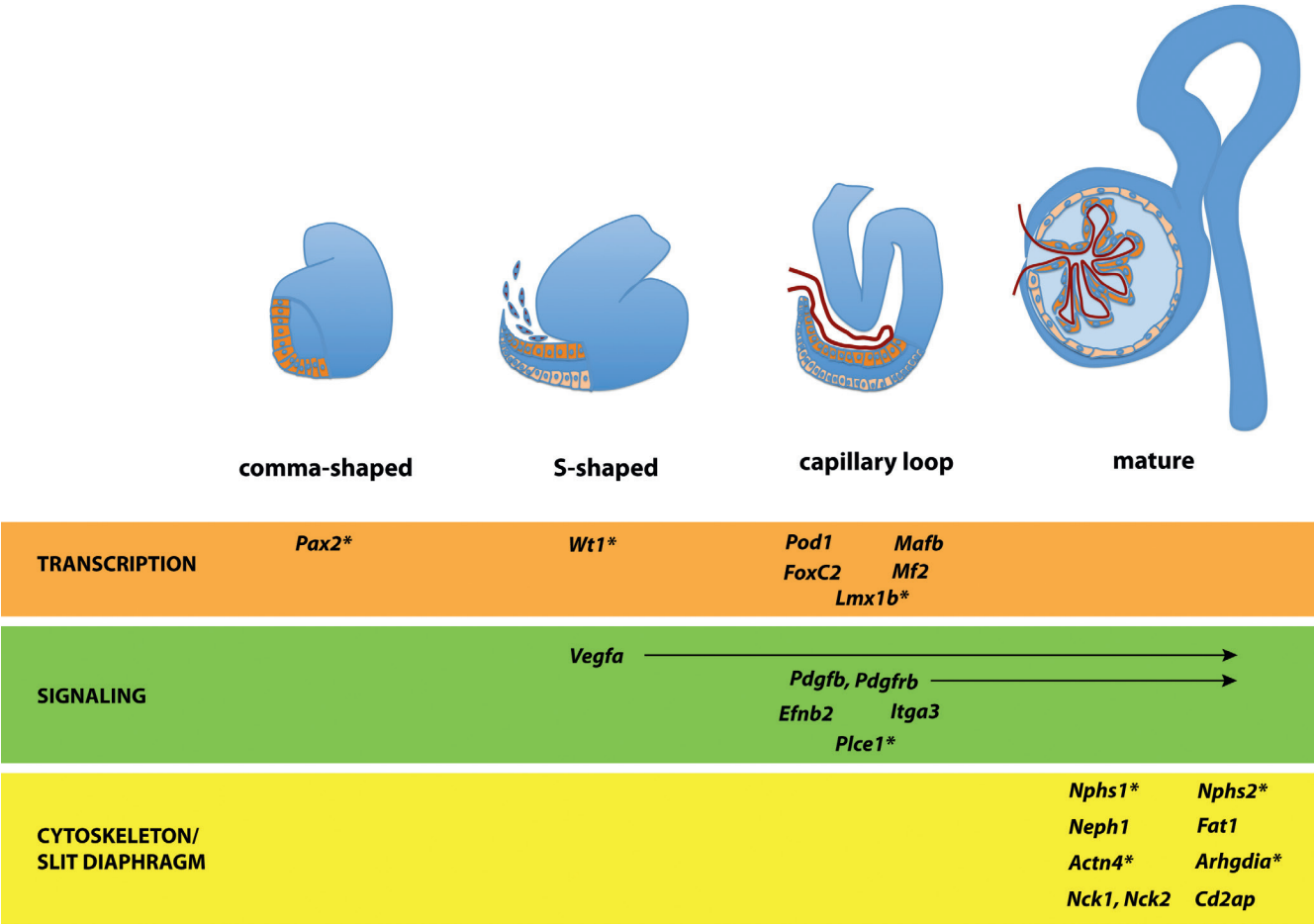


Fig. 1.21 Molecular basis of glomerular development. Key factors are shown along with the time point where major effects were observed in knockout or transgenic mouse studies. Many factors play roles at more than one time point. Genes identified as mutated in patients with glomerular disease are marked by asterisks.

Tead1, *Foxc1*, and *Foxc2*) and genes strongly linked to podocyte dysfunction and nephrotic diseases (*Actn4*, *Arhgap24*, *Cd2ap*, *Col4a3*, *Col4a4*, *Lamb2*, *Nphs1*, *Nphs2*, and *Plce1*).⁴³⁹ Inactivation of *Lmx1b*, *Tcf21*, *Mafb*, and *Foxc2* causes podocytes to remain as cuboidal epithelia and failure to spread on the glomerular capillary bed.^{83,339,434,440} *Tcf21* likely acts upstream of *Mafb* as the latter is downregulated in *Tcf21*-null mice.⁴³⁴ Loss of *Mafb* and *Lmx1b* reduces the expression of *Nphs1* (nephrin) and *Nphs2* (podocin), whereas the absence of *Foxc2* causes the specific downregulation of *Nphs2* and $\alpha3\alpha4\alpha5$ (IV) collagen.^{83,434,441} *Lmx1b* mutations are linked to nail-patella syndrome in humans, with a subset of affected individuals manifesting nephrotic disease.^{435,442} *Wt1*, *Tcf21*, *Mafb*, *Foxc2* and *Lmx1b* are expressed from the S-shape stage onward and remain constitutively expressed in adult glomeruli. Proteinuria develops from loss of these genes, thus underscoring the importance of normal podocyte maturation in the establishment of the glomerular filtration barrier.

Genetic studies have also led to the identification of structural proteins crucial for normal podocyte function and the integrity of the glomerular filtration barrier. The seminal discoveries of the causal link between nephrotic diseases and mutations in podocyte-specific genes *Nphs1* and *Nphs2* set the stage for vigorous investigations that led to the appreciation of the key importance of podocytes in renal filtration.^{443,444}

Mutations in *Nphs1*, the gene that encodes for the protein nephrin, are associated with congenital nephropathy of the Finnish variety (CNF), a serious condition that requires early renal replacement therapy.⁴⁴⁴ Glomeruli obtained from infants affected by CNF are devoid of slit diaphragms. Nephrin, a huge transmembrane adhesion molecule with multiple immunoglobulin-like motifs, was shown to be a structural component of the slit diaphragm. *Nphs2*, whose product is the intracellular membrane-bound protein podocin, and is the first gene identified as being linked to steroid-resistant nephrotic syndrome (SRNS).⁴⁴³ Podocin, which interacts with nephrin in cholesterol-rich membrane microdomains (also called lipid rafts), is also a vital and indispensable component of the slit diaphragm.^{445–449} A number of other genes specifically expressed by podocytes have been associated with proteinuric diseases, including *Cd2ap*, *Kirrel* (*Neph1*), *Fat1*, *Actn4*, *Trpc6*, *Myo1e*, *Arhgap24*, *Arhgdia*, *Rhpn1*, *Inf2*, *Coq2*, *Coq6*, *Plce1*, and *APOL1*.^{77,78,80,82,450–464} The products of these genes are either integral parts of the slit diaphragm complex or direct interacting partners of the complex, while the remainder are important in regulating the development, viability, cytoskeleton, and distinctive morphology of podocytes (Fig. 1.23).

The topologic organization of slit diaphragm components remains unknown, but it is likely that the larger adhesion molecules nephrin and Fat1 could be bridging juxtaposed

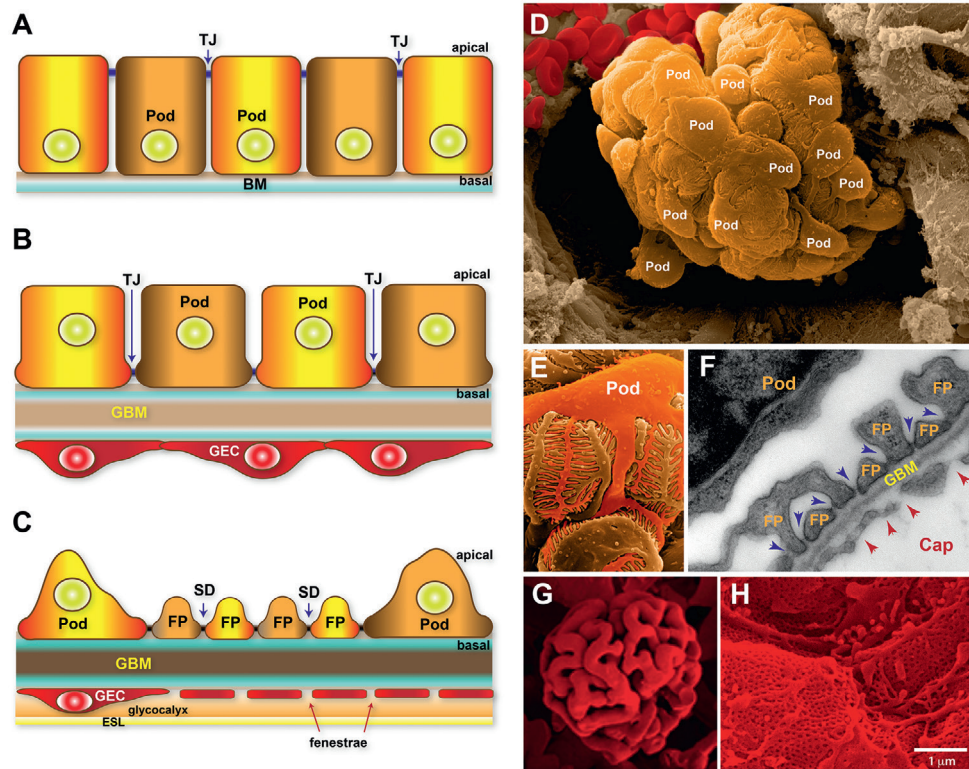


Fig. 1.22 Maturation of the glomerular filtration barrier. (A) In S-shaped bodies, presumptive podocytes (*Pod*) are columnar epithelial cells conjoined by apically localized tight junctions (*TJ*). (B) At the capillary loop stage, the basement membranes (*BM*) of podocytes and glomerular endothelial cells (*GEC*) fuse forming the glomerular basement membrane (*GBM*). During this stage, the podocytes begin to spread, and their cell junctions relocate more basally. (C) As the glomerulus matures, podocytes lose their cuboidal morphology and have developed elaborate foot processes (*FP*) that interdigitate with processes from neighboring podocytes. The podocyte cell junctions have been transformed into slit diaphragms (*SD*) that link juxtaposed FPs. GECs also flatten, develop fenestrae, and become covered by a glycocalyx that, together with absorbed plasma components, form the endothelial cell surface layer (*ESL*). The GBM is now a unified BM between podocytes and the GECs. (D) Scanning electron micrograph (SEM) of a cracked glomerulus exposing the capillary tuft covered by podocytes (gold). (E) Higher magnification view of a podocyte in situ revealing the elaborate FPs interdigitated with FPs from other podocytes. (F) Transmission electron micrograph of a glomerulus showing the SD (blue arrows) between interdigitating FPs and the endothelial fenestrae (red arrows) on the opposite side of the GBM (Cap, glomerular capillary lumen). (G) SEM of a resin cast of the renal vasculature demonstrating the highly convoluted glomerular capillary tuft. (H) A view inside an exposed glomerular endothelial lumen showing its highly fenestrated surface. Adapted from Scott RP, Quaggin SE. Formation and maintenance of a functional glomerulus. In Little, M. H. (ed.). *Kidney Development, Disease, Repair and Regeneration*. San Diego: Academic Press; 2016.

foot processes (see Fig. 1.22).^{459,463–467} Smaller adhesion molecules within the slit diaphragms such as Neph1, Neph3, and P-cadherin may more likely associate in *cis* (within the same foot process surface).^{468–470} Nephrin and the related protein Neph1 are known to interact with the polarity complex proteins Par3, Par6, and aPKC λ/ι , indicating a coregulation between the polarized cell structure of podocytes and the compartmentalized assembly of the slit diaphragm complex along the foot processes.⁴⁷¹ Conditional inactivation in podocytes of aPKC λ/ι or the small GTPase Cdc42, which positively regulates the Par3/Par6/aPKC λ/ι complex, causes proteinuria characterized by abnormal pseudo-slit diaphragms formed between effaced foot processes (Fig. 1.24).^{472–474} It has also been shown that inactivation of aPKC λ/ι can specifically inhibit the localization of nephrin to the cell surface.⁴⁷⁵

Terminal foot processes of podocytes are longitudinally supported by parallel bundles of actin, setting them apart from the larger primary processes that have a microtubule-based backbone.²¹ The stereotypical response of podocytes to injury either through chemical insults or a detrimental gene mutation is effacement of foot processes. In effaced foot processes, the actin cytoskeleton has been remodeled

into a mesh-like network of randomly oriented filaments.⁴⁷⁶ Genetic and biochemical studies provide evidence that the slit diaphragm is functionally coupled to the actin cytoskeleton, and that perturbation of this relationship results in compromised renal filtration and proteinuric disease. Nck adaptor molecules (Nck1 and Nck2) are known to link tyrosine kinase receptors to signaling molecules that regulate the actin cytoskeleton. Podocytes lacking *Nck1* and *Nck2* are effaced and form abnormal slit diaphragms.⁶⁴ Cell culture studies reveal that clustered nephrin is phosphorylated at its cytoplasmic tail by the kinase Fyn, creating distinctive phosphotyrosine sites where Nck1 and Nck2 adaptors can bind directly. The association between nephrin and Nck adaptors consequently recruits N-WASP and the Arp2/3 protein complex to mediate localized polymerization of actin.^{64,477} Loss of Fyn causes congenital nephrosis, whereas podocyte-specific inactivation of *Wasl*, the gene encoding for N-WASP, leads to proteinuric disease.^{478,479} It is also very likely that Nck adaptors can mediate the adhesion of podocytes to the glomerular basement membrane by virtue of their ability to interact with the PINCH–ILK–integrin complex.^{480–482} Cdc42, in addition to its role in podocyte polarization, has also been shown to be

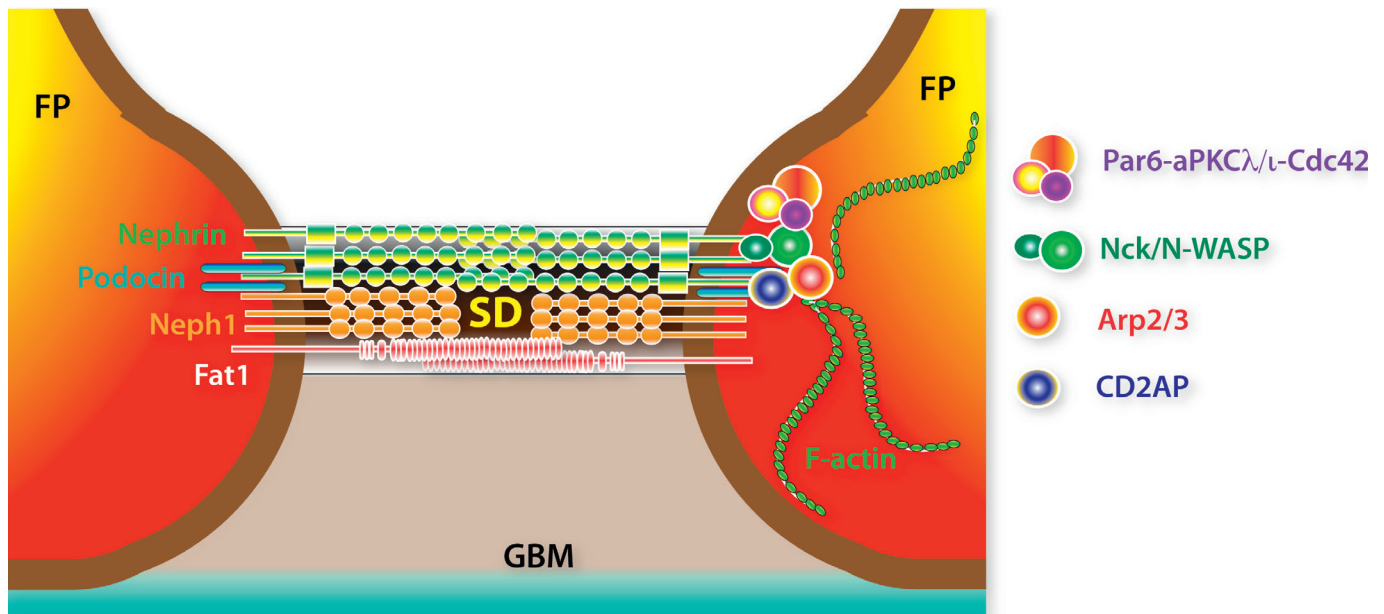


Fig. 1.23 Structural overview of the slit diaphragm. An oversimplified diagram depicting the major adhesion receptors comprising the SD and how they are possibly integrated with the actin cytoskeleton of podocyte foot processes. FP, Podocyte foot process; SD, slit diaphragm; GBM, glomerular basement membrane. Adapted from Scott RP, Quaggin SE. Formation and maintenance of a functional glomerulus. In Little, M. H. (ed.). *Kidney Development, Disease, Repair and Regeneration*. San Diego: Academic Press; 2016.

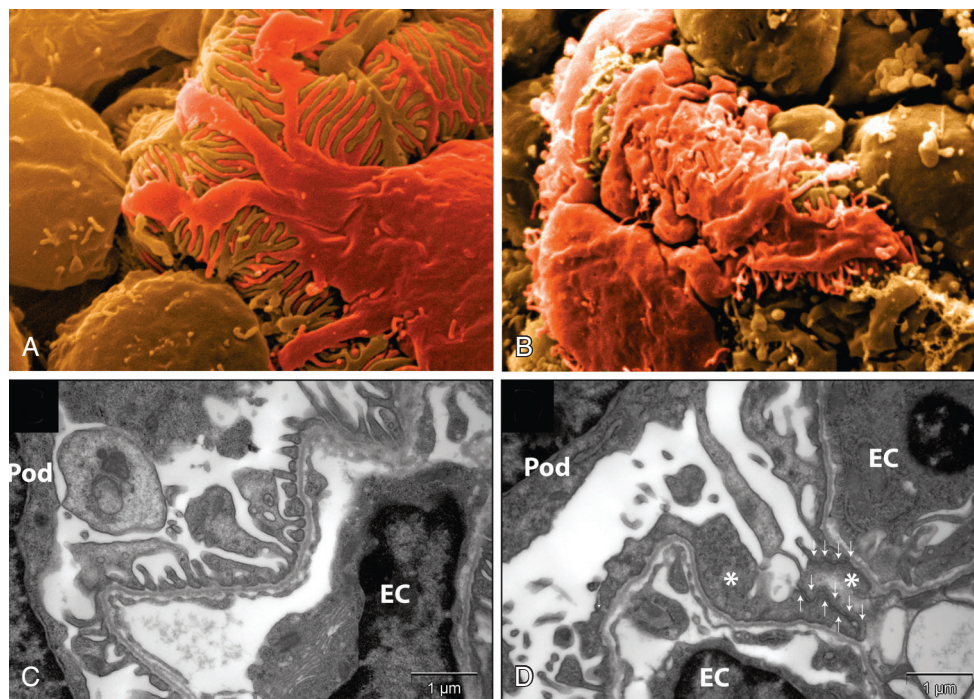


Fig. 1.24 Cdc42 is required for normal podocyte development. (A, B) False-colored scanning electron micrographs of glomeruli from neonatal kidneys: (A) wild-type control podocytes showing normal interdigitated podocytes; and (B) Cdc42-deficient mutant podocytes showing total effacement of foot processes. (C, D) Transmission electron micrographs of sectioned glomeruli: (C) wild-type control showing regularly interdigitating podocyte foot processes and basolateral slit diaphragms; and (D) podocyte-specific Cdc42-deficient mutant showing mislocalized cell junctions (arrows) between effaced foot processes (asterisk). EC, Endothelial cell; Pod, podocyte. Adapted from Scott RP, Hawley SP, Ruston J, et al. Podocyte-specific loss of Cdc42 leads to congenital nephropathy. *J Am Soc Nephrol*. 2012;23:1149–1154.

required for the coupling of actin polymerization to nephrin. CD2AP, a molecule known to stabilize actin microfilaments, is also indispensable in podocytes.^{483–485} Mutations in *Actn4*, *Arhgdia*, *Arhgap24*, *Inf2*, and *Myo1e*, whose protein products are established regulators of the actin cytoskeleton, are also

implicated in pathologic transformation of podocytes and proteinuric diseases.^{77,78,80,451–453,455–458}

It has been proposed that the slit diaphragm likely functions in mechanotransduction in podocytes, allowing them to modulate renal filtration in response to hemodynamic changes

within the glomerular microenvironment.^{486,487} *MEC-2*, the *C. elegans* homologue of podocin, is a component of the touch receptor complex coupled to the ion channel *MEC-4/MEC-10*.⁴⁸⁸ Loss of *MEC-2* in worms leads to insensitivity to touch.⁴⁸⁸ In podocytes, the ion channel *Trpc6* forms an integral component of the slit diaphragm directly interacting with podocin.⁴⁸⁷ In diverse cell types such as myocytes, cochlear hair cells, and sensory neurons, the *Trpc6* channel opening is gated by mechanical stimuli. Human mutations in *TRPC6* have been strongly linked to proteinuria.^{460,489,490}

A novel lipid-dependent signaling pathway involving the VEGF receptor *Flt1* (VEGFR1) has been described recently as crucial for the regulation of podocyte actin cytoskeleton and the maintenance of slit diaphragms. Genetic removal of *Flt1* from podocytes leads to foot process effacement and proteinuria.⁴⁹¹ Intriguingly, a kinase-inactive mutant of *Flt1* is able to support normal podocyte development and function. In vivo, *Flt1* is cleaved, releasing a soluble ectodomain (sFlt1). The secreted sFlt1 has been shown to act as an autocrine factor in podocytes, associating with glycosphingolipids, and mediating podocyte cell adhesion, nephrin phosphorylation, and actin polymerization. It has been proposed that sFlt1 may function physiologically to stabilize the slit diaphragms and the attachment of podocytes to the glomerular basement membrane.

Three groups generated mice carrying a podocyte-specific deletion of *Dicer1*, thereby interfering with the production of functional miRNAs.^{492–494} Podocyte-specific *Dicer1* knockout mice develop albuminuria by 3 weeks of age and rapidly

progress to end-stage renal failure by approximately 6 weeks. A number of potential miRNA targets were identified, but their functional significance in the podocyte in vivo is yet unknown.

From all of these studies, it follows that intrinsic proteins and functions of podocytes play a key role in the development and maintenance of the permselective properties of the glomerular filtration barrier. Podocytes also function as vasculature support cells producing VEGF-A and other angiogenic growth factors. It is likely that endothelial cells also produce hitherto unknown trophic factors that promote terminal differentiation and survival of podocytes.

GLOMERULAR BASEMENT MEMBRANE DEVELOPMENT

The mature glomerular basement membrane (GBM) is a fusion of the extracellular matrices (ECM) of podocytes and glomerular endothelial cells. The GBM is a highly organized and compositionally complex matrix whose abundant components include collagens (types I, IV, VI, and XVIII), laminins ($\alpha 5$, $\beta 2$, and $\gamma 1$), nidogen-1, heparan sulfate proteoglycans (agrin and perlecan), and tubulointerstitial nephritis antigen-like protein (Fig. 1.25).^{495–497} The GBM is an essential part of the glomerular filtration barrier, functioning as an intermediary sieving matrix and a sink for secreted trophic and signaling factors, as well as mediating cellular communication between the glomerular endothelium and podocytes. Furthermore, adhesive cell–ECM interactions among the

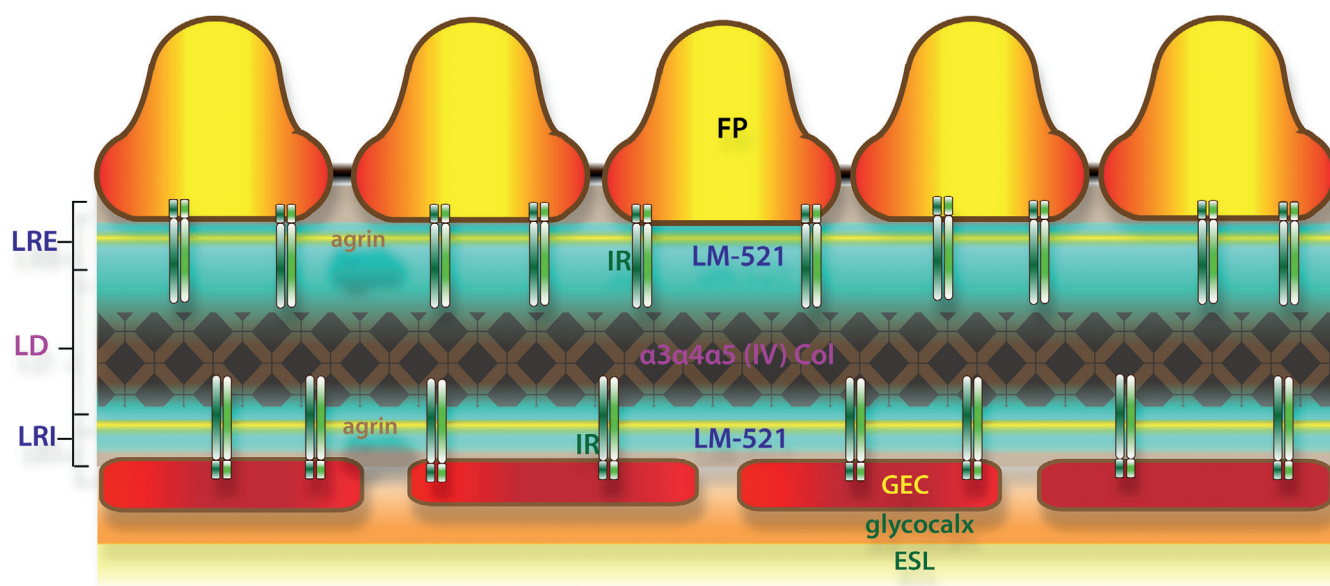


Fig. 1.25 Stratified organization of the glomerular basement membrane. A model of the organization of a mature GBM. The laminin complex LM-521 (pale cyan) and agrin (yellow) form two distinct layers underneath the basal aspects of podocytes and GECs. $\alpha 3\alpha 4\alpha 5$ -Type IV collagens (hatched brown area) are more centrally distributed but are thought to be closer to the GEC. The laminin and type IV collagen complexes have significant overlap. Epitope mapping of integrin- $\beta 1$ suggests that integrin receptors (IR) on podocytes are normally separate and less likely to interact with the type IV collagens in the mature human GBM. The morphologically distinct layers of the GBM in transmission electron micrographs are roughly demarcated. ESL, Endothelial surface layer; FP, podocyte foot process; GBM, glomerular basement membrane; GEC, glomerular endothelial cell; LD, lamina densa; LM, laminin; LRE, lamina rara externa; LRI, lamina rara interna. Based on STORM imaging by Suleiman HL, Zhang L, Roth R, et al. Nanoscale protein architecture of the kidney glomerular basement membrane. *Elife* 2013;2:e01149. Adapted from Scott RP, Quaggin SE. Formation and maintenance of a functional glomerulus. In Little MH (ed.). *Kidney Development, Disease, Repair and Regeneration*. San Diego: Academic Press; 2016.

GBM, the podocytes, and the glomerular endothelial cells maintain the structural integrity of the glomerular filtration barrier.

Among the major components of the GBM, type IV collagens and laminins have been shown to be the most indispensable, highlighted by proteinuric kidney diseases such as Alport syndrome, Goodpasture disease, and Pierson syndrome. Alport syndrome is linked to a growing list of mutations in the genes *COL4A3*, *COL4A4*, and *COL4A5* encoding for type IV collagen subunits $\alpha3$, $\alpha4$, and $\alpha5$, respectively.^{498,499} Maturation of the GBM requires the replacement of juvenile $\alpha1\alpha1\alpha2$ with $\alpha3\alpha4\alpha5$ type IV collagen trimers, a developmental switch improving the structural resilience of the GBM.⁵⁰⁰ In Alport syndrome, the assembly of heterotrimeric $\alpha3\alpha4\alpha5$ type IV collagen complex is compromised while the $\alpha1\alpha1\alpha2$ type IV collagen complex persists. Because $\alpha3\alpha4\alpha5$ type IV collagen trimers constitute about half the total proteins in the mature GBM, it is not surprising that Alport syndrome GBM is severely distorted.⁵⁰¹ The importance of type IV collagens in the GBM is further underscored in Goodpasture disease, an autoimmune disorder targeting $\alpha3$ type IV collagen subunit.⁵⁰² In Pierson syndrome, mutations in *LAMB2* encoding the laminin $\beta2$ subunit impair the assembly of the laminin complex LM-521 (a trimer formed among laminin- $\alpha5$, $\beta2$, and $\gamma1$ subunits).^{503,504} Deformation of the GBM and proteinuria results from loss of *Lamb2* and *Lama5* (laminin- $\alpha5$) in mice.^{234,505–507}

DEVELOPMENT OF THE MESANGIUM

Mesangial cells grow into the developing glomerulus and come to sit between the capillary loops. Gene deletion studies have demonstrated a critical role for PDGF-B/PDGFR- β signaling in this process (Fig. 1.26). Deletion of the *Pdgfb* gene, which is expressed by glomerular endothelia, or the PDGFR- β receptor gene (*Pdgfrb*), which is expressed by mesangial cells, results in glomeruli with a single balloon-like capillary loop, instead of the intricately convoluted glomerular capillaries of wild-type kidneys. Furthermore, the glomeruli contain no mesangial cells.⁵⁰⁸ Endothelial-cell-specific deletion of *Pdgfb* results in the same glomerular phenotype and shows that production of PDGF-B by the endothelium is required for mesangial migration.⁵⁰⁹ PDGF-B-dependent recruitment of mesangial cells into nascent glomeruli also requires the coreceptor neuropilin-1 (*Nrp1*).⁵¹⁰ *Nrp1*-deficiency within the *Pdgfrb*⁺-cell lineage in mice phenocopies glomerular aneurysm seen upon loss of either *Pdgfb* or *Pdgfrb* (Fig. 1.27). In vitro, mesangial cell chemotaxis, but not cell proliferation and survival in response to PDGF-B, strongly requires the mesangial expression of *Nrp1*. Furthermore, loss of the mesangium resulting from *Nrp1* ablation results in the delamination of the glomerular endothelium and inward buckling of the GBM. These findings altogether underscore the key supportive role of the mesangium for the development and maintenance of a highly convoluted glomerular capillary tuft. Indeed, targeted injury to the mesangium, either with snake venom or with antibodies against *Thy1*, results in secondary damage to the glomerular endothelium and glomerular aneurysms.^{511–515} Mesangial cells and the matrix they produce are required to pattern the glomerular capillary system. Loss of podocyte-derived factors such as VEGF-A also leads to failure of mesangial cell ingrowth, likely through the primary loss of endothelial cells and failure of PDGF-B signaling.³⁷⁶

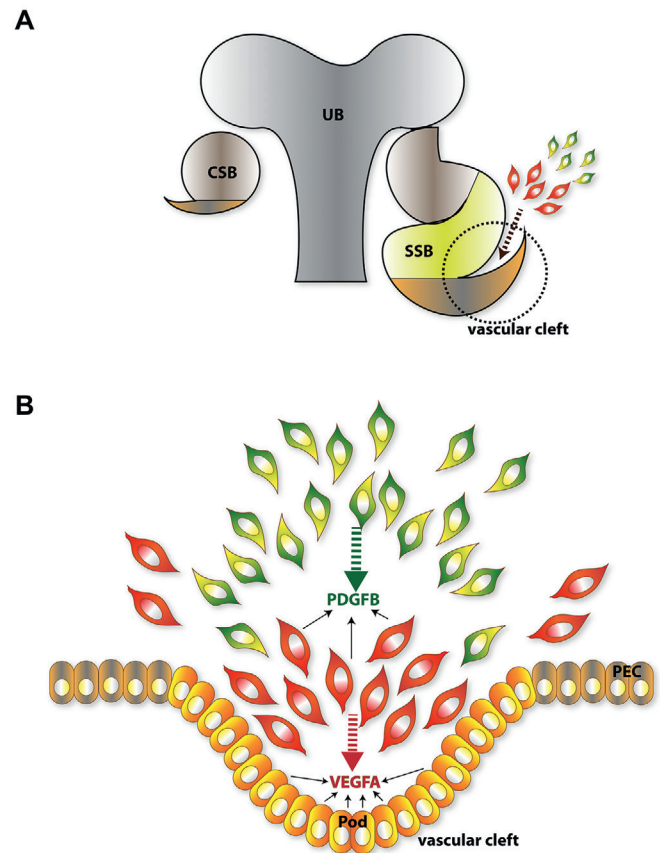


Fig. 1.26 Sequential recruitment of endothelial and mesangial cells into the developing glomerulus. (A) The glomerulus forms around the vascular cleft at the proximal end of S-shaped intermediate nephrons. (B) Within the vascular cleft, presumptive podocytes (*Pod*) secrete VEGFA that attracts VEGFR2-expressing angioblasts (red), which are the precursors of GECs. The angioblasts are followed by mesangial cell precursors expressing PDGFR- β (green), which are attracted by PDGFB secreted by the endothelial cells. UB, Ureteric bud; CSB, comma-shaped body; PEC, parietal epithelial cell; SSB, S-shaped body. Adapted from Scott RP, Quaggin SE. Formation and maintenance of a functional glomerulus. In Little MH (ed.). *Kidney Development, Disease, Repair and Regeneration*. San Diego: Academic Press; 2016.

A number of other knockouts demonstrate defects in both vascular development and mesangial cell ingrowth. Loss of the transcription factors *Tcf21* and *Foxc2* causes defective migration of mesangial cells.^{83,339} Mesangial abnormalities in *Tcf21*- and *Foxc2*-deficient mice are poorly understood in terms of mesangium-specific transcriptional targets of *Tcf21* and *Foxc2*. Nevertheless, these mutant phenotypes highlight the importance of crosstalk between cell compartments within the glomerulus.

DEVELOPMENT OF THE BOWMAN CAPSULE

The outermost cells of the proximal end of the S-shaped nascent nephron are the presumptive parietal epithelial cells that eventually form the Bowman capsule that enclose the glomerular tuft and where primary urinary filtrate collects. Similar to podocytes, parietal precursors are originally cuboidal epithelial cells that progressively flatten and become

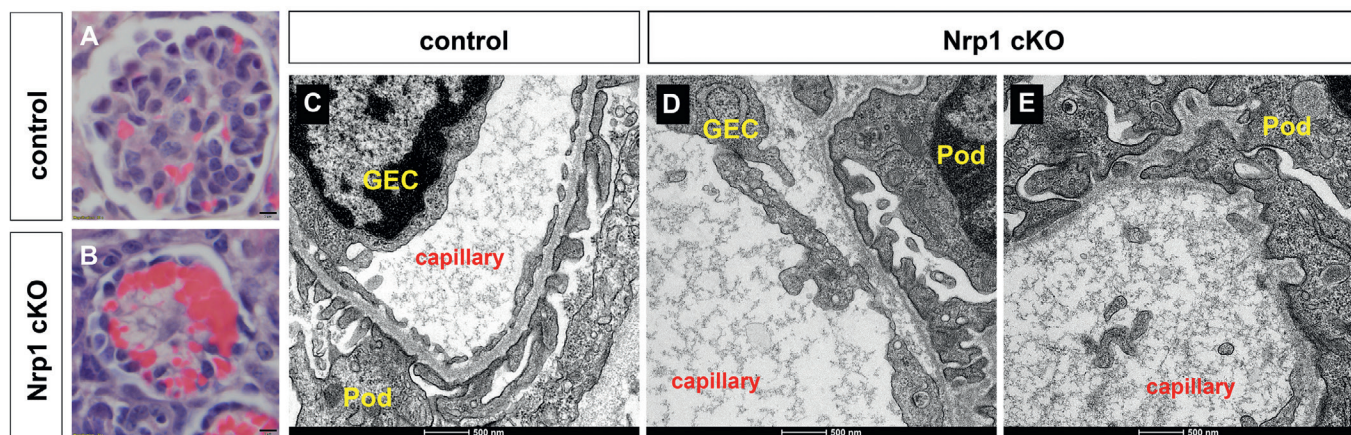


Fig. 1.27 *Nrp1* is required for mesangial cell recruitment in the glomerulus. (A, B) Hypocellularity and glomerular aneurysm due to the absence of the mesangium in *Nrp1*-deficient mice. Representative histological sections of normal control (A) and *Nrp1* conditional knockout (cKO)-mutant kidneys. (C–E) Transmission electron micrographs of glomeruli showing the delamination (D) or complete detachment (E) of the glomerular endothelium from the glomerular basement membrane in *Nrp1*-cKO kidneys. In the absence of *Nrp1* in the *Pdgfrb*⁺ lineage, the glomerular endothelium has impaired development of fenestrae. GEC, Glomerular endothelial cell; Pod, podocyte. Adapted from Bartlett CS, Scott RP, Carota IA, et al. Glomerular mesangial recruitment and function require the co-receptor neuropilin-1. *Am J Physiol Renal Physiol*. 2017;313:F1232-F1242.

squamous. Parietal epithelia express the proteins claudin-1 (*Cldn1*) and claudin-2 (*Cldn2*) that form part of tight junctions, which help contain urinary filtrate.^{516,517} Parietal epithelial cells like podocytes express *Wt1*, although at a reduced level. Whether this differential level of *Wt1* expression affects cell fate commitment to either the podocyte or parietal cell lineage warrants further investigation. *Wt1* is known to attenuate canonical Wnt signaling through an epigenetic mechanism that suppresses *Cttnb1* expression.⁵¹⁸ Specific loss of *Cttnb1* in nephrogenic epithelia from the S-shaped stage causes the formation of glomerular cysts where the Bowman capsule lacks parietal epithelia and are instead formed by podocytes containing notably containing foot processes and slit diaphragms.⁵¹⁹ Parietal cell specification is therefore strongly dependent on canonical Wnt- β -catenin signaling, which sets it apart from podocytes where *Cttnb1* expression is dispensable.⁵²⁰ In the absence of parietal epithelial cells, glomerular capillaries develop poorly while the mislocalized *Vegfa*-expressing podocytes cause formation of ectopic capillaries next to the abnormal Bowman capsule.⁵¹⁹ It can therefore be inferred that parietal epithelial cells may function in compartmentalizing the podocyte–glomerular endothelial cell crosstalk, ensuring that glomerular endothelial cells form a well elaborated capillary tuft covered by an adequate number of podocytes.

NEURAL DEVELOPMENT

Renal vascular tone and urinary functions are regulated by a dense neural network in the kidney that relays bidirectional signals to the brain.⁵²¹ Glomerular filtration rate, renal blood flow, tubular resorption of fluid, electrolytes, and urinary solutes, as well as the secretion of renin, are regulated by sympathetic innervations of the glomeruli, renal tubules and blood vessels.⁵²² Over the past decade, the renal sympathetic innervation has attracted considerable attention after it has been recognized that persistently elevated renal sympathetic nerve activity contributes to the pathogenesis of renal hypertension.^{523,524} In particular, it has been shown that surgical

excision of sympathetic input to the kidneys can alleviate refractory hypertension.^{525,526} The distribution of afferent and efferent nerves to the kidney has been partially mapped but the developmental program involved in establishing them is largely unknown.^{527–529} Fate mapping and molecular studies specifically addressing the origin and development of renal nerves have yet to be reported, although many lessons have been learned about guidance pathways in play in both neural and vascular development in other systems. Neurons, like early blood vessels, appear to closely track the branching UB in cultured embryonic kidneys.^{33,370,530} It can be speculated that the sympathetic innervation and vascularization of the kidney are coordinately synchronized with better known inductive events between the UB and the MM.



Complete reference list available at [ExpertConsult.com](https://www.expertconsult.com).

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BOARD REVIEW QUESTIONS

- Which of the following class of drugs is not recommended for antihypertensive treatment during pregnancy and why? Which aspect of renal development is prominently affected by these contraindicated medications?
 - Adrenergic blockers
 - Calcium channel blockers
 - ACE inhibitors and angiotensin receptor blockers (ARBs)

Answer: c

Rationale: ACE inhibitors/ARBs are teratogens, causing multiorgan congenital malformations (not limited to the genitourinary tract) and fetal death. Angiotensin is important in the normal development of collecting ducts, and its inhibition causes atrophy of the renal papillae and pelvis.

- Fetal vitamin A status is proposed as a factor contributing to variability of kidney size and nephron endowment across populations. Which of the following is not true about the effect of vitamin A on renal development?
 - Vitamin A modulates the pattern of ureteric branching
 - Vitamin A regulates epithelial transformation of the metanephric mesenchyme
 - Vitamin A promotes the conversion of the interstitial stroma into nephrons
 - Vitamin A deficiency can cause paucity in nephrons (oligonephronia)

Answer: c

Rationale: Vitamin A is required for the patterned branching of the ureteric bud (UB) and the epithelialization of the

metanephric mesenchyme (MM) into nephrons. Defective renal branching and nephrogenesis due to vitamin A deficiency can contribute to asymptomatic renal hypoplasia. The interstitial stroma is important for both the synthesis of renal retinoic acid (active vitamin A) and express vitamin A receptors, allowing the stroma to signal to both the UB and the MM to induce their differentiation. The stromal cells are distinct from the precursors of nephrons.

- A patient presenting with recurring urinary tract infection was shown to have duplex kidneys by radiography. Which developmental anomaly is less likely to be the cause?
 - Expanded zone of glial cell-derived neurotrophic factor (GDNF) expression in the metanephric mesenchyme
 - Cystic dilatation of renal tubules
 - Ureteric bud inappropriately induced
 - Misexpression of Ret receptor along the wolffian duct

Answer: b

Rationale: The collecting ducts, the renal calyces, and the ureter are all derivatives of the ureteric bud (UB). The UB is normally induced on a single site on each wolffian duct. Inappropriate emergence of the UB along the wolffian duct, either through abnormal expansion of GDNF expression in the presumptive metanephric mesenchyme or the misplaced expression of Ret along the wolffian duct, can lead to ectopic UBs and duplicated collecting duct systems with attendant errors in ureter-bladder connections, thus causing urinary obstruction and reflux.